MASTER THESIS

Scalable estimation of Retrogressive Thaw Slump occurrence-density across the Arctic on discrete global grids

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Abstract

This thesis presents a scalable framework for estimating Retrogressive Thaw Slumps (RTS) occurrence across the Arctic using Discrete Global Grid Systems (DGGS). By leveraging open satellite and climate data, and employing explainable machine learning models, it addresses the primary research question: identifying key environmental drivers of RTS formation. Doing that, two adjacent research questions are also addressed: evaluating the performance of different ml methods regarding their explainability of feature importances and comparing these models with novel machine learning methods such as embeddings-as-data and AutoML frameworks. A comprehensive analysis of over 190,000 model trainings across multiple DGGS resolutions reveals that winter and spring precipitation, along with decadal trends in thawing and freezing days, are predictors of RTS occurrence. It further shows that the utilization of novel machine learning methods can achieve equal predictive performance as traditional methods with less expert knowledge, however, the explainability of the results especially regarding feature importance drops strongly. The utilization of eSPA, a novel, highly efficient and interpretable cluster-based learner, demonstrates that cheaply learned feature weights correspond to traditional feature importance methods such as SHAP even for large highly multicollinearity datasets.

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Introduction

The Arctic is the fastest-warming land ecosystem on Earth, with surface temperatures increasing nearly four times faster than the global average over the past four decades. (Rantanen et al., 2022). This rapid warming has profound implications for the region's permafrost systems, which are central to the Arctic's hydrological, ecological, and climatic functioning (Biskaborn et al., 2019; Brovkin et al., 2025; Grosse et al., 2011a). Permafrost, which is ground that remains at or below 0°C for at least two consecutive years, is the largest terrestrial carbon reservoir on Earth, storing approximately 1500 Pg of organic carbon (Schuur et al., 2015; Strauss et al., 2025). As permafrost thaws due to rising temperatures, this stored carbon is released into the atmosphere as carbon dioxide and methane, contributing to further global warming (Grosse et al., 2011b; 2011a). This creates a positive feedback loop between climate change and carbon emissions, potentially accelerating climate instability (Schuur et al., 2015).

The widespread thaw of permafrost and the melting of massive ground ice could represent a climate tipping point, a threshold beyond which the system may shift to a new, irreversible state (Turetsky et al., 2019; Winkelmann et al., 2023). While evidence of such a tipping point on a global-scale remains lacking, localized tipping events could be crossed, underscoring the importance of spatial scales in permafrost research (Nitzbon et al., 2024). Current climate models, however, often lack high-resolution representation of the past and present state of permafrost systems, limiting their ability to simulate the dynamics of thaw and associated carbon release accurately (Winkelmann et al., 2023). Closing this data gap through pan-Arctic, high-resolution observations is therefore essential for improving model performance and forecasting future Arctic change.

With the increasing availability of open data global datasets, such as Landsat, Sentinel or ERA5, recent efforts increasingly use remote sensing, climate data, and Machine-Learning (ML) algorithms to observe and model permafrost landscapes at both regional and pan-Arctic scales (Bartsch et al., 2023; Hugelius et al., 2014; Inauen et al., 2025; Nitze et al., 2025; Ran et al., 2021; Zhang et al., 2023). Two trends currently influence the methods of creating such maps and their quality: the increasing open-data access to global datasets and the increasing amount and complexity of numerical methods such as ML or Deep-Learning (DL). However, global datasets often differ in spatial resolution, projection, and temporal coverage, aligning such datasets through reprojection and resampling introduces spatial distortions and biases, complicating large-scale analysis. Discrete Global Grid Systems (DGGSs) offer a robust solution to these challenges by partitioning the Earth's surface into uniquely identified grid cells that are to a certain degree uniform in size and distance to neighboring cells, enabling efficient and unbiased spatial analysis at multiple resolutions, particularly in the Arctic (Sahr & White, 1998).

One of the most important variables for permafrost processes is the ground ice content (Brovkin et al., 2025). Current spaceborne remote sensing technologies cannot penetrate the ground sufficiently to directly observe ground ice at a pan-Arctic scale (Jorgenson & Grosse, 2016; Zwieback et al., 2024). As a result, ground ice content must be inferred indirectly from surface indicators, such as thermokarst landforms, which emerge in response to permafrost degradation. Thermokarst landforms are widespread across ice-rich permafrost regions, particularly where permafrost begins to thaw (Jorgenson & Grosse, 2016). One example for such landforms are Retrogressive Thaw Slump (RTS), which form by the melting of massive ground ice and subsequent thermo-erosion, resulting in large, curved depressions in the ground surface that propagate up slope and can reach depths of several meters (Nesterova et al., 2024). Because RTS only develop when large volumes of ground ice are present, they serve as a direct proxy for the occurrence of massive ground ice and can thus be used to infer ice content and permafrost instability. RTS differ widely in size, shape, depth and optical reflectance and their morphological expression varies regionally, making them difficult to map consistently (Nesterova et al., 2025a; Nitze et al., 2024a). Recent studies utilized DL combined with high resolution satellite imagery to identify the outlines of RTS from space (Dai et al., 2025; Huang et al., 2023; Maier et al., 2025a; Nitze et al., 2025). However, no existing dataset spans the entire Arctic and includes long-term, multi-year observations of RTS occurrence. While initial studies suggest that winter air temperatures and summer precipitation are important climatic drivers of RTS initiation, these findings remain largely based on localized observations at regional spatial scales (Nesterova et al., 2025a). A pan-Arctic understanding of the environmental conditions favoring RTS formation is therefore still missing. **Thus, the overarching research objective of this thesis is to identify potential proxy climate variables for RTS and thus massive ground ice occurrence across various spatial scales.** In order to do this, RTS occurrence maps are created on different DGGS on various resolutions using explainable ML algorithms trained on an comprehensive dataset of environmental variables. The most predictive environmental variables are identified through model interpretability tools, providing insights into the environmental conditions that favor RTS formation.

Permafrost and its mapping

Permafrost is ground, meaning soil, sediment, or rock, that remains frozen at or below 0 °C for at least two consecutive years, with earlier definitions also including the requirement for measurable amounts of ground ice (Everdingen, 1998; Salzmann & Gärtner-Roer, 2017). This frozen layer can be found just below the surface or at depths of several meters and may contain significant amounts of ice. The uppermost layer that thaws during the brief summer period is called the Active Layer. Its depth varies seasonally, growing in the warm months and refreezing in winter, while the permafrost beneath remains frozen year-round (Dobiński, 2020). In the summer, the AL supports forests and tundra vegetation, making the landscape biologically active.

Figure 1 shows a schematic vertical cross-section of a typical permafrost landscape. The diagram illustrates the AL at the surface, the continuous permafrost beneath, and the presence of ice lenses or ice wedges that can form in the frozen ground. The AL thickness can range from a few centimeters in high-latitude tundra to several meters in more temperate permafrost zones (Dobiński, 2020).

Permafrost can be classified as continuous, discontinuous, sporadic, or isolated, depending on the percentage of the landscape it underlies and on the mean annual air temperature (Obu et al., 2019). This classification, however, depends on the scale of observation, resulting in large uncertainties while quantifying and mapping permafrost (Gruber, 2012). Permafrost underlies approximately 13 to 15 million km^2 of the Northern Hemisphere, representing about 14 to 15% of exposed land area, with the broader permafrost region (including areas with some permafrost presence) covering up to 21 to 23 million km^2, or about 22 to 24% of the Northern Hemisphere's land surface (Gruber, 2012; Obu, 2021; Obu et al., 2019). Globally, including the Southern Hemisphere and subsea permafrost, which covers 2.5 million km^2, estimates range from 16 to 21 million km^2 (Gruber, 2012; Overduin et al., 2019).

Recent discussion raised another definition gap regarding permafrost: the missing incorporation of ground water and it's state into the permafrost definition, as it has a major impact on the thermal processes in these regions (Devoie et al., 2024). Ground water and ground ice content, however, are difficult to observe directly because they are occur beneath the surface. Traditional remote sensing techniques like optical and radar imagery can detect surface deformation or vegetation changes which indicate ground ice occurrence, but cannot penetrate the ground far enough to actually measure the ground ice distribution (Jorgenson & Grosse, 2016; Zwieback et al., 2024). Ground-penetrating radar, seismic surveys, and borehole logging provide data from deeper layers but are spatially limited (Boaga et al., 2020; Hilbich et al., 2022; Kawasaki et al., 2017; Zhang et al., 2023). Consequently, there is no single, globally applicable remote sensing method that can map permafrost extent and ice content at high spatial and vertical resolution. Re-

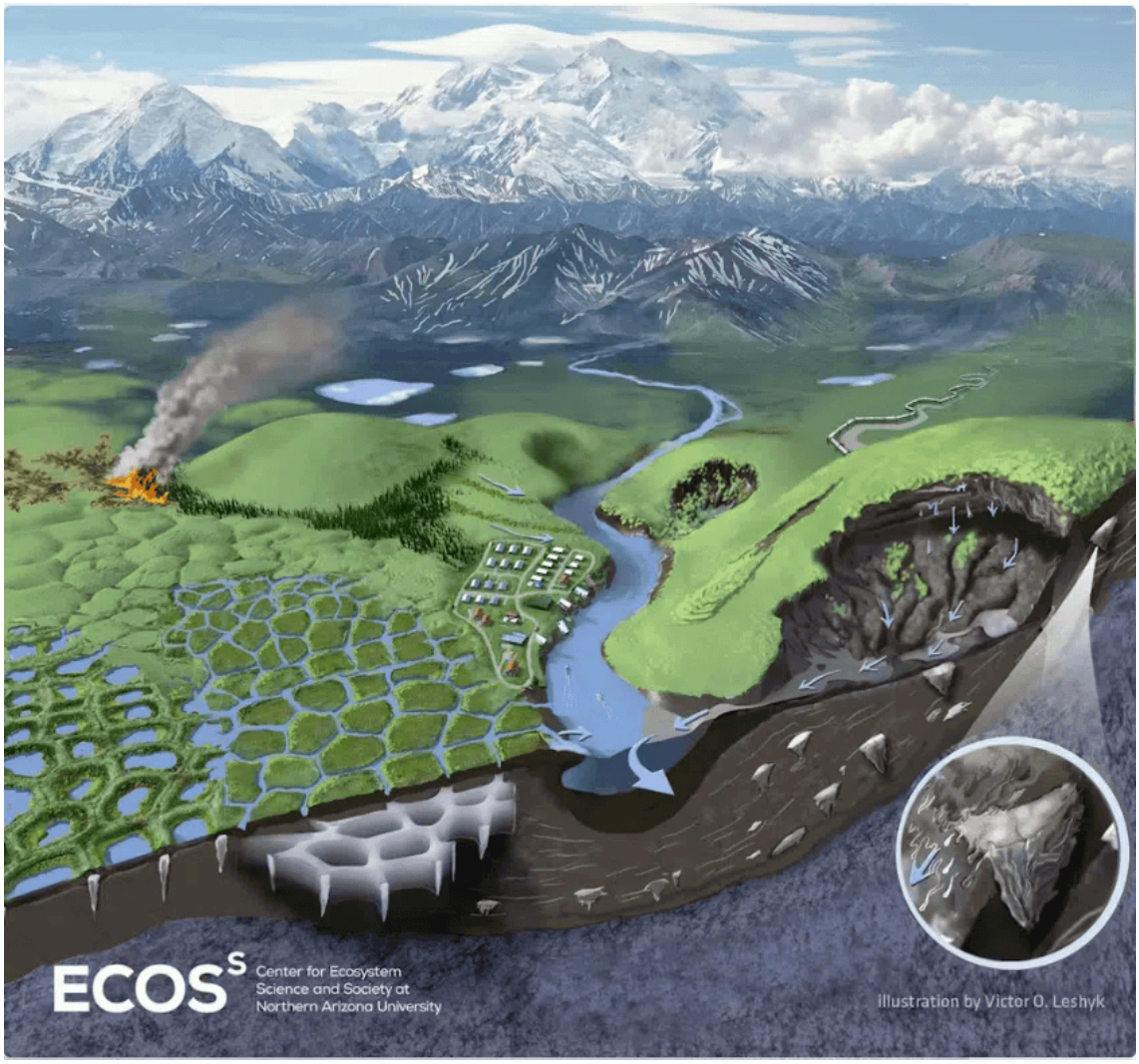

Figure 1 : Schematic vertical cross-section of a typical permafrost landscape. Illustration by Victor O. Leshyk.

searchers are actively developing new approaches, however, comprehensive, long-term datasets remain sparse, especially in remote Arctic regions.

Still, ground ice content plays a crucial role in the permafrost hydrology and ecosystem (Brovkin et al., 2025). Melting of massive ground ice results in a variety of different landscape features, like RTS, thermo-erosional gullies, thermokarst ponds or polygonal patterns formed through the melting of ice wedges (Inauen et al., 2025; Liljedahl et al., 2016; Nesterova et al., 2024). As most landscape features in the permafrost region are formed this way and 16% of the Arctic land surface is occupied by water bodies, it is apparent that major parts of permafrost heat transfer happens through subsurface hydrology (Brovkin et al., 2025; Nitzbon et al., 2024; Vonk et al., 2015). Thermokarst ponds and lakes for example are highly dynamic features, forming and draining within several years or decades, thus dramatically changing the hydrologic flow of the surrounding area (Nitze et al., 2018; Vonk et al., 2015). Taliks, for example, are layers and bodies of unfrozen ground which can form underneath thermokarst lakes and under patches of very deep AL (O'Neill et al., 2020; Vonk et al., 2015). As taliks are not frozen, they allow for subsurface hydrologic flow, potentially connecting the surface with sub-permafrost

layers and therefore playing a crucial role in permafrost heat-transfer (Vonk et al., 2015; Walvoord & Kurylyk, 2016).

A special kind of ice rich permafrost region is the Yedoma domain (Strauss et al., 2017). Yedoma refers to perennially frozen, fine-grained, ice-rich sediments of late Pleistocene age that form distinctive thermokarst landscapes-characterized by alternating hills and lake depressions—across Arctic regions like Siberia, Alaska, and the Yukon (Strauss et al., 2017; Veremeeva et al., 2021). What makes Yedoma special is its exceptionally high and deep ice content (60-95 vol%, up to 50m depth) in the form of large ice wedges, which developed through syngenetic, meaning freezing shortly after deposition (Veremeeva et al., 2021). Through the rapid syngenetic incorporation, Yedoma deposits exhibit high ground ice content and relict organic matter that have persisted for tens to hundreds of thousands of years (Strauss et al., 2021). This combination of age, composition, and landscape impact distinguishes it from other permafrost deposits, but makes it highly vulnerable to the warming climate (Veremeeva et al., 2021).

2.1 Permafrost Extent

As mentioned before, multiple initiatives were made to map permafrost, its extent, properties and landforms on a global, meaning pan-Arctic, scale (Gruber, 2012; Hugelius et al., 2014; Obu et al., 2019). Permafrost related maps are either of high uncertainty or lack spatial continuity and extent (Ran et al., 2021).

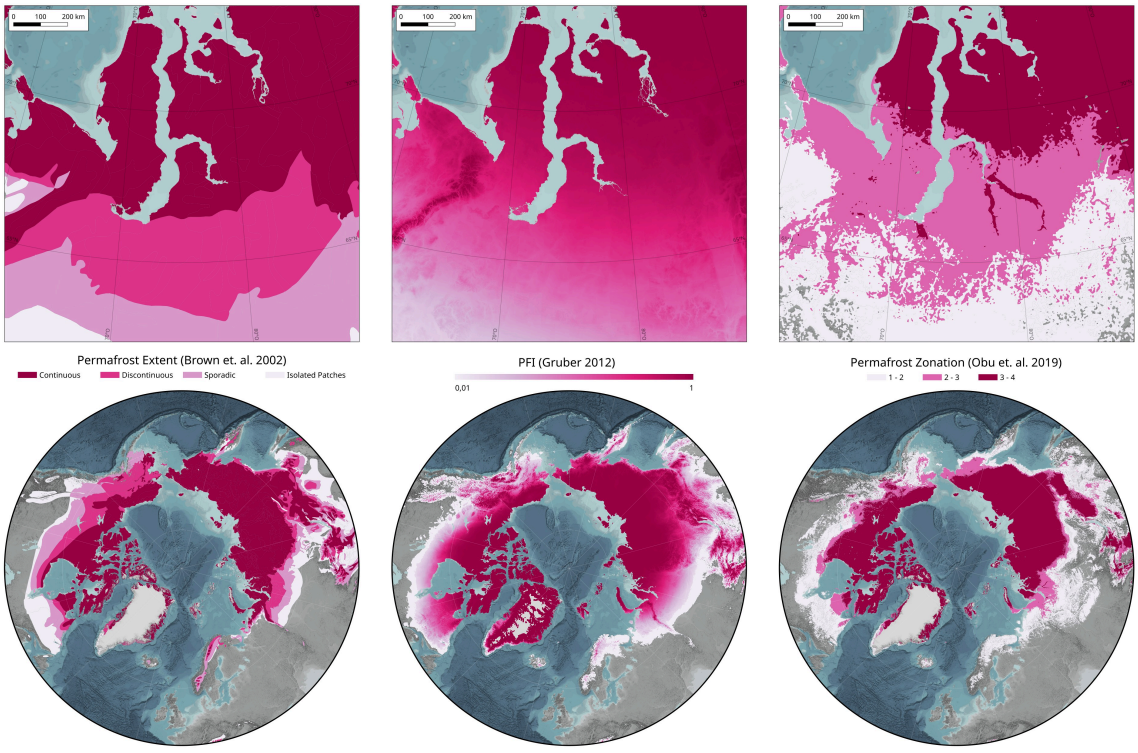

Figure 2 : (Left to right) Permafrost extents from Brown et al. (2002), Gruber (2012), Obu et al. (2019). Top: Closeup of North-West Siberia, Russia. RTopo basemap from " maps@awi 2.0".

The first pan-Arctic map of the permafrost extent where made by J . Brown et al. (1997) nearly 30 years ago, still on paper hand-drawn by regional experts. (IPA-Map) Today the same map is available in digital form and new approaches to correctly model the permafrost extent were made (Brown et al., 2002; Gruber, 2012; Hugelius et al., 2013; Obu et al., 2019).

Gruber (2012) created a model using multiple Mean annual air temperature (MAAT) datasets to compute a Permafrost Zonation Index (PZI), which measures how favorable the conditions for permafrost occurrence are at a less than a kilometer scale. Permafrost occurrence conditions were derived from an estimate over the Mean annual ground temperature (MAGT). Locations with a high PZI do contain permafrost in nearly all conditions, while places with a low PZI only contain permafrost in very favorable conditions, e.g. in shady places within such low PZI locations, where low temperatures and reduced solar radiation help maintain ground temperatures below freezing. The PZI was modelled as the probability of finding MAGT ≤ 0°C using a cumulative normal distribution of the MAAT.

A more sophisticated map was created almost a decade later by Obu et al. (2019). The authors used remotely-sensed land surface temperatures from the Moderate Resolution Imaging Spectroradiometer (MODIS), down-scaled ERA-Interim/Land climate reanalysis data, tundra wetness classes and a land-cover map from ESA Climate Change Initiative (ESA CCI) to employ an equilibrium state Cryogrid v1 model for the Temperature at the top of the permafrost (TTOP) over the 2000-2016 period (Gisnås et al., 2013; Obu et al., 2019; Smith & Riseborough, 1996). From this model MAGT was extracted at the top of permafrost and modelled the permafrost extent as MAGT ≤ 0°C.

While still widely used, the hand-drawn map from J . Brown et al. (1997) lacks any quantitative support and is only of very limited resolution. The simple yet effective approach by Gruber (2012) one and a half decades later greatly improved the situation. However, derived solely by the MAAT without considering hydrological, geomorphological or topographic data it lacks accuracy and certainty on local levels, especially in the transition zones. Utilizing a climate model using multiple remotely sensed data products as inputs can be seen as state of the art in permafrost mapping, with good representation of the permafrost extent patterns observed (Obu et al., 2019). Nevertheless, by utilizing already derived products instead of pure measurements, errors in the input data propagate leaving the model output with MAGT estimation errors. Another problem both model derived maps face is the lack of available borehole coverage to validate and calibrate their models (Gruber, 2012; Obu et al., 2019).

2.2 Climate interaction and tipping

Permafrost ground is the largest terrestrial carbon storage (Schuur et al., 2015; Strauss et al., 2025). It has historically acted as a significant carbon sink through long-term freezing processes that sequestered carbon over millennia, making permafrost not only a legacy store of carbon accumulated over thousands of years (Zech et al., 2011). Although the exact amount cannot yet be measured, recent estimates suggest that the broader permafrost region contains around 1000 Pg of Soil Organic Carbon (SOC) in the upper

3m of soil (Turetsky et al., 2020; Wu et al., 2022). Including highly uncertain 500 PgC deposits in deeper yedoma layers and in river deltas, the total SOC is estimated to be around 1500 Pg (Hugelius et al., 2014; Schuur et al., 2015; Strauss et al., 2025; Turetsky et al., 2020). Compared to the total carbon stored in terrestrial soils, permafrost carbon accounts to 60% (Strauss et al., 2025; Turetsky et al., 2020). Around 800 Pg carbon remains currently frozen in permafrost ground, resulting in 500 PgC to be part in the carbon cycle of the active layers ecosystem (Hugelius et al., 2014).

Thawing of permafrost results in the release of that carbon into the atmosphere through microbiological activity in form of greenhouse gases like carbon-dioxide or methane (Baysinger et al., 2025; Dolle et al., 2025; Laurent et al., 2023). The amounts of carbon released to the atmosphere is not well studied yet, but it is expected that the released greenhouse gases will accelerate the anthropogenic climate change, especially in the already rapidly warming Arctic (Grosse et al., 2011b; 2011a; Schuur et al., 2015; Turetsky et al., 2019). The accelerated warming, results in more and faster thawing of permafrost, thus releasing just more greenhouse gases, resulting in a positive feedback-loop (Nitzbon et al., 2024).

Feedback loops are often used to describe different climate interactions. Climate feedback loops are processes in which a change in a state variable of the Earth system triggers secondary effects that either amplify (positive feedback) or dampen (negative feedback) the initial perturbation (Winkelmann et al., 2025). All climate subsystems are subject to multiple feedback loops, interacting with each-other in a complex manner. Feedback loops can be seen as the essential part of the feedback dynamics which can cause a tipping of the climate system. Tipping of a climate system describes the irreversible and / or abrupt change of the systems state at after a critical threshold, called tipping point.

When a forcing pushes a tipping system toward such a tipping point, the magnitude and sign of the dominant feedbacks determine whether the system will remain in its current state, shift gradually, or undergo an abrupt, irreversible transition (Winkelmann et al., 2025). Such a tipping is an interplay between different feedbacks: Positive feedbacks are particularly relevant for tipping because they can create a self-reinforcing chain of changes, negative feedbacks, by contrast, act to stabilize the system.

There are several systems in the broader permafrost regions which are suspected to have a tipping point: abrupt thaw, permafrost collapse, northern expansion of boreal forests (Winkelmann et al., 2023). A recent study by Nitzbon et al. (2024) claims that there is no global temperature increase below 6 °C at which the global permafrost abruptly retreats. Instead, there appears to be a quasi-linear relationship between global warming and permafrost retreat, and thus a release of CO_2. Accordingly, it is argued that the thaw of permafrost should not be considered a tipping point because a critical threshold and thus the requirement of abruptness is missing in that system. However, permafrost thaw should still be treated as a major issue in the changing climate as it is irreversible on human timescales. Nitzbon et al. (2024) further notes that through talik (unfrozen ground) and hydrologic flow, heat can also spread beyond regional boundaries: “While this suggests the possibility of local-scale tipping due to talik development [...], the tipping at or beyond regional scales is tied to the frequency and spatial abundance of disturbances.” (Nitzbon et al., 2024)

On a global scale, tipping systems are related and interconnected with each other, be it through more direct causes like heat or water transfer or indirect causes like tele-connections (Boaga et al., 2020; Donges et al., 2009; Rahmstorf et al., 2025; Rockström et al., 2023; Vos et al., 2026). Therefore the tipping systems in the permafrost region are influenced by and do influence other tipping systems, while being worse understood than others (Schuur et al., 2015; Winkelmann et al., 2025). This research gap is partially due to a lack of well calibrated climate models focussing on the permafrost area because of an incomplete understanding and observation of the different permafrost variables (Nitzbon et al., 2024). Initiatives like the Global Terrestrial Network for Permafrost (GTN-P) improve the situation by providing standards and databases for measurements of Essential Climate Variable (ECV), which are also incorporated into the Global Climate Observing System (GCOS) (Biskaborn et al., 2019; Bojinski et al., 2014; Irrgang & Lübker, 2026; Starkweather et al., 2022; World Meteorological Organization (WMO) et al., 2016; 2025). Further, first guidelines for unified measurement and observation of permafrost ECVs were created by the WMO ((WMO), 2025). However, with ground ice being hard to observe on a pan-arctic scale ECVs are missed, making the approximation of ground ice an important next step to fully understand our climate. Further, finding possible correlations between ground ice occurrence and environmental variables could even help making projections regarding the change of permafrost system for different climate change scenarios.

2.3 Ground Ice Mapping Initiatives

First statistical estimates on global ground-ice were made in 1999 based on the hand-drawn IPA-Map without contributing a map showing the coverage of ground-ice (Zhang et al., 2000). Only recently first attempts were made to map ground ice based on measurements and proxy variables, however, all attempts focused on single regions instead of the whole pan-Arctic (O'Neill et al., 2019; Zhang et al., 2023; Zou et al., 2024; Zwieback et al., 2024; Zwieback & Meyer, 2021).

A study of ground ice in Alaska by Zwieback & Meyer (2021) showed a potential link between late-season subsidence and top-of-permafrost ground ice (Zwieback & Meyer, 2021). This indicates that excess ground ice can be detected through remote sensing of subsidence, particularly using Sentinel-1 InSAR data, which captures ground deformation during warm summers when the thaw front penetrates previously frozen soils (Zwieback & Meyer, 2021). In ice-rich permafrost, this subsidence can reach 4 to 8 cm, significantly exceeding the low deformation observed in ice-poor areas (-1 to 2 cm), demonstrating high sensitivity and specificity for identifying excess ice at the surface (Zwieback & Meyer, 2021). In a follow-up study, the authors used this approach on remotely sensed InSAR Sentinel 1 subsidence data to create vertical ground ice profile maps across Alaskan landscapes, showcasing that this approach enables automated, large-scale mapping across regions lacking visible ground-ice-indicating landforms (Zwieback et al., 2024).

Recent studies have advanced ground ice mapping through both field-based modeling and machine learning techniques. For instance, Zhang et al. (2023) created a complex model pipeline combining both field-measurements, remotely sensed and modeled soil

parameters and climate data. to estimate ground ice thickness. Geomorphic units were classified across the different horizontal ground layers through borehole data and soil parameters, which then were used together with climate and elevation data to derive the ground ice thickness up to 20m deep (Zhang et al., 2023). Applied to the Northeast China permafrost region, this method achieved 72.3% accuracy and provided regional estimates of ice reserves (Zhang et al., 2023). Similarly, Zou et al. (2024) trained a random forest model on over 600 borehole records in the Qinghai-Tibet Plateau to predict the ground ice content between 2 and 10 m depth across three layers at a spatial resolution of 1 km. The authors showed that for the model predictions of the top layer climate variables had the most contribution to the final score while for deeper layers terrain and soil variables were more decisive (Zou et al., 2024). Another study by O'Neill et al. (2019), estimating ground ice content and type for Canada, distinguished between relict, segregated, and wedge ice, resulting in three separate ground-ice maps. All of the ice models used in the study were decision trees manually constructed based on expert knowledge and paleo-environmental data (O'Neill et al., 2019). This approach emphasized the difference between ground ice types and their influence on permafrost landscapes. Another pan-Arctic ground ice map, specialized on Yedoma, was synthetised by Strauss et al. (2021) using geological and stratigraphical maps at various available scales.

To summarize, a pan-Arctic map of ground ice extent is still missing due to the sparse distribution of field data and the region-specific nature of current mapping approaches. While regional studies have advanced ground ice estimation using borehole data, remote sensing, and machine learning, no single model has been validated across the entire Arctic. Further, pan-Arctic classifications of different ground ice types are also missing.

2.4 Retrogressive Thaw Slumps

RTSs are some of the most dramatic expressions of permafrost thaw. They appear as large, curved depressions in the ground surface, often stretching from a few meters to several hundred meters in length and width (Nesterova et al., 2024). Figure 3 and Figure 4 show a collection of these impressive landscape features. RTSs develop through the excessive melting of massive ground ice or ice-rich permafrost, which leads to erosion of the AL and ground sediments within the ice, making them a proxy for massive ground ice (Nesterova et al., 2024). The continuos erosion results in a retrogression: material is transported downslope, the headwall retreats up slope as the exposed ice continues to melt (Nesterova et al., 2024). Once the massive ground ice is exhausted or the remaining ice becomes thermally insulated by sediments, the slump's advance stops and the feature becomes stabilized (Nesterova et al., 2024).

The exposure of the massive ground ice results in the just mentioned headwall, the most distinguishing feature of an RTS. Headwall heights typically span several meters to 55 m, while lengths may exceed 1 km, Figure 3 shows a photo of an RTS headwall with a human for scale (Costard et al., 2021; Kizyakov et al., 2023). The growth of RTS is often measured through the retreat rate of the headwall, making the tracking of it crucial (Leibman et al., 2021). Exposure can be triggered by both natural and anthropogenic disturbances, including climatic shifts such as prolonged warm periods or heavy snowmelt,

Figure 3 : Photo of an Retrogressive Thaw Slump (RTS) headwall in Yamal, West Siberia, Russia, with a human for scale. Taken by Nina Nesterova in July 2021.

as well as wildfire, which removes protective organic soil layers and accelerates thaw (Nesterova et al., 2024). Additionally, geomorphological processes like thermal erosion, gully development, or active layer detachment slides can expose permafrost and initiate RTS formation (Nesterova et al., 2024). These triggers can also re-expose the massive ground ice below an already stabilized RTS, setting the RTS into an active state again, contributing to their polycyclic nature where cycles of active and stable phases alternate over decades (Nesterova et al., 2024). Simultaneously, many different processes happen concurrently and interacting with each other, making each RTS unique and thus harder to track (Nesterova et al., 2024).

Because of their dependence on a slope in the terrain RTS can be mostly found on hill slopes, coasts, rivers and lakes (Leibman et al., 2021; Nesterova et al., 2024; Nitze et al., 2025). The location of an RTS, together with the type of ice and the terrain, decides it's form, thermocirque formations occurring mostly on hill slopes and lakes while thermoterrace formations mostly near moving waterbodies, especially on coasts (Nesterova et al., 2024). Both formations are shown in Figure 4. For most RTS along coasts and rivers, the additional erosion from waves clears the debris, benefiting the RTS grow and, combined with their spatial continuity, enable formation of terraces (Nesterova et al., 2024). In syngenetic-ice-rich Yedoma regions, RTS are often accompanied by the presence of baydzherakhs, smaller mounds within the RTS floor which developed from sediment-rich polygon centers and the thawing of surrounding large syngenetic ice wedges (Inauen et al., 2025; Strauss et al., 2017; Veremeeva et al., 2021).

A closely related, but distinct feature to RTSs are Active Layer Detachments (ALDs). ALDs are translational landslides that occur in summer on gentle to moderate slopes, where thawing soil overlying permafrost fails along a shallow plane parallel to the surface. While ALDs are distinct from RTSs in its mechanism and scale, they are closely related as both involve the detachment of thawed material from permafrost and are

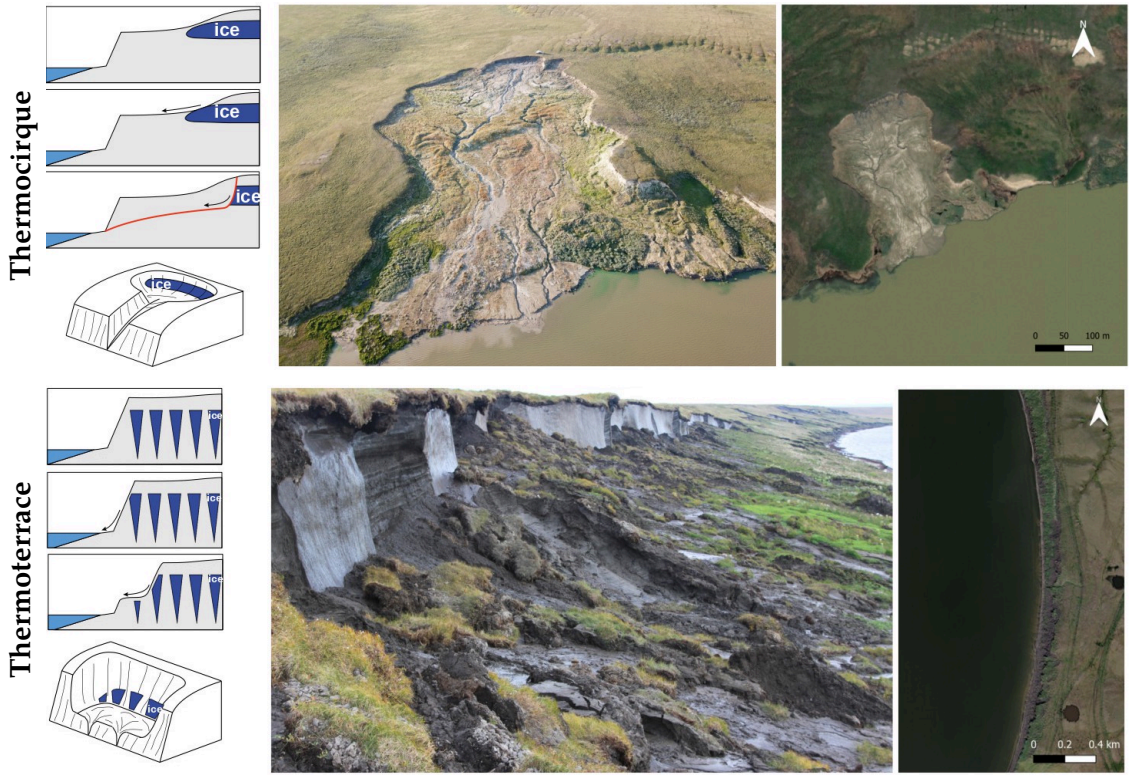

Figure 4 : Two examples of Retrogressive Thaw Slumps (RTSs). Top: Thermocirque RTS in central Yamal, West Siberia, Russia. Left-to-right: Schematic of the formation showing the direction of mass movement the thawed material (black arrow) and the headwall cross-section (red-dashed line), a UAV photo taken in August 2019 by Artem Khomutov, WorldView-2 satellite imagery from July 2018. Bottom: Thermoterrace RTS on the Bykovsky Peninsula, North-East Siberia, Russia. Left-to-right: Schematic of the formation showing the direction of mass movement the thawed material (black arrow), a photo taken on the ground in August 2016 by Alexander Kizyakov, WorldView-2 satellite imagery from August 2020. Schematics and images where taken from Nesterova et al. (2024).

driven by active layer thawing (Lewkowicz & Harris, 2005; Nesterova et al., 2024; Nitze et al., 2025). In some cases, an ALD can initiate or contribute to the development of an RTS by destabilizing slopes and promoting further ground ice melting (Nesterova et al., 2024). Since both RTS and ALD leave sediment-rich scars in the landscape, they can exhibit similar spectral properties, making automated discrimination on satellite or aerial imagery challenging (Nitze et al., 2025).

Scientists track RTSs using a combination of remote-sensing, GPS monitoring, and field surveys (Maier et al., 2025a; Nitze et al., 2025; Rettelbach et al., 2024; Swanson & Nolan, 2018; Ward Jones et al., 2019). Satellite imagery can reveal changes in the landforms outline over time, while ground-based measurements record subsidence rates and ground-temperature profiles (Maier et al., 2025a; Nitze et al., 2021). Recent studies have also used unmanned aerial vehicles (UAVs) to produce high-resolution digital elevation models, allowing researchers to quantify the volume of material displaced in each slump cycle (van der Sluijs et al., 2018). While climate change is becoming more intense, with the Arctic being the fastest warming land-ecosystem on earth, it was measured that the prevalence and intensity of RTSs have increased (Nesterova et al., 2025a). As Arctic warms and precipitation patterns intensify, the rate of ground ice melt is assumed to

accelerate, feeding the retrogressive process (Schuur et al., 2015). Moreover, the resulting water drainage patterns can influence local hydrology and even accelerate the thaw of adjacent permafrost (Brovkin et al., 2025). In this sense, RTSs are both a visible indicator of permafrost degradation and a potential feedback mechanism that could further amplify regional warming (Brovkin et al., 2025). However sediment and debris deposition as well as ecological succession constitute negative feedbacks, allowing for stabilization on multi-year timescales (Nitzbon et al., 2024). While first studies suggest that winter warming and summer precipitation are the driving variables behind RTS creation, the exact environmental influences related to RTS occurrence are not well studied yet (Nesterova et al., 2025a).

2.5 RTS Datasets and Inventories

Because RTSs are quite disruptive landscape features clearly visible from space, RTS mapping initiatives where more successful than for ground-ice content or permafrost extent. The vastness of the regions where RTS are known to occur, however, limits the ability to manually map RTS (Nesterova et al., 2025b). Thus, several datasets containing only manually detected RTS are fairly limited regarding their spatial extent (Barth et al., 2025; Lewkowicz, 2025; Lewkowicz & Way, 2019; Nesterova et al., 2025b; Sluijs & Kokelj, 2023). Sluijs & Kokelj (2023), Lewkowicz & Way (2019) and Lewkowicz (2025) manually mapped RTS across several patches in Northern Canada and Alaska. An earliest of these studies, Lewkowicz & Way (2019), mapped centroids of RTS on Banks Island, a RTS hotspot located in Northern Canada. The Lewkowicz (2025) dataset consists of the centroids of manually identified RTS, derived from a 30 m resolution Landsat time-series imagery spanning the period from 1984 to either 2016 or 2018, depending on the specific patch. While covering fewer patches and the shorter time period between 2011 and 2016, Sluijs & Kokelj (2023) also provides delineations of their mapped RTS. However, a study from Nitze et al. (2024a) showed that delineations of RTS are even for domain expert very subjective, with inter-rater Intersection over Union (IoU) varied wildly between 16% and 92%, showing that RTS delineations are very arbitrary. To address this problem Yang et al. (2025) proposed semantic guidelines for delineating RTS; however, these guidelines have not yet been applied in a large-scale mapping effort.

In a manual mapping approach by Nesterova et al. (2025b), RTS points were mapped for the two peninsulas Gydan and Yamal in the West Siberian Arctic. For this mapping effort, multiple datasets containing automatic detected RTS where manually filtered and supplemented by comparing ESRI Wayback (Nesterova et al., 2025b). In a follow-up study the authors utilized additionally declassified imagery from the spy-satellites Hexagon and Corona delineate RTS dating back to the mid 1960s, resulting in a 60-year time span of observed RTS for a few key-sites (Nesterova et al., 2025a). Through this large temporal coverage Nesterova et al. (2025b) were able to show that summer precipitation is a short-term driver for RTS initiation, while winter warming is the dominant long-term driver (Nesterova et al., 2025a). Another manual mapped inventory in the Russian High Arctic was created by Barth et al. (2025). The authors used high resolution optical

imagery from various satellites to delineate RTS at five study sites over an 11-year time span from 2011 to 2020.

With the rise of ML and DL and especially advances in the field of image segmentation, more and more automated approaches were published in recent years. Each approach is different in their use of data sources, spatio-temporal resolution and ML method, with all except for one focusing on few regions instead of the complete pan-Arctic. While most approaches created some sort of RTS delineations with image segmentation methods, some classified each pixel individually, interpreting the RTS mapping problem as a tabular ML problem where each pixel is handled as an individual row of a table.

First non-DL approaches utilized trend and change data, such as Landsat Trends or change in elevation based on TanDEM-X observations (Bernhard et al., 2020; Runge et al., 2022). Runge et al. (2022) used a simple change-detection algorithm based on the

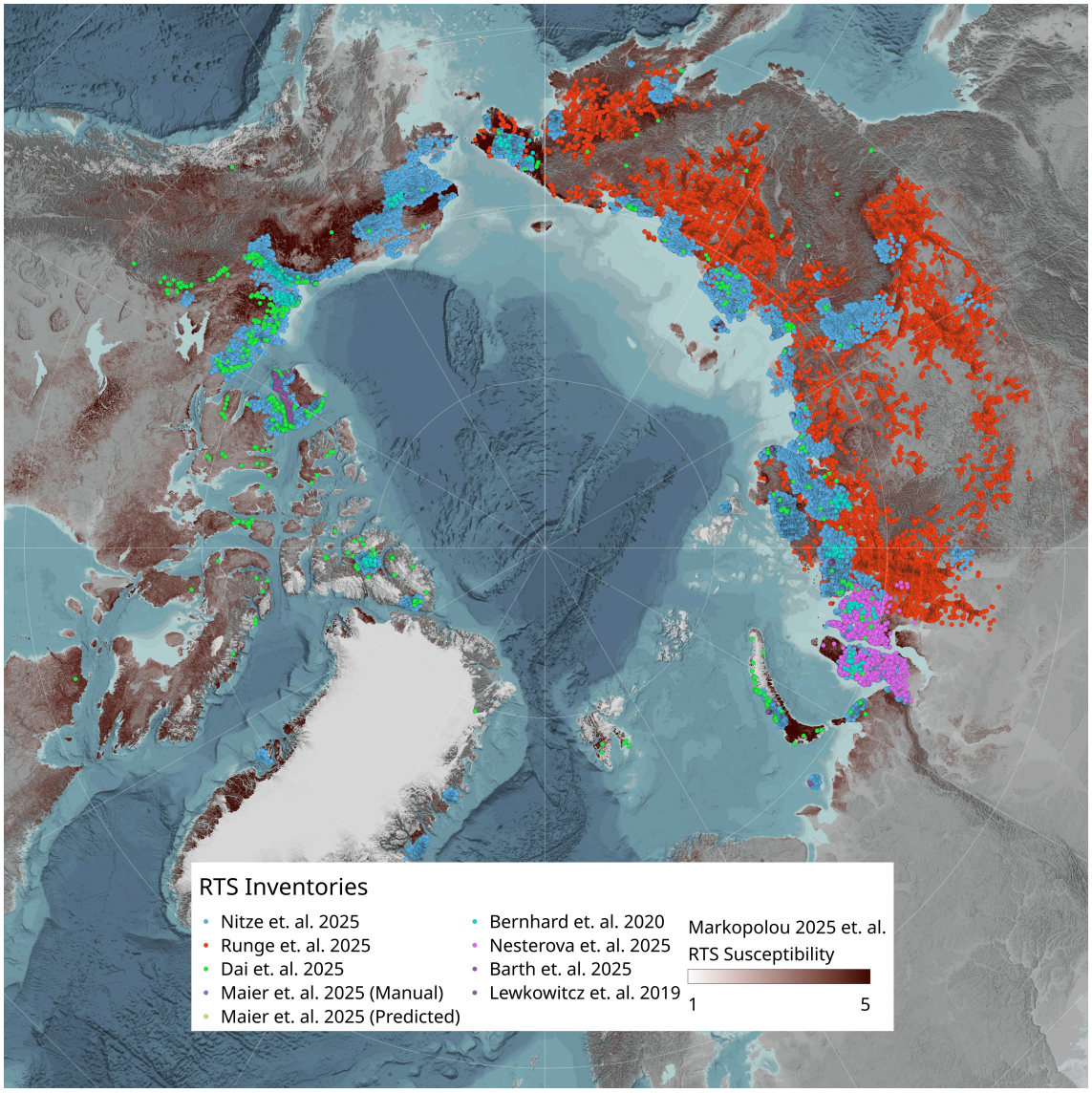

Figure 5 : All discussed Retrogressive Thaw Slump (RTS) datasets, except for Huang et al. (2023) and Yang et al. (2023) because of Data availability issues at the time of writing. RTopo basemap from " maps@awi 2.0".

Landsat-based detection of Trends in Disturbance and Recovery (LandTrendr) method, followed by various filtering techniques to filter out false-positives (Kennedy et al., 2010). Their dataset consists of RTS delineations across the Russian Arctic between 2000 and 2019, however, despite the several filtering steps still consisting of many false-positives (Nesterova et al., 2025b; Runge et al., 2022). For RTS detection, elevation data holds a critical benefit over optical imagery with the ability to detect also volumetric changes. Simple ML methods, to be exact linear threshold models, random forests and Support Vector Machine (SVM), where used in the study from Bernhard et al. (2020) to detect RTS on elevation change data. The elevation change was calculated based on single-pass TanDEM-X observations spanning several years between 2011 and 2016 (Bernhard et al., 2020). In a follow-up study, this dataset was utilized to estimate volumetric changes across their key-sites across the Arctic (Bernhard et al., 2022).

Elevation data was also used in approaches utilizing DL methods (Dai et al., 2025; Huang et al., 2023; Maier et al., 2025a). Similar to Bernhard et al. (2020), the more recent work from Maier et al. (2025a) utilized TanDEM-X elevation data to train a U-Net on the retrieved differential DEM. With their method the authors were able to match the domain expert's inter-rater accuracy on the RTS delineations, however, only for RTS where a large change in elevation happened, similar to the other method utilizing elevation data. In a follow-up study this method was further refined to also quantify volume changes and the resulting mobilization of carbon on the Tibetean Plateau (Maier et al., 2025b). Another multistep approach was utilized by Dai et al. (2025) where a simple threshold classification on elevation change from co-registered ArcticDEM strips was used to detect potential RTS in a first step, followed by a filtering and delineation step based on an Convolutional Neural-Network (CNN). The ArcticDEM was also utilized by Huang et al. (2023) in a similar multi-step DL setup where heavy feature engineering was involved to ultimately train a YOLOv4 object detection algorithm.

A DL approach based on the widely used U-Net architecture was utilized by Nitze et al. (2021) to detect RTS on 3m PlanetScope optical imagery. Alongside optical imagery, three Tasseled Cap indices of a Landsat Trend dataset were utilized, the slope and Topographic Position Index (TPI) calculated from the ArcticDEM and the Normalized Difference Vegetation Index (NDVI) to train two models which where then used in an ensemble to retrieve the final predictions (Nitze et al., 2017; 2021). The authors continued their work by scaling up this approach to detect RTS over various areas across the pan-Arctic from 2018 to 2023, resulting in the Database of AI-detected Arctic RTS footprints (DARTS) dataset (Nitze et al., 2025). The resulting delineations of the DARTS dataset are, however, quite noisy because of the noise in the optical imagery. Further, DARTS also lacks consistency along the temporal dimension, as only some regions contain observations from all years. A very similar approach was done by Yang et al. (2023), where alongside the ArcticDEM also optical imagery from Maxar and Sentinel 2 was utilized to train an augmented version of the U-Net. In a follow-up study the authors created the Arctic Retrogressive Thaw Slumps dataset (ARTS) dataset, which contains a standardized collection of multiple RTS datasets across the Arctic (Yang et al., 2025).

A more statistical approach was utilized by Makopoulou et al. (2024) to estimate RTS susceptibility across the complete pan-Arctic. The authors utilized multiple published RTS databases to retrieve large inventory of RTS locations, which was then used as

input for a boosting classifier together with CHELSA Climate data, slope data from Geomorpho90m and sediment data from SoilGrids250m 2.0 as part of a ML workflow (Makopoulou et al., 2024). A big drawback of their method is their handling of unknown locations: To train a classifier both positive and negative labels are required, thus empty space outside the inventories coverages were fills with negative labels, potentially assigning false-negatives to new locations.

To summarize, RTS are mapped across the arctic with diverse methods. All datasets discussed are shown in Figure 5 and further compared in Table 11 (Appendix B.3). Recent years showed an increase in the use of automated mapping methods, mostly backed by DL techniques. Although the broad availability of methods and datasets, a high-accuracy and robust map spanning the complete pan-Arctic is still missing.

2.6 Novel techniques in Earth Observation

Recent efforts were taken to create a new way of mapping environments: Remote-Sensing Foundation Models (RSFMs) (Feng et al., 2025). RSFMs, sometimes also called geospatial foundation models, are large DL models trained using self-supervised pretraining methods, often supporting multi-modal and multi-sensor inputs (Forgaard et al., 2026; Jakubik et al., 2023; Szwarcman et al., 2024).

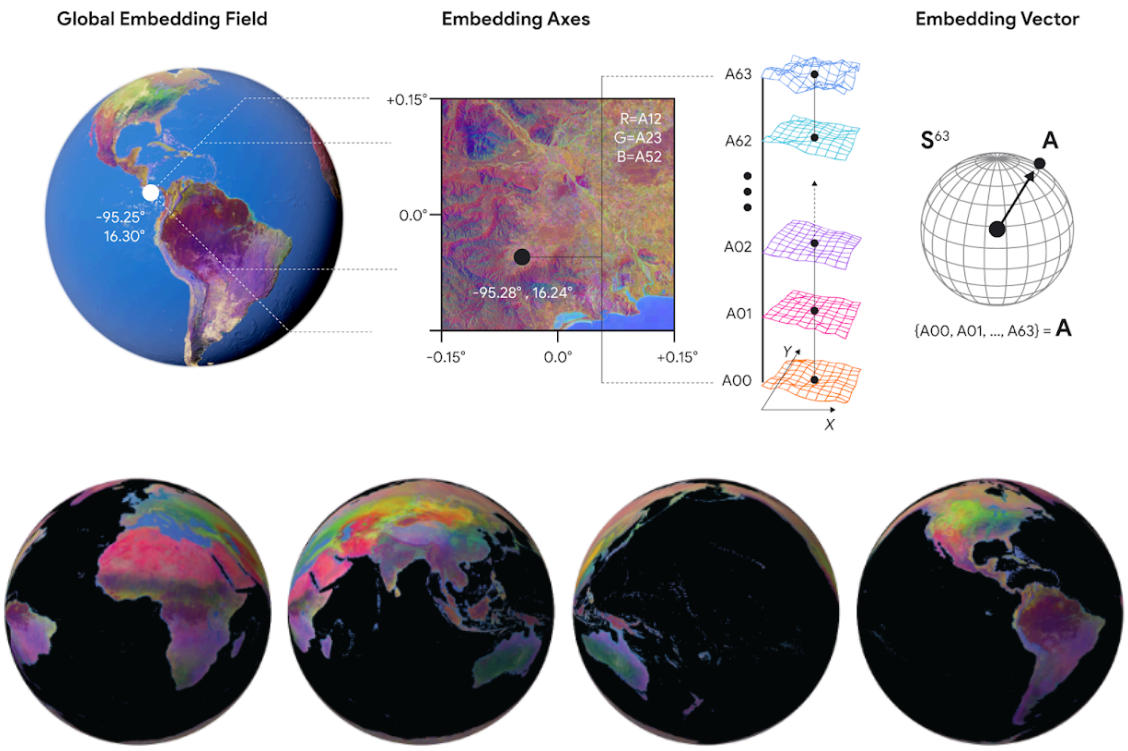

Figure 6 : Top: "Diagram showing a global embedding field broken down into a single embedding, from left to right. Each embedding has 64 components which map to coordinates on a 64-dimensional sphere." (Brown et al., 2025) Bottom: "Complete 360° view of 2023 annual embedding field covering Earth's land surface." (Brown et al., 2025) Graphics taken from Brown et al. (2025).

The main idea behind RSFMs is to train a extremely large model once on a generic set of different data inputs and then fine-tune it for various downstream applications on task-specific data (Xiao et al., 2024). This drastically reduces the need for large task-specific datasets and intensive compute, as the model already has the generalized knowledge from the pretraining. The fine-tuning and inference of those models, however, is still compute intensive because of this extreme model size. This resulted in a new approach of utilizing the learned generalized knowledge of RSFMs: embeddings-as-data (Brown et al., 2025; Feng et al., 2025). Embeddings-as-data, sometimes also called satellite embeddings, are precomputed, generic, task-agnostic features that can be easily utilized by low-cost ML models on various downstream applications (Feng et al., 2025). These features are high-dimensional vector representations that capture the multi-modal environmental state of a given location in an abstract high-dimensional vector space. They are conceptually similar to embeddings used by large language models, which are used to represent text in an numerical vector space. The idea behind this similar to RSFMs, to train an extreme large model on a diverse set of inputs once, but instead of fine-tune the complete model on a downstream task, feature embeddings are extracted from the model to train smaller ML models (Brown et al., 2025; Feng et al., 2025; Szwarcman et al., 2024; Xiao et al., 2024). Since the extracted features are much smaller in size and the information is more compressed than in the foundation model, also small ML models are able to solve complex downstream tasks with these embeddings. While similar, this approach is therefore more compute-efficient that the fine-tuning of large foundation models on such downstream tasks (Gomes et al., 2025; Jakubik et al., 2023; Szwarcman et al., 2024).

AlphaEarth Foundations is an "an embedding field model yielding a highly general, geo-spatial representation" (Brown et al., 2025) created by Google Deepmind doing exactly that. The derived satellite embeddings are 64-dimensional vectors representing a complete year, with years between 2017 and 2024 on a 10m by 10m per pixel grid available. Figure 6 shows how these embeddings look like on a global, regional and pixel scale. The model was trained on a vast collection of diverse datasets, including Sentinel-1 SAR, Sentinel-2 and Landsat 8/9 optical imagery, GEDI LiDAR data, ERA5 climate reanalysis, and other auxiliary sources. While the AlphaEarth Foundation satellite embeddings are openly available via the Google EarthEngine, the underlying model and training pipeline remain proprietary closed-source.

Being entirely open-source, TESSERA emerges as a fully transparent alternative, trained exclusively on Sentinel-1 and Sentinel-2 data to produce 128-dimensional yearly embeddings, as shown in Figure 7 (Feng et al., 2025). While only data for 2024, current plans include to backfill the years from 2017 to 2023 within the coming months. Its commitment to openness contrasts with AlphaEarth's closed architecture, democratizing geo-spatial AI for researchers, and policymakers.

A countervailing trend in ML is the use of smaller, extremely efficient and explainable algorithms, such as the entropy-optimal Scalable Probabilistic Approximations algorithm (eSPA) or Sparse Probabilistic Approximation for Regression Task Analysis (SPARTAn) (Groom et al., 2025; Horenko, 2020; Horenko et al., 2022). These are ML algorithms used for classification and regression tasks, thus only useable in tabular ML problems. Another novel genre of ML methods is AutoML, frameworks which handle everything

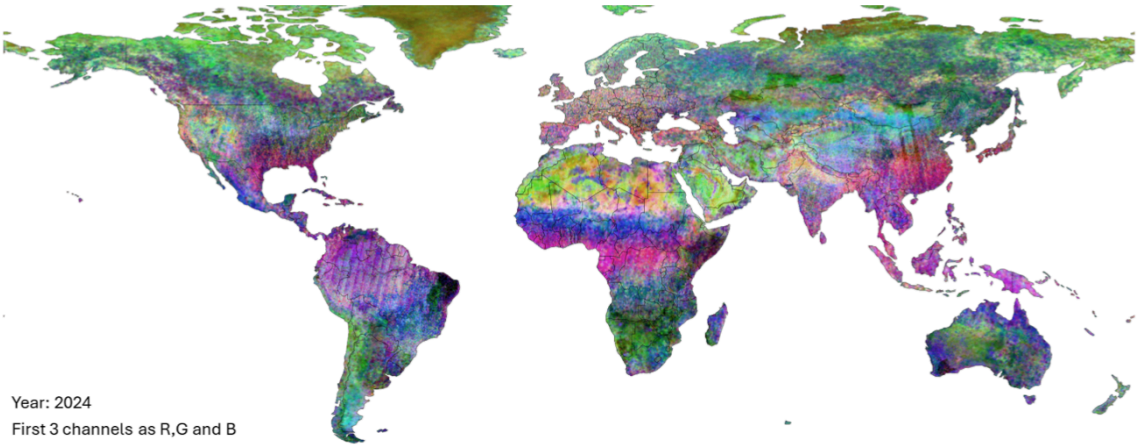

Figure 7 : First three channels of the TESSERA embeddings for 2024. Graphic taken from Feng et al. (2025).

ML related only needing a (tabular) dataset as input (Ali, 2020; Erickson et al., 2020). These frameworks automatically clean and standardize the dataset and atomically train a handful of well-known ML algorithms without any human interaction needed, drastically reducing the time and knowledge needed for such a task. However, none of the RSFMs, the embeddings-as-data, the extremely efficient eSPA and SPARTAn models, nor such AutoML frameworks are currently used in permafrost Earth-Observation (EO), opening the possibility of an methodological improvement over the current situation.

2.7 Open Questions

As discussed before, climate change has a severe impact on the Arctic and its warming leads to widespread permafrost thaw. Because the exact permafrost extent and volumes of ground ice are not yet accurately mapped, predictions regarding carbon release, landscape and hydrology changes come with large uncertainties. With RTSs serving as proxies for massive ground-ice, a global inventory could lead to further studies and climate model improvements regarding ground-ice coverage across the pan-Arctic. However, a pan-Arctic map of RTS occurrence across spatial scales is still missing. Moreover, climatic conditions promoting RTS occurrence are also not well studied yet, which prevents accurate modelling of RTS processes across spatial-scales and the creation of accurate projections regarding changes in the permafrost system due to climate change. This opens up the primary research question of this thesis:

> **Q1:** Which environmental variables are important for RTS occurrence and do these importances change over spatial scales?

In order to avoid expert bias, these variables should be identified solely through data without relying on prior knowledge. To achieve this, various ML models are trained to predict RTS occurrence, with the aim that they learn or approximate the complex interactions between these environmental variables. Afterwards the most predictive variables of the models are identified through model interpretability tools. Here, permafrost extent and ground ice information are intentionally excluded to ensure that robust proxies for

RTS are found and can be used in future studies to further contribute to the mapping of permafrost extent and ground ice content. This marks the second question of this thesis:

> **Q2:** Which Machine-Learning methods are suitable for a high-quality pan-Arctic RTS occurrence prediction and can simultaneously identify which variables are relevant?

State-of-the-art ML and DL techniques, such as embeddings-as-data and AutoML frameworks, are more complex but also less transparent than traditional ML models. Consequently, their use represents a trade-off between explainability and predictive performance. This trade-off opens the third question of this thesis:

> **Q3:** How do novel Machine-Learning methods, such as AutoML frameworks or embeddings-as-data, compare to traditional variable selection and model training?

Data and Methods

To answer the three research questions, a uniform methodology was developed, enabling quick but comparable and reproducible exploration of ML methods and datasets. Figure 8 shows this setup. First, multiple large datasets were created, consisting of many climate and topographic variables, as well as novel RSFM embeddings. These datasets were aligned onto a DGGS grids of various resolutions, enabling an easy ensembling of variables from different sources without further spatial alignment necessary while also allowing analysis across spatial scales.

Then various ML models were trained to predict RTS occurrence on different dataset combinations using different sets of variables. Among traditional ML algorithms such as k-Nearest-Neighbors (kNN), Random Forest (RF) and eXtreme Gradient Boosting (XGBoost), also the novel eSPA algorithm and the AutoGluon AutoML framework were utilized in this step. Here, the problem of RTS occurrence prediction is seen as a tabular ML problem: each cell of the DGGS grids are seen as rows of a table while the different variables are columns. This drastically reduced the complexity of this approach and further allowed for the use of less complex models, because no advanced DL image segmentation models were necessary. After each training the respective model was evaluated regarding its predictive performance using common performance metrics such as the F1-score, and further the feature importances were identified using in four different feature importance metrics.

First, research question Q2 was answered by comparing the predictive performances of the trained models. The stability of the feature importance metrics were compared between the models to assess which models and feature importance measuring methods are suitable to identify relevant variables. Simultaneously the third research question Q3 was answered also by comparing the predictive performances of the trained models grouped by novel versus traditional methods. Finally, the primary research question Q1

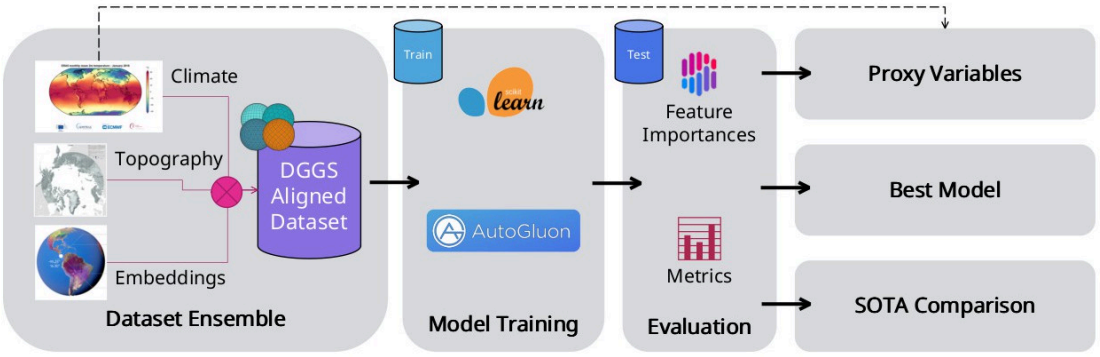

Figure 8 : Overarching method used to answer the three research questions.

was then answered through the analysis of the feature importances of the best model trainings which used only climate and topographic variables. Here, the different resolutions of DGGS came in handy as they allowed for analysis across different spatial scales.

Overall, this approach introduced four novelties in permafrost EO: (1) the use of a large amount of explanatory variables as input features for the ML models, (2) the use of highly efficient probabilistic approximation algorithms, (3) the use of satellite embeddings and (4) the use of DGGSs for coherent analysis of spatial scales.

3.1 Data preparation and ensembling

Before the variables of the different datasets could be used for training, the datasets were enriched and aligned to a selection of DGGS at different levels. Figure 9 shows the processing steps of the four datasets used in this work.

3.1.1 Discrete Global Grid Systems

Large scale EO remote sensing applications usually use pixel-based projections or geographic coordinates, which introduce re-projection errors when aligning multiple datasets and biases regarding area or distance depending on the projection or coordinate format. DGGSs are spatial reference frameworks solving these problems by partitioning the Earth's surface into a hierarchical, multi-resolution grid of uniquely identified cells, enabling efficient organization, integration, and analysis of geo-spatial data at global scales. (Sahr & White, 1998) These grids tessellate the globe into evenly spaced cells,

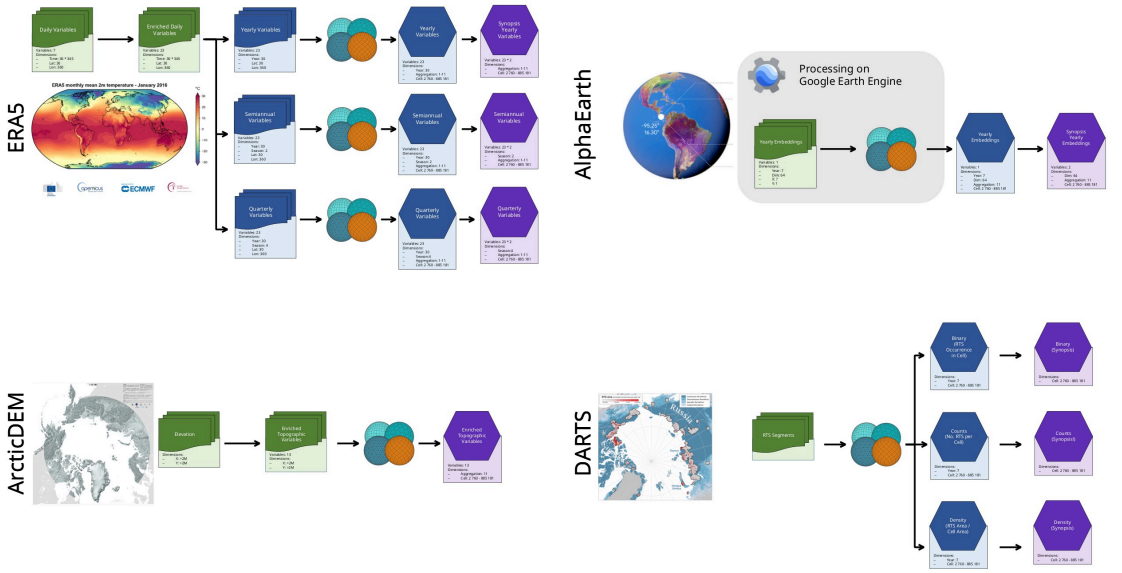

Figure 9 : Processing steps of each dataset. Hexagons represent the Discrete Global Grid System (DGGS) aligned dataset version, with blue ones having a temporal dimension and purple ones do not. Graphics of the datasets taken from the original authors: Hersbach et al. (2020), Brown et al. (2025), Porter et al. (2023) and Nitze et al. (2025).

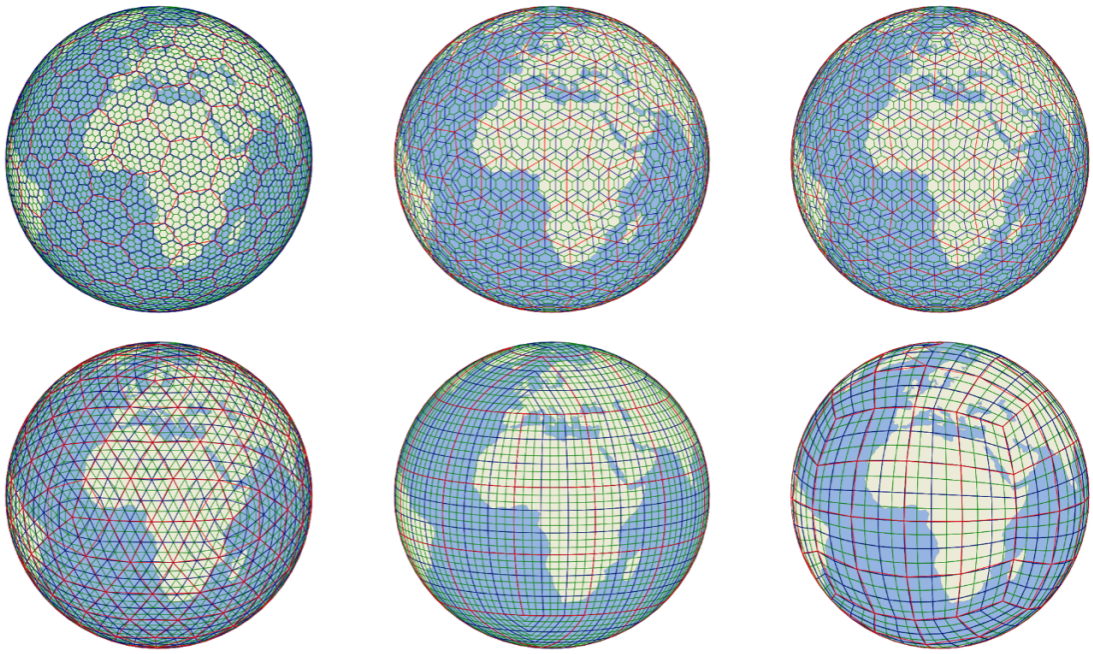

Figure 10 : Discrete Global Grid System (DGGS) grids. "Most widely implemented tessellation schemes (left to right, top to bottom): aperture 7 hexagons (7H), 3H, 4H, aperture 4 triangles (4T), aperture 9 mixed (rHEALPix), aperture 4 squares. The last two instances have a cube as a base solid, the former ones use an icosahedron." (Kmoch et al., 2022a) Graphic taken from Kmoch et al. (2022a).

often based on subdivisions of regular polyhedra such as the icosahedron, tetrahedron or cube, with cells shaped as triangles, hexagons, or quadrangles to approximate the Earth's surface with minimal distortion, as shown in Figure 10 (Kmoch et al., 2022b; 2022a; Sahr et al., 2003). DGGSs support multiple discrete scales, also called levels, which often follow a hierarchical structure, as shown in Figure 11 (Kmoch et al., 2022b). This makes data processing and alignment across different resolutions easier and, depending on the tessellation, increase the spatial-statistical correctness (Béjar et al., 2023). Depending on the tessellation and cell shape used, a DGGS has different properties regarding equal-area and equal-distance preservation or hierarchical preservation, shown in Table 1 for a selection of DGGS (Kmoch et al., 2022b; Liao et al., 2025). Overall, DGGSs provide a powerful alternative to traditional geographic coordinate systems by enabling scalable, consistent spatial data management on global and local scales.

When aggregating a dataset onto such an DGGS different methods are possible depending on the resolution of the original dataset and the resolution of the desired grid (Bousquin, 2021; Béjar et al., 2023). For somewhat equal resolutions, interpolation method like nearest-neighbor or cubic spline interpolations are the most practical ones, as they avoid data gaps (Bousquin, 2021). If the resolution of the grid is slightly sparser than the resolution of the original dataset, so that one cell contains a couple to a few tens of pixels from the original dataset, a median aggregation for each cell can be feasible. For large resolution differences where a cell contains many pixels, a histogram or Kernel Density Estimate (KDE) could be created for each cell, aggregating the cells into a distribution instead of single values. Since such aggregations, but also traditional re-projections, can introduce several biases and unwanted patterns within the data, it is

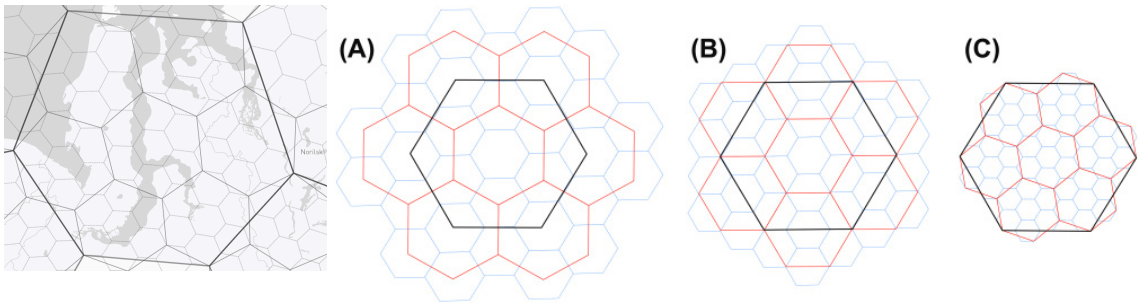

Figure 11 : Left: Different resolutions of the hexagonal Uber H3 grid over North-West Siberia, Russia, projected on latitude-longitude coordinates. Screenshot taken from the Uber H3 website. Right (A, B, C): "Hierarchical hexagon grid resolutions scaling using (A) aperture-3, (B) aperture-4, and (C) aperture-7. dggridR generated the grids for apertures 3 and 4, H3 generated grids for aperture-7." (Bousquin, 2021) Right graphics taken from Bousquin (2021).

DGGS	Geometry	Polyhedra	Area Variance	Cell distances	Hierarchy
HEALPix	Quadrangles	Rhombic Dodecahedron	$\lessapprox 20\%$	2D Grid	Strict
H3	Hexagons	Icosahedron	$\lessapprox 60\%$	~Equal	Approx.
A5	Pentagons	Regular Dodecahedron	$\sim 0\%$	~Equal	Approx.

Table 1 : Properties of different Discrete Global Grid Systems (DGGSs).

important to always calculate features on the original data, thus before aggregations or re-projections.

Healpix and Uber's H3 DGGS were chosen as base-grids for this thesis because they fulfill the requirement of almost equal cell distribution across the Arctic and of their availability in the software library "xdggs" used to create the grids (Kmoch et al., 2024). Two different DGGSs were chosen to examine the influence of the geometries on the training results. Various resolutions where chosen so that the range of resolutions to also compare results across spatial scales. Table 2 shows the chosen grid levels which where categorized into "low", "medium" and "high" resolution grids. Please note that the terms "low", "medium" and "high" to describe different resolutions were arbitrary chosen and do not comply with various non-standardized descriptions of different sensors used across the field of remote sensing. The "low" resolutions were chosen, as single cells of these grids typically cover a complete region, thus regions are represented by a single value. The "high" resolutions represent the landscape on a more local scale, as typically a single cell can cover for example a large lake. Higher levels were not used, as the number of cells of these grids is hard to manage on consumer grade hardware and would open the need for HPC, which would be unreasonable for the scope of this work. E.g. H3 level 7 has a minimum cell area of around 3.1km^2, which would compare to a $1.7\frac{\text{km}}{\text{px}}$ side length in a pixel based grid and would result in more than 6 million grid cells.

The levels 3-6 where chosen for the H3 DGGS and levels 6-10 where chosen for the Healpix DGGS, trying to mimic the range of H3. H3 level 3 is meant to be somewhat

Grid	Level	No. Cells	Avg. Area	Area Variance
H3	3 (*low*)	2 760	12 512.0 km^2	12.1%
H3	4 (*low*)	18 699	1 788.8 km^2	12.0%
H3	5 (*medium*)	128 174	255.5 km^2	12.0%
H3	6 (*high*)	885 181	36.5 km^2	12.0%
HEALPix	6 (*low*)	3 328	11 158.5 km^2	6.0%
HEALPix	7 (*low*)	13 105	2 802.8 km^2	6.2%
HEALPix	8 (*medium*)	51 211	701.8 km^2	6.2%
HEALPix	9 (*medium*)	201 487	175.6 km^2	6.2%
HEALPix	10 (*high*)	799 113	43.9 km^2	6.2%

Table 2 : Selected grid levels with their number of cells, average cell area and the variance of the cell area. The levels of the grids are grouped into low, medium and high resolutions, the color-coding applied to their underlines are later used in various plots.

comparable with Healpix level 6, H3-4 with Healpix-7 and 8, H3-5 with Healpix-9 and H3-6 with Healpix-10, however, the percentual differences in area between comparable grids are inconsistent and not small. Further, the percentual differences between cell areas within a grid can also be quite high as shown in Figure 12, with H3 having for example a difference of almost 40%. Nevertheless, this relative area difference is still a lot lower than usual latitude-longitude grids.

For later use as spatial baselines, cell-ids together with their geometries where created for all grids with the xdggs library and then transformed into GeoPandas DataFrames, resulting in nine reference geometry datasets (Jordahl et al., 2020; Kmoch et al., 2024).

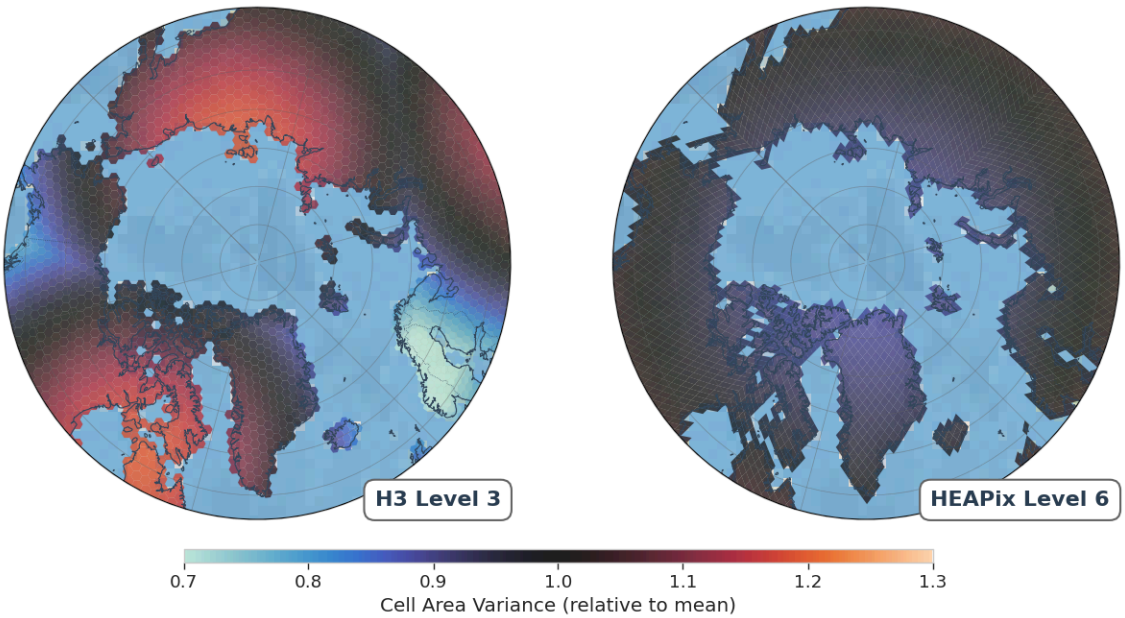

Figure 12 : Relative variance of cell areas across the the H3 and HEALPix grids.

Each reference geometry dataset was filtered to only contain cells which bounds are within 50-80° latitude and are covered by at least 10% land, and finally projected to the North-Polar-Stereographic Projection (EPSG:3413). To determine the land-water ratio the covered water area was calculated with the Open-Street-Map simplified watermask[1] (OpenStreetMap contributors, 2017). The cells themselves were not clipped to the water-mask, it was only used for the water area calculation and filtering. Finally, the reference geometry datasets where stored as GeoParquet files for use in the following steps. The creation of grids did not take up any significant amount of memory or compute. The time taken to create a grid was between few seconds for smaller levels up to a few minutes for the high levels.

3.1.2 RTS target dataset

Five different ground truth datasets (target datasets) were used in the experiments: (1) The hand-labeled RTS labels "initze/ML_training_labels: v1.0" from Nitze et al. (2024b), (2, 3) Level-2 and 3 of DARTS v1.2 from Nitze et al. (2025) and (4,5) Level-2 and 3 of DARTS v2 (unpublished). The experiment framework and the code behind it are able to support experiments also with other target datasets, as long as they provide an extent shapefile and some sort of binary feature shapefiles filled with only polygons.

All target datasets were spatially aligned, meaning aggregated, onto the 9 different grids and stored as Zarr arrays xDGGS encoded, a format which is an Xarray accessor specialized on DGGS. For each geometric cell the number and density of observed RTS intersecting with respective cell, as well as the percentual coverage of the original datasets extent were calculated. This calculation method theoretically would allow for duplicate counted RTS if one would span across two or more cells, however, through the large scale deficit between the size of a RTS and the size of the cells this should be accounted for as insignificant.

The hand-labeled RTS labels (in the following just "DARTS-labels") are a collection of RTS labels hand-drawn by multiple authors over a time period of 5 years. Foundation of these labels are PlanetScope satellite imagery with a 3.25m/px spatial resolution. The original dataset consists of multiple feature and associated extent shapefiles. However, the total spatial extent of this dataset is, compared to the size of the largest grid cells, quite small. In order to create grid-aligned datasets (1) with enough samples, both the labels and the extents were overlayed and spatially combined by taking the maximum extent, thus removing any temporal information from the dataset.

DARTS v1.2 is an RTS dataset which was created by a machine-learning model trained on the DARTS-labels dataset with PlanetScope imagery. Only some minor post-processing manual filtering was applied by the authors, so this dataset must be seen as a model-derived dataset. In this work, the Level-2 data of the DARTS v1.2 dataset was used to calculate two grid-aligned target datasets. The Level-2 data consists of yearly maximum-area shapes based from all observations within a single year. The spatial extent of DARTS v1.2 is larger than the one from the DARTS-labels and partially covers the years

[1] https://osmdata.openstreetmap.de/data/water-polygons.html, accessed 20. November 2025

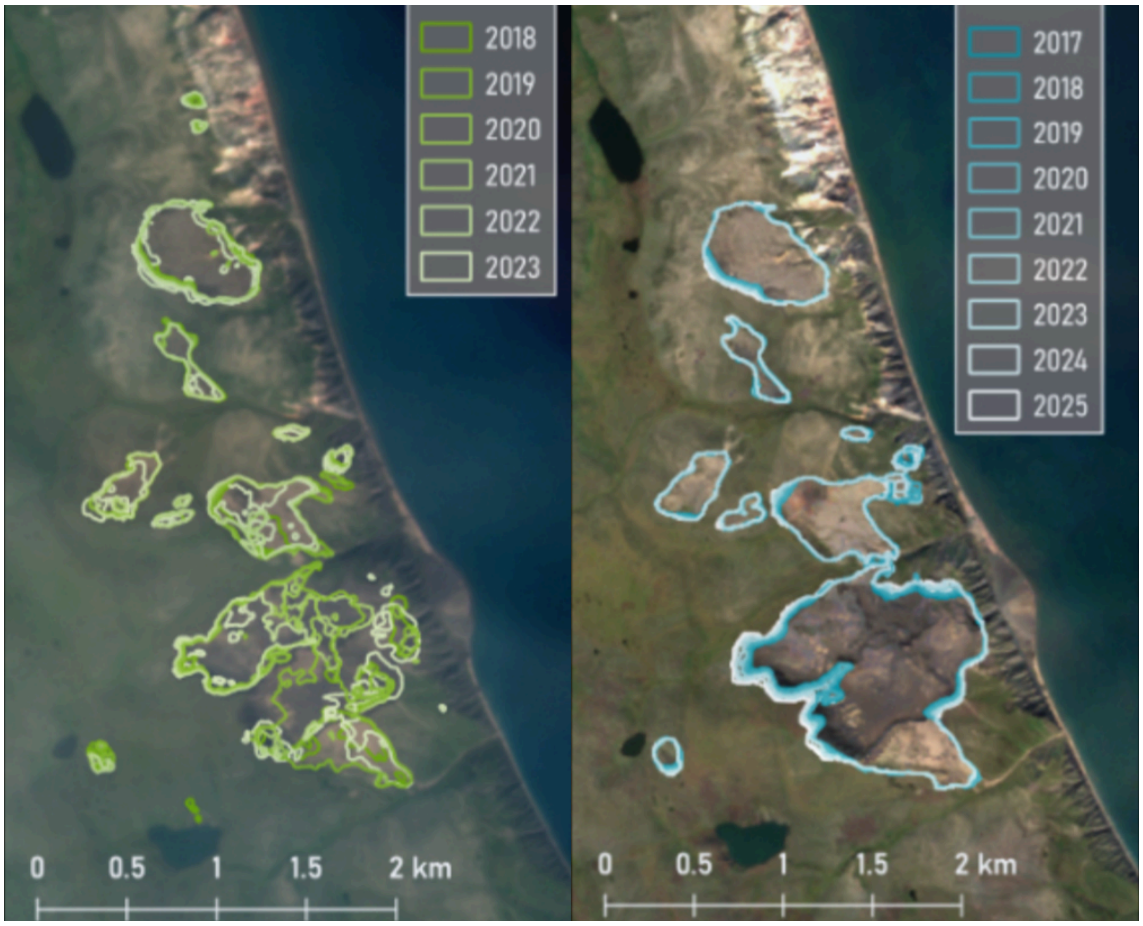

Figure 13 : Retrogressive Thaw Slump (RTS) delineations of the Database of AI-detected Arctic RTS footprints (DARTS) v1.2 and v2 datasets. The v1.2 delineations are visible noisy with smaller segments present within larger ones, because the model used is unsure about the "activeness" of these parts. The v2 delineations are much more stable in comparison, with less noise. Graphics taken from (Küpper, 2026).

2018 to 2023. Grid-aligned datasets (2) were created by simply grouping the extents and delinations by years, resulting in temporal affine grids.

Since the spatial extent vary between the years and focuses on predefined core-regions instead of the complete pan-arctic, a second non-temporal version of the grid-aligned DARTS v1.2 datasets were created, again by overlaying the delineations and taking their maximum extent. This version is referred to as "DARTS v1.2 synopsis" by this work.

Having access to the unpublished follow-up dataset of DARTS, namely DARTS v2, two more grid-aligned target datasets were created. DARTS v2 used the same ML pipeline as DARTS v1.2 does with some tweaks. Instead of PlanetScope imagery, DARTS v2 is based on 10m Sentinel-2 mosaics, allowing for a spatial extent of the complete northern arctic for all years from 2017 to 2025. Further, the quality of the delineations are much more stable across the years, thanks to more models used in the ensemble, better image quality and improved training of the models. This quality improvement is clearly visible in Figure 13. Like before, grid-aligned datasets (4) were created by grouping the extents and delineations by years. Even if the spatial coverage of DARTS v2 is large and also equal across all the years, non-temporal grid-aligned datasets (5) were created, similar to the versions before.

For each of these target datasets, the density and the count per cell was calculated. Further, classification target variables were calculated by creating binned categories for the density and count and a binary target for RTS occurrence in a cell.

Similar to grid creation, no significant amount of memory, compute or time was necessary to create the target datasets. In total, 45 (5 source-dataset × 9 grids) grid-aligned target datasets were created and are available in the framework, each containing five different target variables. However, only the datasets based on the two DARTS v1.2 variants were used for the experiments, as the DARTS-labels variants contained not enough samples to calculate statistical correct evaluations and the DARTS v2 dataset was not ready in time. Table 10 (Appendix B.2) shows the sample counts (number of cells) of each variant. Further, only the binary and density target variables were used in the final experiments, because the count target was statistical unsound for the darts_v1 dataset because of noise and first training explorations targeting the binned categories didn't succeeded.

3.1.3 ERA5-Land

The ERA5 datasets are a set of reanalysis products of the Copernicus Climate Change Service, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) under the European Space Agency (ESA) Copernicus programme (Hersbach et al., 2020). They combine a state-of-the art weather model with an extensive archive of observations, filling observation gaps with simulated data to generate a physically consistent, globally complete dataset. ECMWF Reanalysis v5 (Land dataset) (ERA5-Land), one specific dataset, delivers hourly fields on a 0.1° (≈10 km) grid for many atmospheric, land-surface and oceanic variables, such as 2-meter air temperature (t2m) and total precipitation (tp) which are widely used for climate and impact studies. While the sparse observational network in the high-latitude Arctic can lead to occasional biases, the reanalysis's temperature and wind fields remain reliable across the polar region, making ERA5-Land an important data-source for Arctic climate research (Graham et al., 2019).

ERA5-Land hourly data was downloaded and aggregated to obtain climate describing data. The data was downloaded from the Earth Data Hub, because of their easy-to-use and efficient Zarr store (Destination Earth, 2025; Zarr Community, 2025). Table 9 (Appendix B.1) shows the variables used and engineered by this work. Because the ERA5-Land dataset consists of hourly data on a 0.1° grid, many different temporal and spatial aggregations were made to create more suitable representations of the seasonal regimes, and to fit the data onto the DGGS grids. Three versions of DGGS grid-aligned ERA5-Land datasets were created to represent different seasonal regimes: (1) annual representations, (2) semiannual (6 months) representations and (3) quarterly (3 months) representations.

To save storage, the hourly data was aggregated "on the fly" into daily data while downloading. When aggregating or extracting data from ERA5-Land, multiple methods can be feasible thanks to the high temporal resolution of the dataset. Because different variables need different temporal aggregation approaches to preserve climatological

correctness, ERA5-Land differentiates between two types of variables: instants and accumulating variables. Temperature, for example, is an instant variable, which describes the "state" at a specific location and time, thus aggregations along both temporal and spatial dimensions need to preserve this state-fulness, making a summation for example meaningless. Accumulating variables, like precipitation, on the other hand describing a quantity, thus temporal aggregations like summations are viable and useful. However, for spatial aggregations the viability of an aggregation method depends on the unit of the aggregation, e.g. precipitation is measured in m ($\frac{m^3}{m^2}$) by the ERA5-Land dataset, having already a spatial component. Thus, summing the precipitation over area is equally meaningless and wrong like summing instant variables.

Because of the large amount of available variables in the ERA5-Land dataset, only a subset of 7 variables was selected for download and further use. The hourly 2m air temperature (t2m) was aggregated into minimum, maximum and mean daily temperature. With snow-cover (snowc), snow-depth (sde) and lake-ice-bottom-temperature (lblt) three more instant-type variables were chosen. The mean snow-cover and snow-depth were used as daily representation of both variables, as they are not expected to change drastically over a single day. With lake-ice-bottom-temperature, a highly speculative variable was used in the hope that the estimations are good enough to find pattern with ML in it. As for lake-ice the thawing related temperatures are interesting for permafrost dynamics, e.g. in talik formations, the maximum lake-ice-bottom-temperature was chosen as a daily aggregation. Precipitation (tp), snow-fall (sf) and surface-sensible-heat-flux (sshf) were summed over the day, to retrieve daily total amounts. The ERA5-Land dataset stores precipitation in an accumulating way on a daily basis: instead of hourly precipitation, the total amount of precipitation is accumulated over a day and resetted at the first hour of a day, at 1am. Thus, taking the sum over the hours would result in a way to large amount of precipitation. Instead, the first value of the next day 0am or 12pm was taken as the total daily precipitation. As temperature, precipitation and the snowfall related variables are obvious choices, the surface-sensible-heat-flux was chosen in the hope that the (simulated) heat transfer could also matter to permafrost dynamics. As there is a lack of heat-flux measurements not only across the arctic, but in general, this variable is highly speculative and solely based on simulation data.

Based on the daily variables, engineered variables where derived in a second step, as formulated in Equation 1. As temperature being the most relevant climate variable for permafrost, three variables where directly calculated for 2m above-ground temperature to further express the temperature behavior in further aggregations. The temperature range was calculated by substracting the daily minimum temperature from the daily maximum temperature. This range was then used to calculate the average daily temperature by dividing it by 2. This average daily temperature is different than the mean daily temperature, which is derived from all hourly values of a day. This is important for the calculation of the daily temperature skew, a measure which shows how far off the actual mean is from the expected average. The temperature skew was calculated by substracting the minimum temperature from the mean temperature and then normalizing this difference by dividing it by the temperature range. Thus, the temperature skew is always a value between 0 and 1, with values larger than 0.5 express a day which was relatively more hot than cold, having a cold "spike" at some point of the day, while values smaller than 0.5 express the opposite.

$$\begin{aligned} \mathrm{t2m}_{\mathrm{range}} &= \mathrm{t2m}_{\mathrm{max}} - \mathrm{t2m}_{\mathrm{min}} \\ \mathrm{t2m}_{\mathrm{avg}} &= \frac{\mathrm{t2m}_{\mathrm{range}}}{2} \\ \mathrm{t2m}_{\mathrm{skew}} &= \frac{\mathrm{t2m}_{\mathrm{mean}} - \mathrm{t2m}_{\mathrm{min}}}{\mathrm{t2m}_{\mathrm{range}}} \end{aligned} \tag{1}$$

Because the state of water changes at 0°C, 273.15°K, this value marks an important aspect of permafrost dynamics. Hence, four more variables where calculated to express the environmental state for each day: freezing-days, freezing-degree-days, thawing-days and thawing-degree-days, following the Canadian glossary of climate variables (*Glossary of Climate Related Terms*, n.d.). Freezing-days were defined as days which have an average (2m air) temperature below 0°C, while freezing-degree-days are defined as degrees below zero. For thawing-days and thawing-degree-days, the definition and calculation is similar, just with temperatures above 0°C. For a single day, these values are not very telling compared to the average temperature, however, summed over a longer time-period they express how much thawing or freezing potential existed in that period.

Apart from the temperature related engineered variables, precipitation-occurrence, snowfall-occurrence and a naive-snow-isolation metric where calculated. A day counts a precipitation- or snow-occurrence if there was more than 1mm of rain or snow respectively. The naive-snow-isolation metric was calculated by multiplying the snow-cover by the snow-density.

After the data engineering based on the daily variables, the data was aggregated into their final respective temporal representations. For the annual representation an aggregation period was defined as the period from 1st October of the previous year to the 30th September of a year to not split the winter season. For example, values aggregated into the year 2020 were taken from the 1st October 2019 to the 30th September 2020. Similarly, a winter season of the semiannual representation starts at the 1st October and ends on the 31st of March while the summer season starts at the 1st April and ends on the 30th September. The quarterly seasons where defined as January to March (JFM), April to June (AMJ), July to September (JAS) and October to December (OND). All accumulating variables were summed, instant variables were aggregated with the median over the respective period, except for 2m air temperature minimum and maximum, which were aggregated using minimum and maximum values respectively. Additional to the aggregations, further variables were derived: the snow-cover minimum and maximum of the period and the effective-snow-depth based on Slater et al. (2017). The effective-snow-depth is a highly engineered variable which estimates how much the snow layer thermally isolates the ground over a (winter) season. It is calculated with the Equation 2, with M being the months of the observation season and S_m being the snow depth measured in this month. For the annual aggregations, the day-of-first-thaw, day-of-last-thaw, day-of-first-freeze, day-of-last-freeze and the thawing-period-length was also derived.

$$S_{\mathrm{depth,eff}} = \frac{\sum_{m=1}^{M} (S_m \cdot (M + 1 - m))}{\sum_{m=1}^{M} m} \tag{2}$$

Since the quarterly, semiannual and annual aggregations were still on the 0.1° grid, different aggregation methods where used to project the data onto the DGGS grids. Because of

the low resolution of the 0.1° grid, it was not feasible to use the same aggregation methods across all DGGS levels. A single cell of the H3 grid for example, depending on the latitude, contained around 235 pixels on level 3 while on level 6 some cells only contained two pixels. For the "low" resolution grids, meaning H3 levels 3 and 4 and Healpix levels 6 and 7, many aggregations where drawn from the large amount of pixels contained by a cell: mean, median, standard deviation, minimum, maximum, and the 1st, 5th, 25th, 75th, 95th and 99th percentile. For the "medium" resolution levels H3 level 5 and Healpix level 8 and 9 only the mean was drawn, since only a handful of pixels were contained by each cell. For the "high" resolution levels (the highest used by this work), data was interpolated by the nearest neighbor method. All spatial aggregations relied heavily on CPU compute, since Cupy lacking support for nan-quantile. Simultaneously, the number of pixels within a cell of the higher resolution grids was too small to overcome the data movement overhead of using a GPU, when aggregations where performed on a per-cell-basis. As the dataset was too large to be moved completely onto the GPU, a vectorized approach would have been more suitable for such kind of operations. However, as the time of writing, no vectorization GPU implementation for such geometric aggregations existed in any of the libraries used. Still, through the use of 40 parallel CPU processes, this task was doable within few hours of compute. As the time to compute these aggregations scales linear with the number of processes, this task is theoretically also possible to do on consumer grade hardware, it would just take more time.

In order to create timeless variants alongside the quarterly, semiannual and annual datasets, further temporal aggregations were applied on these datasets by calculating the mean and the linear trend. Similar to the DARTS targets, these variants are referenced as "synopsis" datasets in the following chapters, as they are meant to describe an overview or synopsis of the data along the temporal axis. While using the mean and trend for these synopsis variants, drastically reduces the amount of features in a tabular ML setting where otherwise each year would be handled as a separate feature. Non-linear and event information get lost through these operations, however, adding further temporal

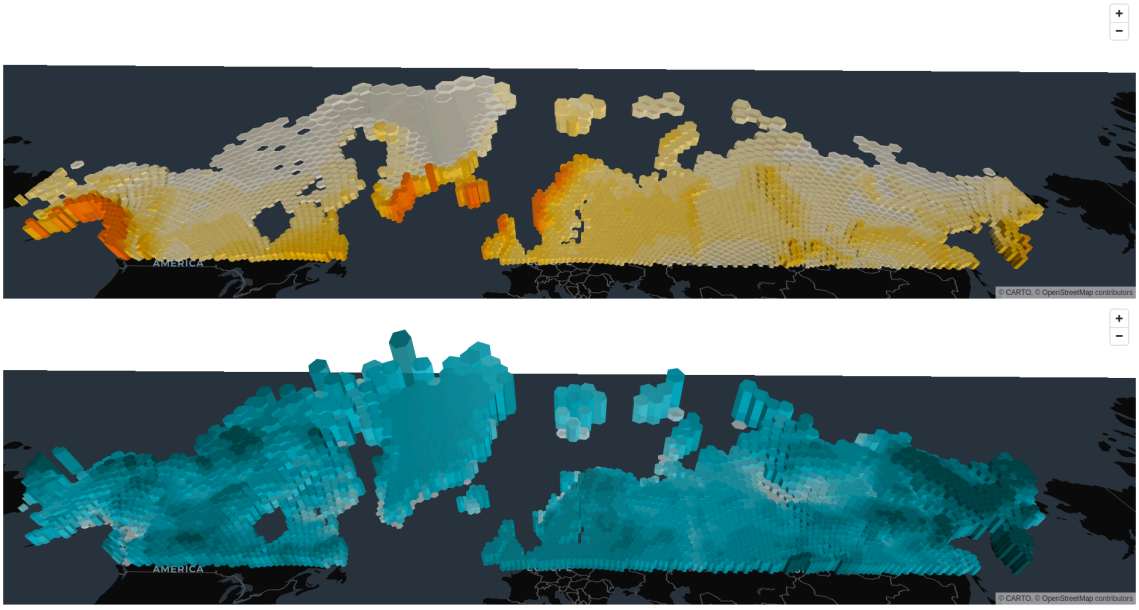

Figure 14 : Top: H3 level 3 aligned mean annual total precipitation. Bottom: Trend of the effective snow depth. Visualized in the Streamlit Dashboard.

aggregation metrics like the Mutual information (MI), autocorrelation, maximum or the mode would increase the number of features.

3.1.4 AlphaEarth Satellite Embeddings

As mentioned before, the use of generated feature embeddings have become more and more common in various fields. To incorporate such embeddings in this work, the satellite embeddings from AlphaEarth Foundations were chosen because of their temporal completeness compared with TESSERA. Instead of downloading the complete dataset, wich is more than 200TB large, Google Earth Engine was used to crop and aggregate the data onto the DGGS grids before download, drastically reducing the size of the dataset. To speed-up the calculations and prevent out-of-memory errors within Google Earth Engine, the scale parameter was used to use only a subset per cell for all aggregations. The scale was set individually for each grid, so that each cell subset had approx. 10 000

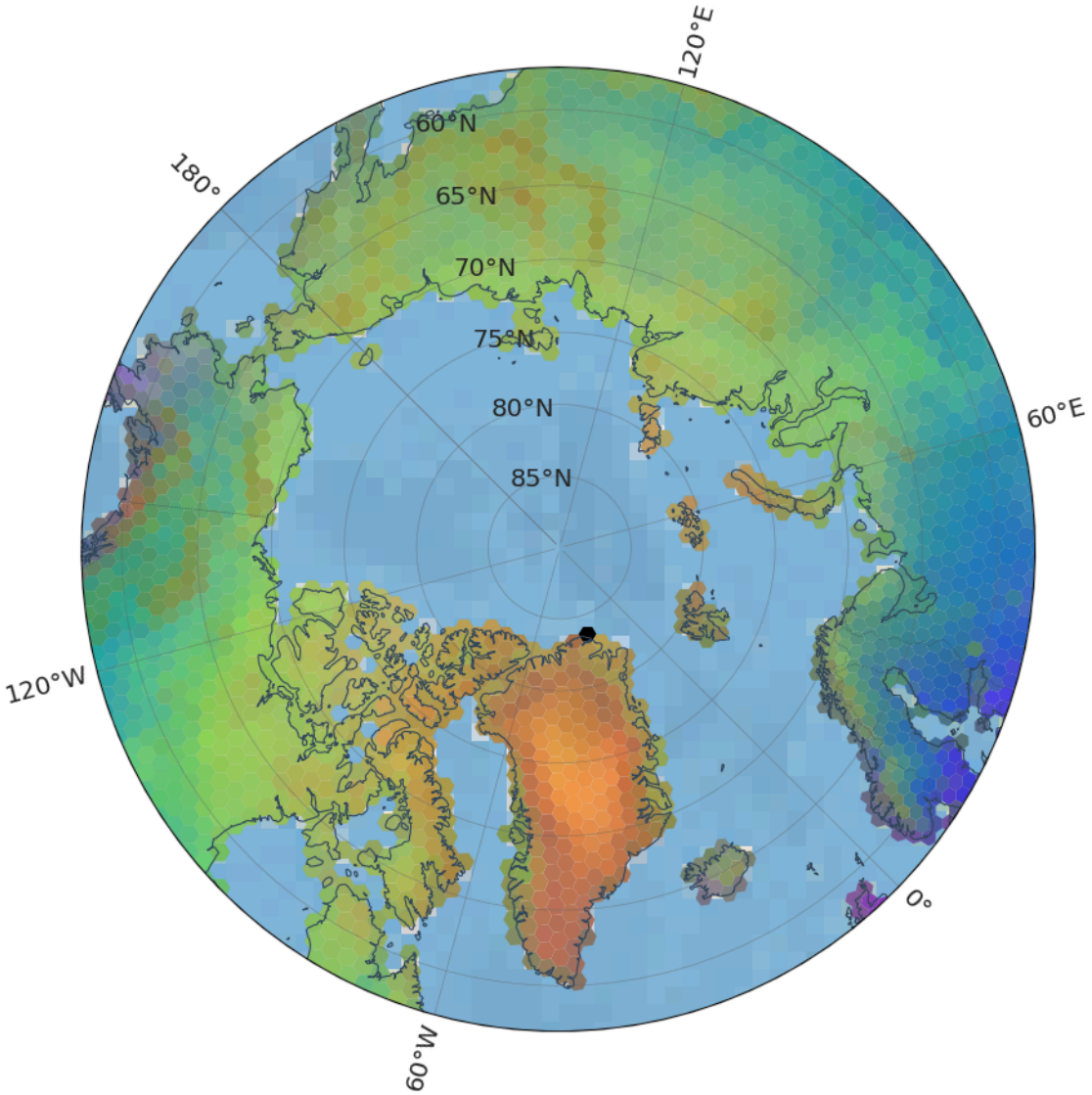

Figure 15 : H3 level 3 aligned AlphaEarth Satellite Embeddings. Band A40, A50, A48 were used for the RGB representation.

pixel to aggregate over. For example, this 10 000 pixel subset is equivalent to a 1600 $\frac{\text{m}}{\text{px}}$ scale for the low resolution grids and approx. 100 $\frac{\text{m}}{\text{px}}$ for the high resolution grids. Further, the computation and download were done in 100-cell batches to further ensure that compute limits of Google Earth Engine were not reached. The same aggregations as for the "low" resolution levels of the ERA5-Land where applied for all grids: mean, median, standard deviation, minimum, maximum, and the 1st, 5th, 25th, 75th, 95th and 99th percentile. While the data of the low resolution grids was quickly computed and downloaded, the high resolution grids took a week to download. Because Earth Engine was used this step can also be done with consumer grade hardware. Similar to the ERA5-Land dataset, a "synopsis" variant was created by calculating the mean and the trend across the very short, but existing temporal dimension.

3.1.5 ArcticDEM

To add comprehensive terrain data, the 32-meter resolution ArcticDEM v4.1 mosaic was also downloaded and aligned with the DGGS grids. The ArcticDEM is a high-resolution digital elevation model that covers the entire Arctic region and is offered at three native resolutions: 2m, 10m and 32m (Porter et al., 2023). It is a Digital Surface Model (DSM), "that portrays first-return elevation values that include vegetation, tree canopy, buildings, and other man-made surface features" (Porter et al., 2023). A global mosaic and data strips are provided in Sea Ice Polar Stereographic North projection and referenced to WGS84 horizontal datum (EPSG:3413) for all resolutions, with strips being available for the years 2007 to 2024. The elevation is derived from stereoscopic optical imagery acquired by the WorldView constellation, processed with a TIN-based Search-Space Minimization photogrammetric pipeline that automatically extracts 3D point clouds and interpolates them into a seamless DEM. Because the source imagery is optical, the product inherits some of its limitations: cloud cover, haze and other atmospheric effects can introduce noise, which is propagated into the elevation data of the strips. As the generation process of the elevation data is strictly automatic, no human corrections where made on the effected parts. Because the mosaic is based on strips collected over several years, temporal inconsistencies may appear where surface change has occurred. Moreover, the remote-sensing nature of the acquisition means that ground-control points are sparse in many high-latitude locales, leading to occasional horizontal and vertical mis-alignments of up to 4m. Despite these constraints, ArcticDEM remains one of the most detailed and widely used elevation datasets for Arctic geomorphology, infrastructure planning, and climate-impact assessments.

The dataset was acquired using the SmartGeocubes library, with the around 270GB download spanning approximately eight hours. (Hölzer, 2025) The SmartGeocubes library highly optimized the download, simultaneously downloading and writing the downloaded data to the drive, using approx. 10GB of memory. The resulting compressed Zarr store had a size of approx. 160GB. Afterwards, a set of topographic metrics was computed from the elevation data: Slope, aspect, and curvature were calculated using the xrspatial library, while own CuPy implementations were use to calculate three topographic ruggedness indices: Terrain Ruggedness Index (TRI) following Riley et al. (1999), Vector Ruggedness Measure (VRM) following SAPPINGTON et al. (2007), and

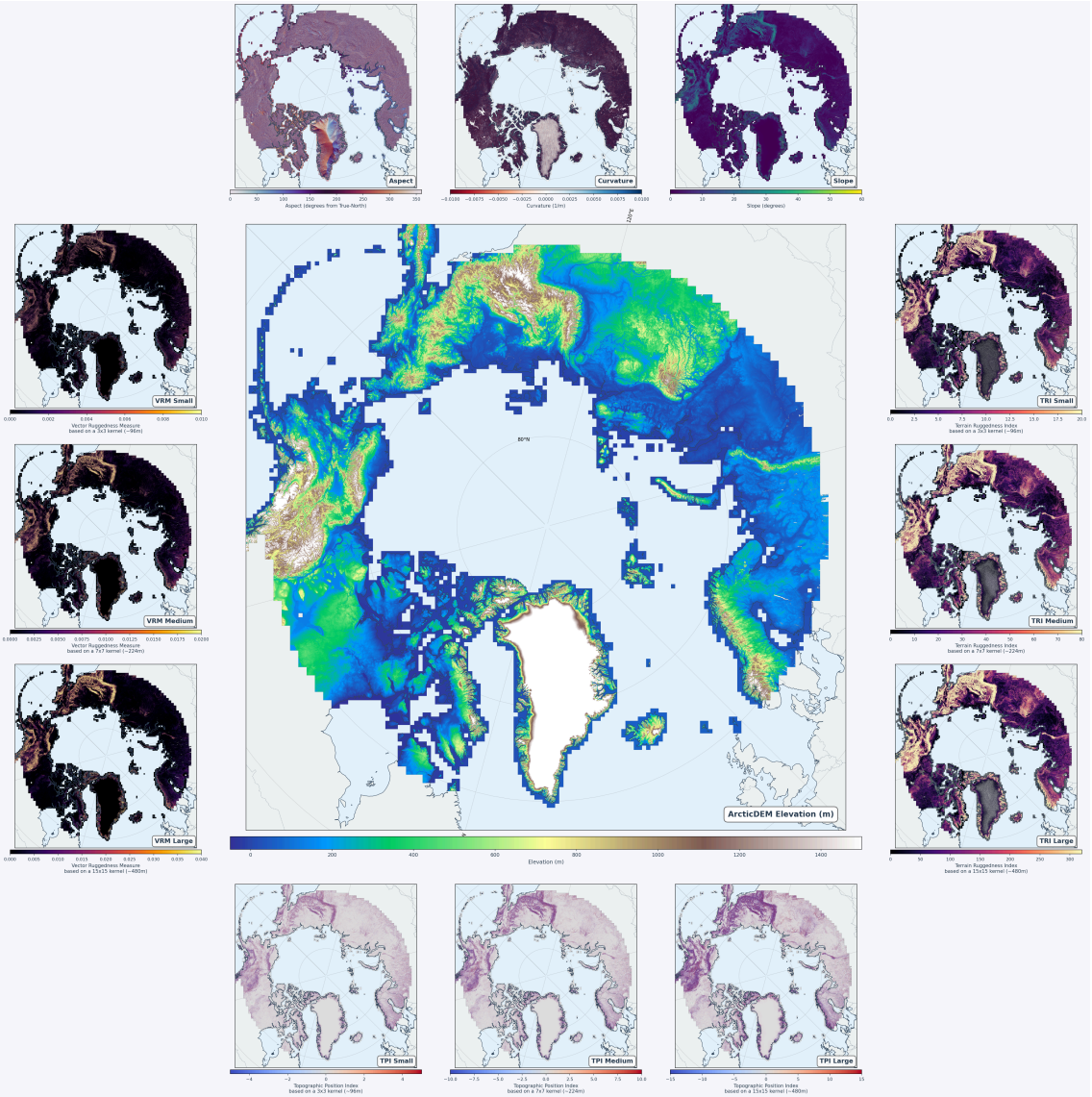

Figure 16 : ArcticDEM elevation and derived topographic variables, before DGGS alignment.

TPI, also known as relative elevation. To capture multi-scale terrain variability, three kernel sizes were employed: 96m (3×3 pixels), 224m (7×7 pixels), and 480m (15×15 pixels), representing small, medium, and large spatial resolutions, respectively. Figure 16 shows the DEM and computed topographic variables for the complete Arctic. The data enrichment took around 2h on a single A100 GPU node (40GB) with a local dask cluster consisting of 7 processes, each using 2 threads and 30GB of memory, making up a total of 210GB of memory. The enrichment scripts can be configured easily to consume less memory by reducing the number of processes or threads. More processes or thread could not be used to ensure that the GPU does not run out of memory.

For the DGGS alignment, the same aggregations were calculated as for the AlphaEarth embeddings: mean, median, standard deviation, minimum, maximum, and 1st, 5th, 25th, 75th, 95th, and 99th percentiles. Similar to the ERA5-Land aggregations, this dataset alignment relied heavily on CPU compute. To effectively reduce the compute time of the aggregations to approx. 30 minutes per grid, 40 concurrent CPU processes were used utilizing a total of 300GB of memory on an HPC cluster. Nevertheless, these

aggregations would also work on customer grade hardware without problems, just would take more time.

3.1.6 Dataset ensembling

With all source datasets aligned to the DGGS and stored as Zarr arrays, a wrapper utility for dataset ensembling was developed to enable efficient and flexible construction of composite datasets. This utility allows experimenting with different combinations of multiple environmental variables, spatial-aggregation methods and temporal spans. Each resulting ensemble dataset is uniquely defined by its configuration of variables, aggregation method, and time range, ensuring reproducibility and deterministic outcomes. The resulting datasets are fully compatible with both training and inference workflows. For training purposes, the target dataset and the task as defined in Section 3.1.2 can be specified.

This wrapper utility heavily leverages the lazy nature of Zarr arrays, which allows on-the-fly access to only the relevant spatial cells and variables during the ensembling process. To further optimize workflow efficiency, generated ensembles are cached on disk upon first access, reducing computational overhead for subsequent reads and accelerating iterative development cycles.

3.2 Machine-Learning methods

As discussed before, the use of supervised ML and DL methods have emerged as standard tools in remote sensing, earth-observation and mapping. Unlike traditional physics-based models that rely on explicit equations and assumptions, ML approaches are data-driven, allowing for modelling without knowledge about the underlying processes. An ideal setup for RTS occurrence or ground-ice content estimation, where underlying processes and material quantities are not well known.

3.2.1 Machine-Learning vs. traditional Environmental Modelling

The key distinction between ML and traditional environmental models is presence of prior knowledge. Traditional environmental models are typically mechanistic or process-based, abstracting physical, chemical, or biological laws to simulate the behavior of the real world under varying conditions (Razavi et al., 2022; Willard et al., 2022). Such models vary in size and complexity, and even models of relatively low conceptual complexity can become computationally expensive, for example because of repeated numerical integration or because the computational cost grows very rapidly (often exponentially) with increasing temporal, spatial or dimensional resolution (Bröchin et al., 2024; Wang et al., 2022). ML on the other hand, seen from a probabilistic view, abstract over observed data by trying to find patterns within it by utilizing the data to adjust parameters of a (statistical) model in order to minimize an objective function (Bishop, 2019). This becomes especially helpful if underlying processes are not well studied or

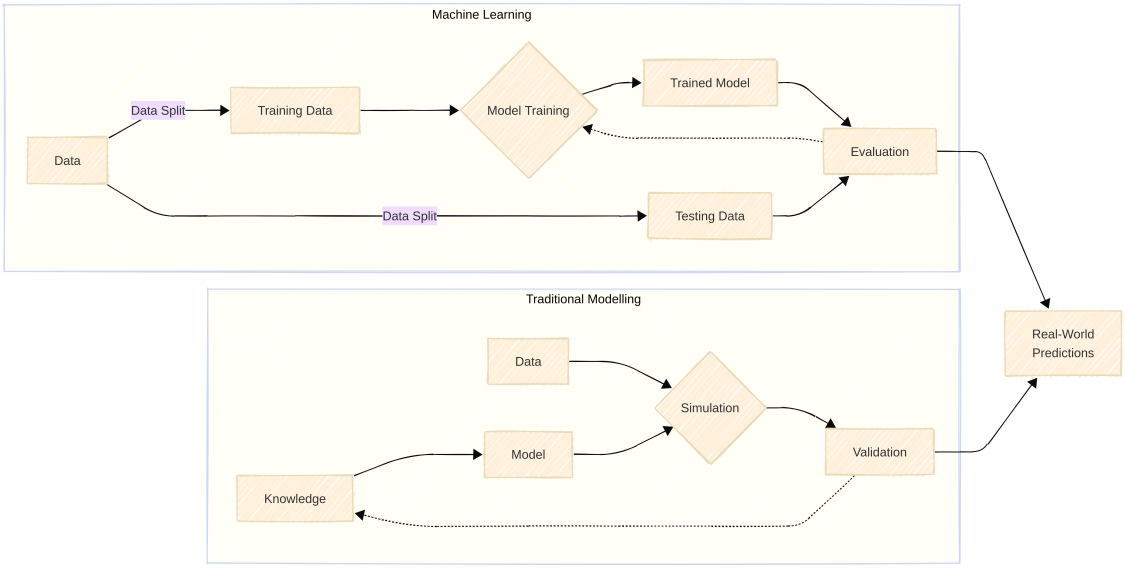

Figure 17 : Abstract workflow of machine-learning in contrast to traditional simulations.

generally hidden and not observable (Bishop, 2019; Pichler & Hartig, 2023; Razavi et al., 2022). However, ML and especially DL approaches are inherent computationally expensive through large amount of trainable parameters or many optimizer steps (Bishop, 2019). Both approaches can sometimes be difficult to calibrate, traditional models often require some sort of parametrization of unknown or unmeasurable variables used in their formulas, while ML and especially DL approaches often need to search for the best hyperparameter combination, which further increase compute needs (Bishop, 2019; Razavi et al., 2022; Willard et al., 2022).

While consent in the scientific community state that traditional process-based models offer a better interpretability, it was recently argued that this statement truthiness depends on the definition of the term interpretability considering the use case of the model (Lipton, 2016; Razavi et al., 2022; Willard et al., 2022). Uncontroversially they offer better algorithmic transparency than ML models, however, depending on their parametrization, engineered inputs and algorithmic complexity their outputs can be just as untransparent and nonintuitive as ML models (Lipton, 2016). Still, opposed to ML model they offer causality, making them intrinsical suitable for extrapolation to novel scenarios, a big shortcoming of ML approaches where out-of-distribution data plays a crucial role in experiment setup and validation because of the data-dependence of the training steps (Bernardini et al., 2024). It was shown that under the right conditions, meaning the presence of many training samples covering the complete possible distribution of the underlying process, ML models outperformed process-based models (Bernardini et al., 2024). This said, ML on sparse or incomplete datasets is still an unsolved problem (Horenko, 2020; Vecchi et al., 2022). Another shortcoming of ML methods is the inability of analytical Uncertainty Quantification (UC), meaning the derivation of model and output uncertainties without Monte-Carlo runs (Bonas et al., 2024; Franzen et al., 2024). Recent approaches created hybrid approaches which incorporate ML into process-based models or vice-versa to combine the best of both worlds, however this field of computer science is not well-studied yet (Razavi et al., 2022; Shen et al., 2023; Willard et al., 2022).

3.2.2 Validation and Evaluation

As ML and DL methods are data-driven, a robust and rigorous model evaluation and validation is even more essential for these methods to achieve usable and reliable results (Pichler & Hartig, 2023; Tsamardinos et al., 2015).

Depending on the task and distribution of the data different metrics can be used to evaluate ML models, Table 3 shows a selection of such metrics (Thieu, 2024). For classification tasks, both binary and multi-class, widely used metrics are accuracy, precision, recall, the F1-score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) (Rainio et al., 2024; Thieu, 2024). The precision is a metric used to tell how accurate a model is regarding a positive prediction, while recall is used to tell how likely true positives will be predicted as positives. Analogue to recall, the specificity tells how likely true negatives will be predicted as negatives. The F1-score is a mix between them and is a good first indication about the overall performance of a model. For datasets with an uneven distribution of target classes, the accuracy is often biased towards the

Metric	Measures	Formula
Accuracy	Proportion of correct predictions	$\frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}$
Precision	Proportion of positive predictions that are correct	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$
Recall	Proportion of actual positives correctly identified	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$
Specificity	Proportion of actual negatives correctly identified	$\frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}$
F1-Score	Harmonic mean of precision and recall	$2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}$
AUC-ROC	Discrimination ability	Integral of TPR vs FPR
MAE	Difference between predictions and actual values	$\left(\frac{1}{n}\right)\sum_{i=1}^{n}\lvert y_i-\hat{y}_i\rvert$
MSE	Sensitivity on outliers	$\left(\frac{1}{n}\right)\sum_{i=1}^{n}(y_i-\hat{y}_i)^2$
RMSE	Square root of MSE; original units of target variable	$\sqrt{\left(\frac{1}{n}\right)\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}$
R^2	Proportion of variance in target explained by model	$1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$

Table 3 : A selection of classification and regression metrics.

class with the most occurrences. The AUC-ROC is a more complicated metric showing how well a model can separate between classes by creating a function between the recall and the specificity based on multiple thresholds used to determine the predicted class based on the predicted probabilities returned by the model. In a multi-class setup, these classification metrics can also be calculated per class and by micro- or macro-averaging. Micro-averaging aggregates the total true positives, false positives, and false negatives across all classes before computing precision, recall, or F1, resulting in weighted metrics by class frequency. Macro-averaging on the other hand first computes each class's metric independently and then averages them, giving equal importance to every class regardless of size. In regression tasks, metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) assess how closely the predicted values match the continuous ground truth (Botchkarev, 2019; Plevris et al., 2022; Thieu, 2024). These metrics differ in sensitivity to outliers and interpretability, with for example MSE penalizing larger errors more heavily than MAE. Objective functions, also known as loss functions, differ from metrics as they are used as to guide the optimization process at model training by quantifying how well the model performs on the training data in respect to the given objective (Terven et al., 2025). Not all metrics can be used as objective functions, as some ML and especially DL methods depend on an differentiable loss function, a property which not all metrics possess.

Table 4 shows a different set of metrics which can be used to assess the similarity between two variables, often used to measure correlations between two features. The Pearson correlation coefficient, also known as Pearson's r is widely used to assess linear relationships between two variables, ranging from –1 to +1. (Pearson, 1900) A value of –1 indicates a strong negative correlation, 0 implies no correlation, and +1 signifies a strong positive relationship. In a similar fashion like Pearson's r, the Spearman rank correlation coefficient, also known as Spearman's ρ is used to assesses monotonic relationships between two variables, also ranging from –1 to +1. (Spearman, 1904) Unlike Pearson's r, Spearman's ρ also captures non-linear relationships if they are non-recurring. For recurring relationships, e.g. the relationship of x and $f(x)$ values of a sine curve, the MI is a more suitable measure. (Shannon, 1948) It expresses the amount of information shared between two random variables.

Metric	Measures	Formula
Pearson's r (Pearson correlation coefficient)	Linear correlation	$\frac{\mathbb{E}[(X-\mu_X)*(Y-\mu_Y)]}{\sigma_X * \sigma_Y}$
Spearman's ρ (Spearman's rank correlation coefficient)	Monotonic correlation	$\frac{\mathrm{cov}[\mathrm{R}[X],\mathrm{R}[Y]]}{\sigma_{\mathrm{R}[X]} * \sigma_{\mathrm{R}[Y]}} \mid \mathrm{R}:\mathrm{Rank}$
Mutual information	Mutual dependence	$I(X;Y) = H(X) - H(X\|Y)$

Table 4 : A selection of similarity metrics.

Overfitting occurs when a model learns noise or irrelevant patterns in the training data instead of underlying generalizable relationships, leading to poor performance on unseen data. This is particularly problematic in spatial datasets where nearby observations are highly correlated, increasing the risk that the model memorizes local patterns rather than capturing true environmental drivers. In order to assess over-fitting, usually a portion of the data is reserved as a test set on which the model is evaluated after training (Tsamardinos et al., 2015). However, using a single train-test split can lead to overly optimistic or unstable performance estimates, especially when data is limited or highly structured, e.g. for spatial datasets (Pang et al., 2023). Cross-Validation (CV) solves this by repeatedly training and testing on different subsets of the data and combining the metrics of the tests, resulting not only in more robust evaluation metrics but also in estimates about performance variance. Standard CV can be adapted to specific settings, by changing the method used for splitting between training and testing subset. For example stratified k-fold preserves class proportions in imbalanced classification, or spatially-aware splits respect spatial autocorrelation by grouping nearby observations into the same fold. Such strategies reduce optimistic bias and better reflect real-world deployment conditions. Figure 18 shows a collection of common folding strategies.

Most of the algorithms used to train ML and DL models and the architecture of the models themselves can be tweaked through setting so-called hyper-parameters (Bergstra & Bengio, 2012). As they heavily influence model performance, it is crucial to find a good or even the best configuration of hyper-parameters for a given model and dataset, which results the problem of Hyperparameter-Optimization (HPO) (Bergstra & Bengio, 2012; Florek & Zagdański, 2023). Multiple approaches for HPO exist, as are shown in Table 5,

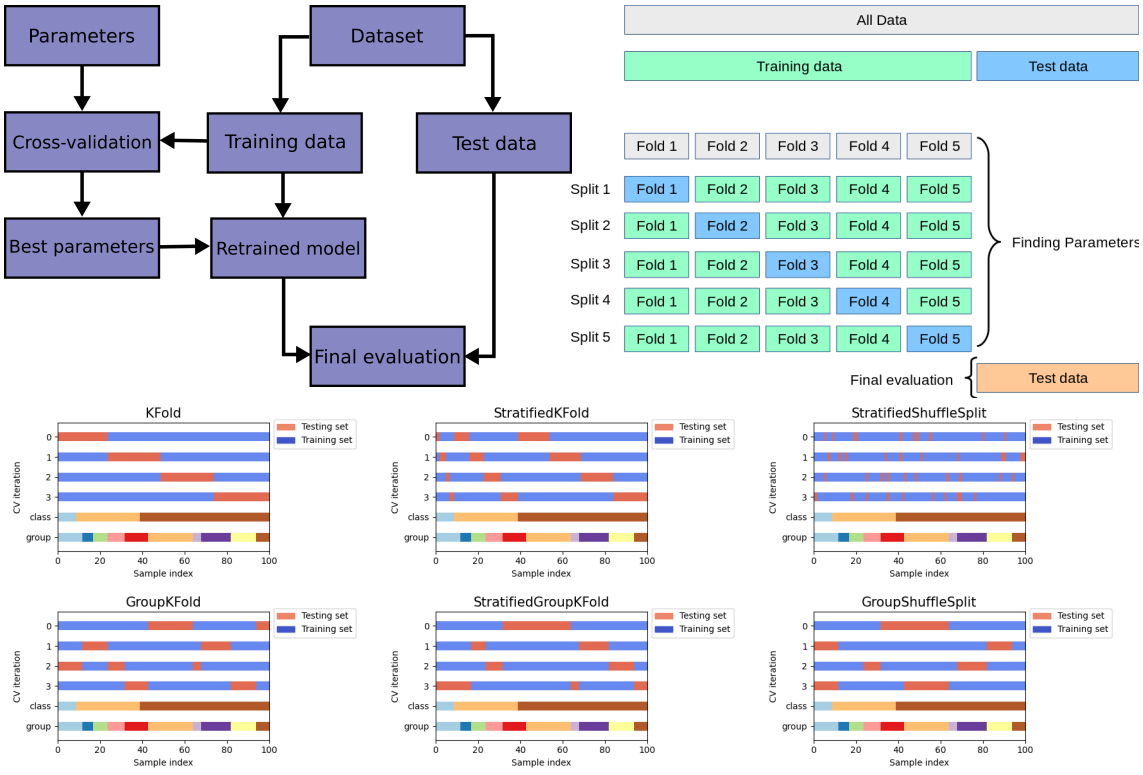

Figure 18 : Top: Cross-Validation (CV) in a Hyperparameter-Optimization (HPO) setup. Bottom: Different fold-split methods. Graphics taken from the SciKit-Learn 1.8.0 Documentation. (Pedregosa et al., 2011)

	Space Exploration	Method
Grid Search	Discrete & Complete	Brute-force
Random Search	Continuous & Complete	Brute-force
Bayesian Search	Continuous but incomplete	Exploration and Exploitation
Successive-Halving Genetic Search Hyperband	Continuous & Complete	Partial training

Table 5 : Hyperparameter Optimization approaches.

for example grid-search, random-search and Bayesian-search which are the most known ones (Akiba et al., 2019; Tsamardinos et al., 2015). Grid-search is a brute-force method that systematically evaluates all possible combinations of hyper-parameter values within a predefined search space, expecting a discretization of continuous parameters (Alibrahim & Ludwig, 2021; Bergstra & Bengio, 2012). With grid-search the predefined search space is searched in it's complete extent, however, the amount of trainings necessary grows exponentially with the number of parameters and parameter values, as each combination needs to be trained and evaluated (Alibrahim & Ludwig, 2021). Further, the discretization of continuous parameters, e.g. the learning-rate of a Deep Neural-Network (DNN), creates holes in the parameter space which are not searched, resulting in potentially missed optimums (Bergstra & Bengio, 2012). This drawbacks are resolved by the random-search, which explores the search space by picking hyper-parameter by random, resulting in a faster exploration of the search space (Bergstra & Bengio, 2012).

Bayesian Search (or Bayesian Optimization) uses probabilistic surrogate models to estimate the relationship between hyper-parameters and model performance (Alibrahim & Ludwig, 2021; Jones et al., 1998; Turner et al., 2021). It continuously builds a model, e.g. a Gaussian process, of the objective function based on previously evaluated points (Alibrahim & Ludwig, 2021; Jones et al., 1998). New hyper-parameter configurations are proposed using an acquisition function, e.g. expected improvement, that balances exploration and exploitation (Alibrahim & Ludwig, 2021). This is much more efficient than a grid-search, since a suitable hyper-parameter configuration is often found quicker and with more certainty regarding continuous parameters (Turner et al., 2021). However, grid- and random-search can be seen as more intuitive as Bayesian-search and better interpretable regarding their search-space because they cover the search space continuously while the Bayesian-search focuses on the region around potential minima (Stuke et al., 2021). With successive-halving, genetic-search or Hyperband, recent approaches try to further reduce compute costs by training many models only partially, then iteratively continue to train only the most promising ones (Akiba et al., 2019; Alibrahim & Ludwig, 2021; Li et al., 2016). They follow the example of Bayesian-search of being faster, but more complex and less intuitive than the conventional grid- and random-search.

Because HPO can be seen as a meta problem over the ML and DL problem of finding best model parameter, single data split for evaluating the model on unseen data is not sufficient: if the test-set would be used to determine the score of a hyper-parameter configuration, the HPO would potentially overfit on the test-set (Tsamardinos et al.,

2015). This is also called data leakage and can be avoided by either creating a second split, thus further reducing the amount of training data available, or by performing a CV for each configuration of hyper-parameters, as shown in Figure 18 (Pedregosa et al., 2011). Combining HPO with CV allows for the optimization on the CV score, which is independent of the test-set while also using as much data as possible for the training (Pedregosa et al., 2011).

While CV is a useful technique to prevent data-leakage while training, it is also important to prevent data-leakage for post-training analysis. If, for example, predictions of a model trained on the ERA5-Land climate dataset are analyzed on correlations with climate patterns also derived from ERA5-Land, the risk of introducing a bias is high. This especially holds true for models trained on embedded data like embeddings from AlphaEarth Foundations, which are created based on, among others, ERA5-Land data.

3.2.3 Common Machine-Learning methods for tabular data problems

The most common form of structured data in real-world applications is tabular data, thus tabular ML problems like classification and regression are most prominent (Erickson et al., 2020; Jiang et al., 2025; Somvanshi et al., 2024). In tabular data problems each sample is represented as a row in a table with shared features as columns, and each cell contains a numeric value forming a feature vector (Arik & Pfister, 2021; Jiang et al., 2025; Somvanshi et al., 2024). This structure creates a vector space that distinguishes tabular problems from more complex input spaces like time series or image segmentation, which involve higher dimensionality (Jiang et al., 2025).

A very common tabular-ML method used across various works across all environmental sciences are Decision Trees and RFs, which are shown in Figure 19 (Bernardini et

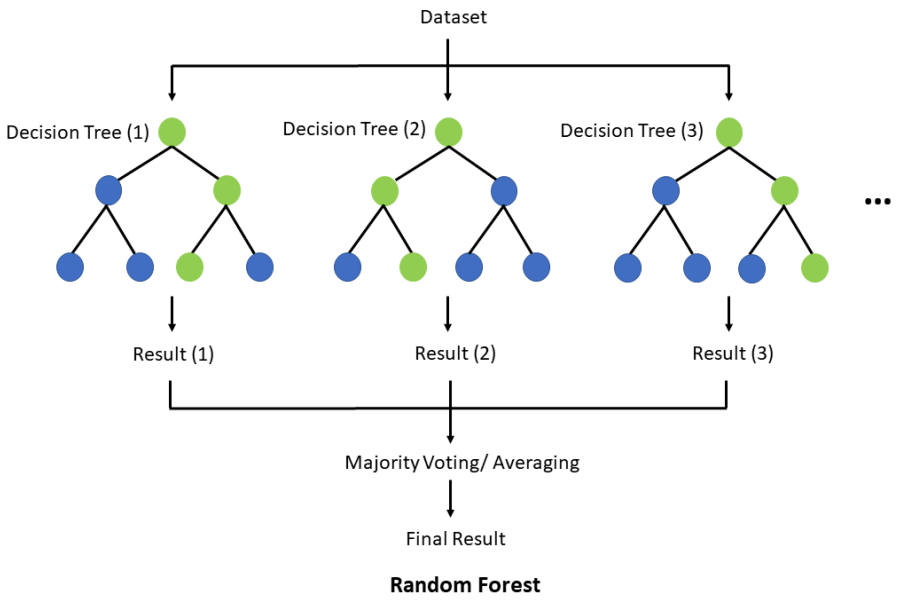

Figure 19 : Decision process of a Random Forest (RF). Graphic taken from Tse-KiChun at Wikipedia. https://de.wikipedia.org/wiki/Random_Forest#/media/Datei:Random_forest_explain.png. Licensed under CC BY-SA 4.0. Accessed February 13, 2026.

al., 2024; Pichler & Hartig, 2023). Decision trees form a hierarchical structure of sequential decisions, learned by recursively partition the feature space based on the available data (Fratello & Tagliaferri, 2019; Mienye & Jere, 2024). The partitioning, or "decision", can be applied on various data types and is able to capture also nonlinear feature interactions (Mienye & Jere, 2024). Each leaf of the tree is assigned to a prediction, either a class label or a continuous value, making them applicable for both classification and regression tasks (Mienye & Jere, 2024). Thanks to their algorithmic transparency, decision trees can be seen as interpretable ML model, however, single trees are unstable and prone to overfitting, as small changes in the data can drastically change the tree structure (Fratello & Tagliaferri, 2019; Mienye & Jere, 2024; Pichler & Hartig, 2023). Random Forests address this instability by constructing an ensemble of uncorrelated decision trees, each trained on a random subset of features and samples (Mienye & Jere, 2024; Pichler & Hartig, 2023). The final prediction is then made based on some sort of vote, e.g. majority vote or averaging (Mienye & Jere, 2024).

Gradient boosting model (GBM) go a step further by sequentially constructing an additive model of many shallow trees, similar to RFs, while fitting each new tree to the negative gradient of a specified loss function, allowing flexible optimization for regression and classification (Florek & Zagdański, 2023). There are many different implementations available, each of which with it's own strengths and drawbacks, such as XGBoost (eXtreme Gradient Boosting) or LightGBM (Chen & Guestrin, 2016; Florek & Zagdański, 2023; Pichler & Hartig, 2023; Shi et al., 2022). While GBM can be extremely powerful, the method can be computationally expensive and requires additionally a dense HPO to achieve the best results, but thanks to GPU implementations of these algorithms they offer a great training-time-to-performance ratio (Florek & Zagdański, 2023).

Ensemble models, like RFs combine multiple individual learners to improve predictive performance by reducing variance, bias, or both, with two common ensemble strategies being bagging and stacking (Ganaie et al., 2022). Bagging, or "bootstrap aggregation", describes the training on multiple homogeneous models all based on different subsets, also called "bootstrap samples", of the data (Erickson et al., 2020; Ganaie et al., 2022; Mienye & Jere, 2024; Mienye & Sun, 2022). If, for example, the individual decision trees of a RF are trained on such subsets, the RF can be seen as a bagging ensemble. (Ganaie et al., 2022; Mienye & Sun, 2022) Figure 20 shows how AutoGluon, an AutoML framework, trains different models through such a bagging mechanism in combination with a cross-validation (Erickson et al., 2020). In this case, five train-validation splits of

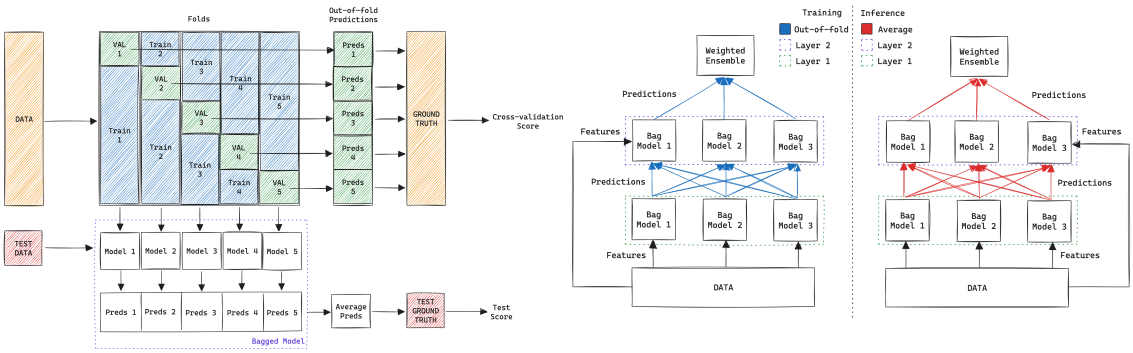

Figure 20 : Left: Bagging method used by AutoGluon. Right: Stacking method used by AutoGluon. Graphics taken from the AutoGluon 1.5.0 documentation. (Erickson et al., 2020)

the training data are created and models are trained and validated on each split, resulting in five scores which are then further aggregated to a single cross-validation score indicating how well the single models perform on unseen "out-of-fold" data. To retrieve a test score on how well the ensemble of models perform on unseen data, a held-out test dataset is given into all models of the ensemble, resulting in five predictions which are then averaged and compared with the ground truth of the test set. In contrast to bagging, stacking combines heterogeneous models by training a meta-learner to optimally integrate their predictions, often achieving higher accuracy by leveraging complementary strengths of diverse algorithms (Erickson et al., 2020; Ganaie et al., 2022; Mienye & Sun, 2022). Also shown in Figure 20 is the stacking approach from AutoGluon, where multiple layers of model stacks are trained, passing the outputs of each model of a layer to each model of the next layer (Erickson et al., 2020).

With the increase of hardware being able to handle highly parallelized computation, Feed-forward multi-layer perceptrons, or shallow "Neural-Networks" became more and more viable for complex tasks as they are universal approximators, capable of representing a wide class of nonlinear functions given sufficient number of nodes (Kiranyaz et al., 2020; Schmidhuber, 2015). Shallow networks are, similar to GBM, trained via gradient-based optimization and are particularly effective when the underlying mapping is relatively simple or low-dimensional (Barbier et al., 2025; Schmidhuber, 2015). However, they may require many units to model highly complex or hierarchical patterns and can be sensitive to initialization and optimization dynamics (Barbier et al., 2025; Kiranyaz et al., 2020; Pichler & Hartig, 2023). DL describes the training of "DNNs", which are Neural-Networks with many layers, enabling the learning of hierarchical feature representations from data (Schmidhuber, 2015). While being able to represent more complex relationships in data and thus often performing very good thanks to their massive parameter counts, they suffer even more from over-fitting in out-of-distribution situations letting them require even larger datasets (Bernardini et al., 2024; Schmidhuber, 2015). Combined with specialized mechanisms such as convolutions, long-short-term-memory or self-attention DNN become very powerful and versatile, enabling DL for tasks such as image classification and segmentation, natural-language processing or time-series forecasting (Hochreiter & Schmidhuber, 1997; Lecun et al., 1998; Ronneberger et al., 2015; Vaswani et al., 2017).

Through the extensive availability of diverse ML and DL methods trying out every method on a problem to find out the optimal one which performs well with as less compute needed as possible became more and more time consuming while increasing the surface for errors made by the researchers. Thus, this process of finding and training an optimal model was automated by so called AutoML frameworks, such as PyCaret or AutoGluon (Ali, 2020; Erickson et al., 2020). These frameworks implement different models, ensembling methods, such as stacking and bagging, HPO, cross-validations and several data preprocessing methods, while making it easy to utilize them by just supplying a dataset to the framework allowing for automated training and validation (Ali, 2020; Breiman, 1996; Erickson et al., 2020; Wolpert, 1992). AutoGluon provides a set of presets, which contain model setups known to work well on most datasets, varying on model complexity. The most simplest presets utilize mainly RFs and GBMs while the most complex one uses tabular foundation models such as TabPFNv2, TabICL, and Mitra

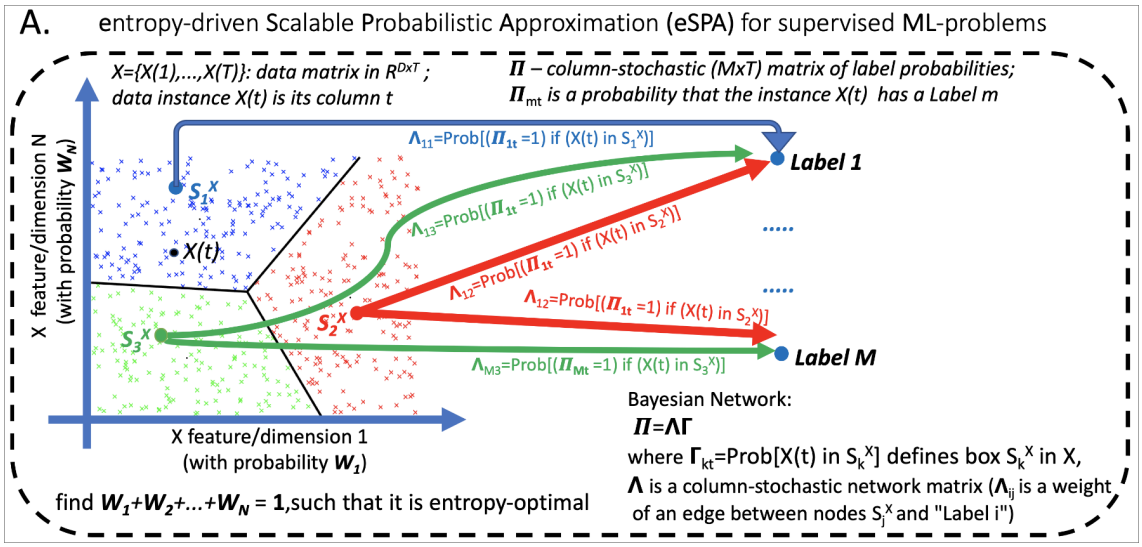

Figure 21 : Architecture of the entropy-optimal Scalable Probabilistic Approximations algorithm (eSPA). Cluster-centers alongside feature weights are trained such that the sum of the weights equal 1 and are entropy-optimal. Simultaneously, each probabilities for all cluster-center to label associations are also trained. At inference time the probabilities that the sample is assigned to each cluster is calculated utilizing these weights. Graphic taken from Horenko (2020).

or the DL model TabM (Gorishniy et al., 2024; Grinsztajn et al., 2025; Qu et al., 2025; Zhang et al., 2025).

Apart from commonly used ML and DL approaches this work also utilizes eSPA+, an ML algorithm of a novel set of sparse, probabilistic learners consisting of eSPA eSPA+ and SPARTAn (Horenko, 2020; Horenko et al., 2022; Vecchi et al., 2022). These are algorithms which train weighted Bayesian networks coupled to clusters based on an entropy based loss-function trying to reduce the entropy within the model ultimately resulting in a concave loss-landscape (Horenko, 2020). This results in an extremely fast convergence of the algorithm while finding always the optimal solution thus reducing the compute cost of the training drastically (Groom et al., 2025; Horenko, 2020; Vecchi et al., 2022). Figure 21 shows the architecture of eSPA. This is a major improvement over conventional DL gradient-descent approach, which are prone to optimize into a local loss-minimum (Horenko, 2020). Another benefit of these probabilistic learners is their robustness against over-fitting, allowing for training on small datasets with many features (Horenko, 2020). Further, thanks to the implementation of feature weights into the Bayesian network, the algorithm automatically detects important features and filters irrelevant features, enabling further analysis of relevant features (Groom et al., 2025; Horenko, 2020). A major drawback of these methods is the need for extensive HPO, similar to RF or GBM methods.

Another, but much simpler cluster based algorithm is kNN. This algorithm predicts by selecting the majority class or average value of the k closest training samples in the feature space (Fix & Hodges, 1989). This algorithm has only two tunable hyperparameter: k and the weighting metric used to adjust the majority vote or average of the k neighbors.

3.2.4 Feature Engineering

As ML approaches are data-driven, their effectiveness strongly depends on the data used for model training. Feature engineering describes a class of approaches to increase data quality before training and consists of several methods (Verdonck et al., 2021).

A quite straightforward method is the reduction of features enabling models to focus on important features by filtering them by some sort of feature relevance metric, such as variability, F-statistic, MI or any other feature-importance metric (Venkatesh & Anuradha, 2019; Verdonck et al., 2021). Table 6 shows a selected set of feature selection methods. While the variability can be a good indicator on the spread of values, it is only slightly correlated with the actual separability of a feature. Utilizing the F-statistic on the

	Order	**Measure**
Variability	Prior to training	Value spread
F-statistic	Prior to training	Linear correlation
Mutual information	Prior to training	Non-linear similarity
Hierarchical Clustering	Prior to training	Linkage
Recursive Feature Elimination	Post training	Feature importance

Table 6 : Feature selection methods.

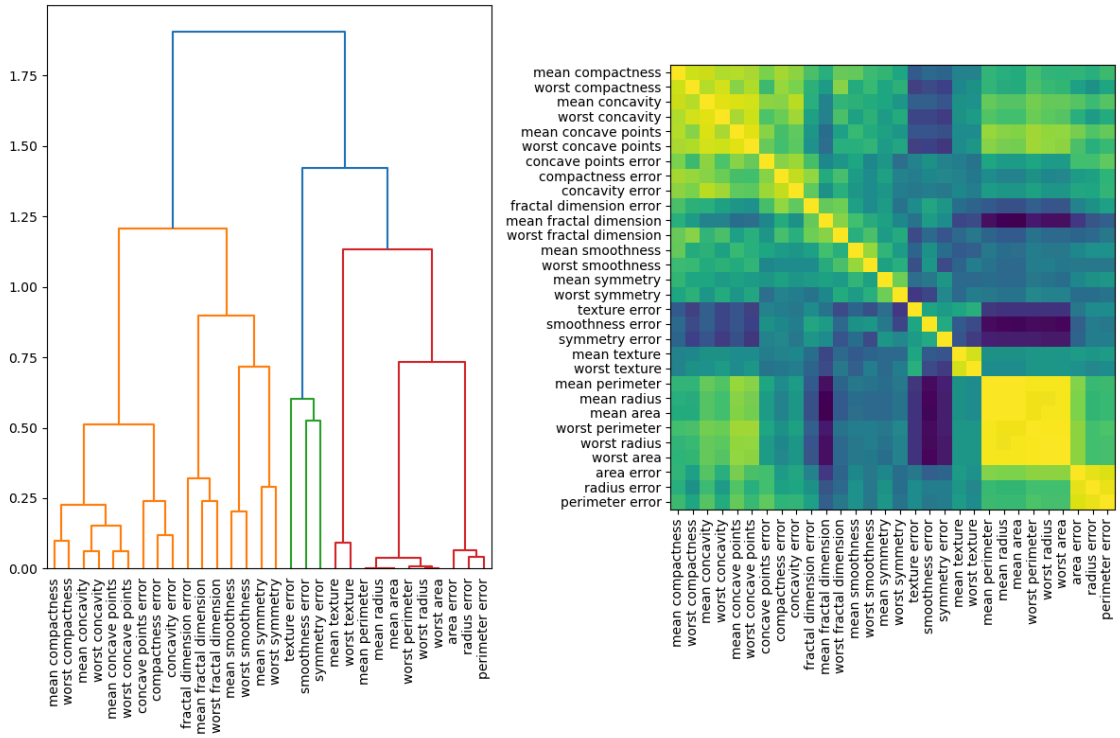

Figure 22 : Hierarchy dendrogram and correlation heatmap of a multi-collinear dataset in one of the SciKit-Learn Examples. Graphic taken from the SciKit-Learn 1.8.0 documentation (Pedregosa et al., 2011).

target values is a better indication on the separability of a feature regarding these target values, however, still fails for non-linear and recurring features, making MI a second viable option as separability measure (Lomax, 2007; Shannon, 1948). Another method, very suitable for datasets with highly correlated features, is the selection of features based on a hierarchical clustering method. First, the similarity or distance between all features is calculated and a hierarchy graph is build, as shown in Figure 22. Ward linkage is a commonly used metric for this, as it's created clusters are more homogenous than these from other methods (Großwendt et al., 2019; Ward, 1963). Then, a threshold is selected to group the features into clusters and a feature from each cluster is chosen to be kept. Feature importance metrics which are calculated post-training open the possibility of special approaches, such as recursive feature elimination (Venkatesh & Anuradha, 2019). Recursive feature elimination trains a model, calculates the least important feature and then retrains the model without this feature until the desired number of features is eliminated. This method, of course, is much more computationally expensive, especially for large feature counts, and further strongly depends on a good feature importance metric.

Another simple, yet important method, for example, is feature scaling. Feature scaling is relevant for all ML which utilize any distance measure in feature space, thus expecting it to be normalized. For unnormalized feature spaces, e.g. where the value range of one feature is between 0 and 0.1 and the value range of another feature is between 0 and 1000, distances would incorrectly be measured within two different unit systems, thus having a bias on smaller valued features. This heavily effects clustering and neighborhood algorithms, such as kNN but also eSPA. Figure 23 shows how standard scaling and normalization can be applied to unnormalized data to achieve more suitable data distribution and properties. The standard scaling forces each feature to have a mean of zero and a standard deviation of one, removing the bias regarding different sized feature spaces. Normalization forces the data to have a one-unit length, making clustering approaches less feasible, however, improving correlation analysis.

Featurization describes the process of transforming raw, potentially unstructured data into structured features suitable for ML algorithms. This becomes especially interesting in the climate or environmental domain, since there are many ways to further model certain meaningful variables, e.g. effective-snow-depth or thawing-days (*Glossary of Climate Related Terms*, n.d.; Slater et al., 2017). Beside the calculation of meaningful variables, the feature-space can also be transformed into a new abstract space through variability increasing methods, such as Principle Component Analysis (PCA) (Lever et al., 2017). Similar, but more complex than featurization methods are methods for feature space embedding. Embeddings can, similar to PCA, be seen as vectors in a new abstract feature-space, however, they are usually derived from large foundation models (Brown et al., 2025).

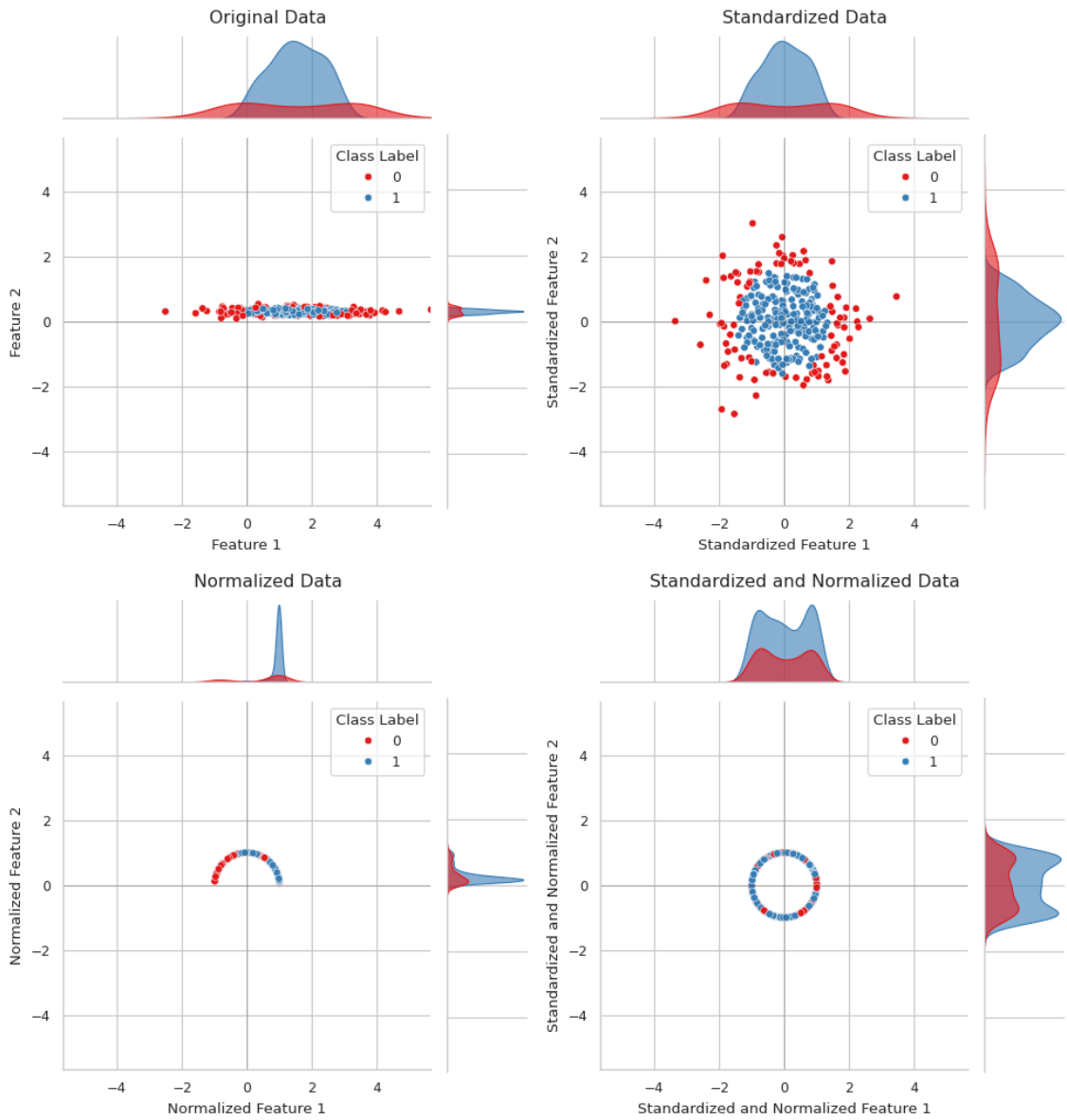

Figure 23 : Different feature scaling methods applied on a mock dataset.

3.2.5 Explainability and interpretability

Apart of metrics, further interpretations of the outputs and decisions of an ML model are often necessary for various reasons, such as increasing trust or increasing the informativeness of the model outputs (Horenko, 2020; Lipton, 2016; Lundberg et al., 2020; Lundberg & Lee, 2017). In general, one can differentiate between two different kinds of interpretability: model transparency and post-hoc interpretations (Lipton, 2016). Transparency typically denotes how easy a human can understand the algorithm of the decision finding process behind the model, opposed to opacity or black-boxiness. Through the vast grow of data, compute and complexity of ML models over the last decade, almost all ML can be seen as black-box models which inner workings are hard to understand, thus having poor transparency (Lundberg & Lee, 2017). Post-hoc interpretations are trying to work around that fact by analyzing learned models, instead of the

algorithm behind it, through diverse methods, such as text explanations by a another model trained in parallel, the visualization of certain layers of a deep neural-network or by varying the inputs and analyzing the respective change of the outputs (Lipton, 2016). The results delivered by these methods vary strongly in complexity with most of them being prone to human mis-interpretation.

The most common class of post-hoc interpretation methods is the estimation of feature importances, meaning how much one feature influenced the models decision for a single sample (Lundberg & Lee, 2017). Thanks to the simplicity of the concept "feature importance" the risk of mis-interpretation is quite small. One common technique of calculating global feature importances is permutation feature importance, introduced by Breiman (2001). This method permutates the input data by shuffling a single feature and lets the model predict the target again based on the permutated input. For predictive features, this should increase the error between the target and the predicted, for un-predictive features this should have no effect. Repeating this for every feature returns measures of feature importances in form of decrease in score or increase in error. A key advantage of that technique is the possibility to apply it to any model, even total opaque ones (black boxes). However, since the measure of permutation feature importance is not uniform and strongly depends on the predictive power of the trained model, permutation feature importance is not suitable comparing different models nor does it provide meaningful features importances for poorly performing models (Pedregosa et al., 2011).

With the framework of Shapley additive explanations (SHAP) values a unified method to estimate feature importances was developed by Lundberg & Lee (2017). SHAP values are based on game theory and describe how much an feature influenced the prediction of a model, as shown in Figure 24. For complex models it is not feasible to compute the true SHAP values, thus only estimated of these values can be made. The estimation on SHAP values is, like the permutation feature importances, model-agnostic, meaning they can be estimated for any model. For some model types there are model specific approximation methods, such as Linear-SHAP for linear regressors, Tree-Explainer for Decision-Trees, Random-Forests and gradient boosted trees, or Deep-SHAP for Deep Neural-Networks (Lundberg et al., 2020; Lundberg & Lee, 2017). These model-specific methods are less computative expansive and provide better approximations on the SHAP values. A major benefit of SHAP values is that they can be robustly estimated on a per-sample basis. While this is theoretically also possible for permutation feature importances, SHAP was designed on a sample basis.

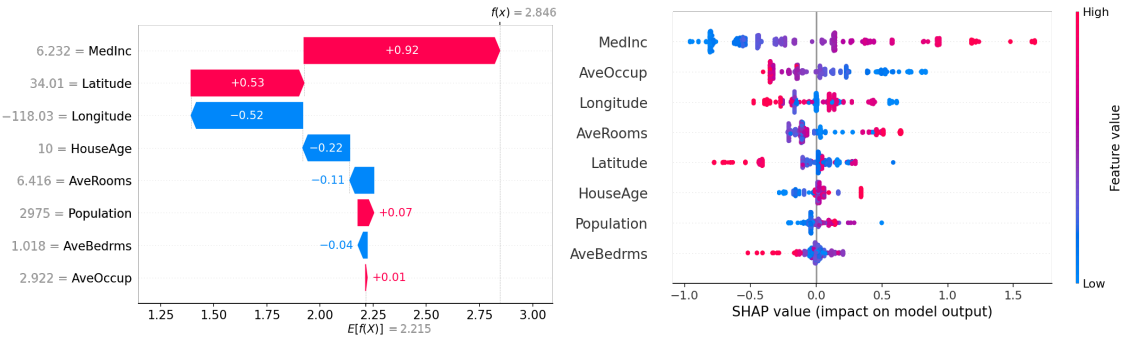

Figure 24 : Left: SHAP values for a single explanation. Right: SHAP values for all explanations of a dataset. Graphics taken from the SHAP 0.50.0 documentation (Lundberg & Lee, 2017).

Through the estimation of feature importances it is possible to infer properties or generate hypotheses about the causal relationships between the input features and the target variable of a model (Lipton, 2016). The idea is quite simple: if a complex model can correctly predict / estimate a real world phenomena based on the given set of input features, the feature importances measured by e.g. SHAP values would indicate how influential a respective feature is in regard to the real world phenomena. However, this of course only shows a potential correlation between the features and the target variable and causal relationships are not guaranteed (Lipton, 2016). Still, combined with a large set of features this may provide experts with sufficient hints to come up with a causal driven model or hypotheses about yet hidden relationships.

3.3 Experiments

In order to answer the primary research question about the importance of environmental variables for RTS occurrence and suitable ML methods, a framework for rapid experimentation was created. This framework allowed the combination of the extensive dataset configurations as explained in Section 3.1 together with the different ML methods and model architectures described in Section 3.2.

3.3.1 Exploratory experiments

Before structured experiments were conducted, exploratory experiments tried to gain "quick wins" by training several models on various hyper-parameter combinations and dataset configurations. The knowledge, or rather intuition, about the dataset gained by these experiments was then used to pin-down the configurations of the structured experiments used to answer the research questions of this work. An interactive dashboard with Streamlit was developed to quickly visualize different dataset configurations and the results of different model trainings, Figure 25 shows two pages of this dashboard.

At first, the hyper-parameter space of the eSPA was analyzed in respect to different dataset combinations. While the exact best combination of hyper-parameters varied for different dataset combinations, a clear pattern emerged: the optimal number of initial

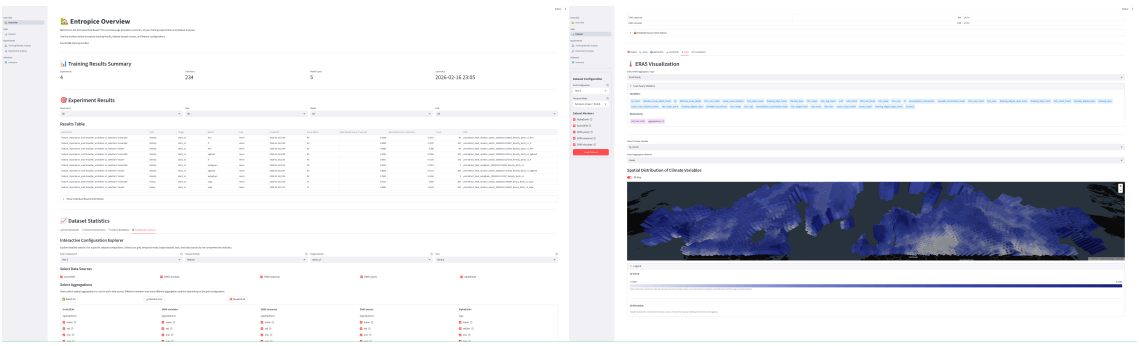

Figure 25 : Two pages from the developed Streamlit dashboard. Left: Overview about all experiments and dataset configuration. Right: Dataset preview. The dashboard was created for quick exploration of the different dataset configurations and experiment results.

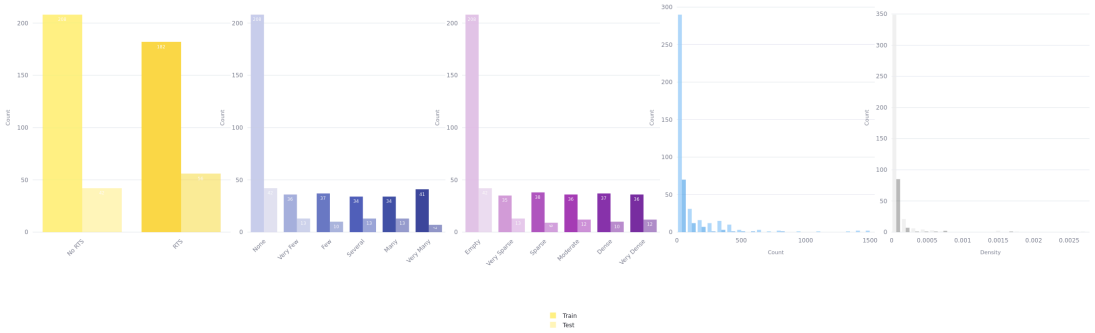

Figure 26 : Histograms of the different target variables derived from the H3 level 3 DARTS v1.2 synopsis dataset. Histograms are spitted into their train (left) and test (right) splits.

clusters K increased with the grids resolution. This is expected, as with higher resolutions more, complicated patterns are expected to emerge and need to be represented by more groups. Further, it was found that ε_{cl} needs to be in the range between 10^{-11} and 10^{-7} while ε_{e} needs to be in the range between 10^{4} and 10^{8} for optimal results.

Handling each year of the temporal agnostic variables as their own feature proved counter-productive for model training due to the high co-similarity between many resulting features. For the most extreme dataset configurations this resulted in over 60 000 features, which lead to strong overfitting as the number of training samples, especially for the low resolution grids, was too small in comparison. This was the main motivation for the creation of the "synopsis" variants of the datasets. For some model trainings, however, some years seemed to be especially relevant regarding eSPA feature weights, however, no statistical support was found for this. The same problematic applied to the different spatial aggregations used to align the original datasets into the DGGS grids, as the number of features exploded when treating each aggregation (per year, per variable) as own, independent feature.

Only the binary (RTS occurrence) and density (RTS area) target variables derived from the DARTS datasets seemed to resulted into promising model trainings. Trainings targeting the two binned targets, where the count and the density where binned into five equally sized categories, couldn't surpass trainings targeting the binary target variable in predictive performance. The bins were very distorted, as there were only very few high density or high count cells, but very many cells with only one or two RTS present, which further damped training performance as such low-count and low-density cells were very prone to false positives. Similarly, the RTS count target variable was also prone to noise in the underlying DARTS dataset, clearly visible in Figure 13. Figure 26 shows the distribution of the different target variables, further showing the skewness of the count and density distributions.

3.3.2 Structured experiment setup

A structured experimental design was created to address the two central research questions simultaneously in a single, cohesive workflow, visualized in Figure 27. To reduce the complexity and number of features of the dataset, the median spatial aggregated "synopsis" versions of the variables were used, if more than one aggregation was present

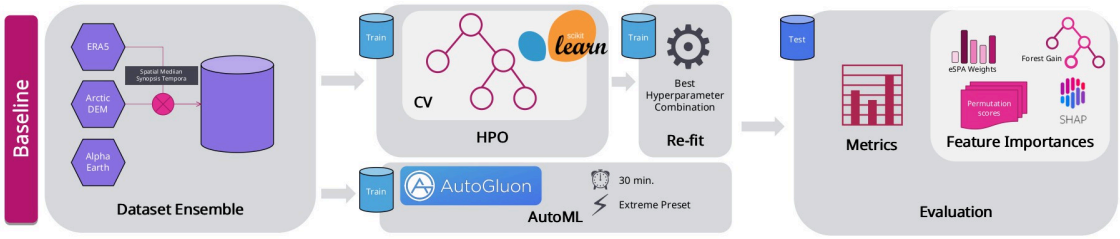

Figure 27 : Baseline experiment setup.

for a cell. This still left a large amount of 179 features in the dataset ensemble, which is important for the experiment design: the models were meant to learn and select which features are most relevant for RTS occurrence out of that large pool of features. This is contrary to other ML approaches used to detect RTS, as these typically only used a small selected of features chosen by experts, thus potentially ignoring potential relevant, but yet unknown or unexpected features.

All nine grids were used, shown in Table 2, both the H3 and the Healpix grids were used to ensure the training results were independent on the spatial griding method used. If the training results, regarding model predictive performance, were equal the griding method would not be relevant. DARTS v1.2 RTS occurrence was used as the target variable for a binary classification task and DARTS v1.2 RTS area-density was used as the target variable for a regression task, because the other possible target variables did not prove to be useful in the exploratory experiments. Other DARTS versions were not used, because the number of samples from the "DARTS-labels" dataset was to small and it was not possible to validate the target variables created based on the DARTS v2 in time.

XGBoost, RF and kNN were used as baseline ML models as they are common across various research fields, with XGBoost and RF being robust tree-based models, needing little data preprocessing or further feature engineering and kNN being conceptually similar to eSPA being a clustering-based method. No DNN or other DL models were utilized, as these models do not provide accessible insights into their inner state and their respective feature weighting. HPO was performed using a 300 combination random search with a stratified-shuffle CV for the binary task and a k-fold CV for the density task, both utilizing five folds. Random search was preferred against Bayesian search to analyze the hyperparameter space after the training. No advanced folding strategy were used, as the target variables were already unevenly distributed and the number of samples especially for the low resolution grids very small.

Table 7 shows the the hyperparameter space searched by the HPO. As mentioned earlier, the hyperparameter for eSPA were already estimated by exploratory experiments. With XGBoost being the most complicated model used by these experiments, the hyperparameter space explored was also the largest one among these models. The number of estimators, thus trees, was set to somewhat reflect the number of cluster centers of the eSPA model, however, big compute performance drops were observed for more than 500 estimators. For the RF performance drops were already observed for more than 300 estimators. To account for this the trees of the RF were set to be deeper than these of the XGBoost. Because the RFs were so slow to train, the number of randomly selected hyperparameter combinations was set to 100 instead of 300. kNN also showed compute performance drops for more than 200 neighbors. Because the hyperparameter space

Hyperparameter	Distribution	Low	High
			eSPA
ε_{cl}	log-uniform	10^{-11}	10^{-6}
ε_{e}	log-uniform	10^{4}	10^{8}
$K_{initial}$	random-integer	50	800
			XGBoost
Learning rate	log-uniform	0.01	0.3
Number of Estimators	random-integer	100	500
Maximum Depth	random-integer	3	10
Column samples by Tree	uniform	30%	100%
λ_{reg} (L2 Regularization)	log-uniform	10^{-8}	10
			Random Forest
Number of Estimators	random-integer	50	300
Maximum Depth	random-integer	10	50
Maximum Features	uniform	10%	50%
Minimum sample Split	random integer	2	20
			kNN
Number of Neighbors	random-integer	10	200
Weights	categorical	{“uniform”, “distance”}	

Table 7 : Hyperparameter space used in the Hyperparameter-Optimization.

of kNN effectively only contained a single dimension, the number of hyperparameter combinations used by the random search was set to 40.

The best hyper-parameter combination was determined based on the F1-score for the binary classification and the r^2 metric for the density regression, after which the models were refitted on all available training data without CV. The eSPA model was trained identically to the baselines but was restricted to the binary classification, as it is a classifier and the regressor counterpart SPARTAn has is not implemented in Python yet.

For a more automated approach, AutoGluon was employed representing novel AutoML techniques. The “extreme” preset was used to train the AutoGluon models over 30 minutes. No further HPO or CV setup was utilized, relying solely on its internal stacking and bagging mechanisms as it is recommended by their documentation (Erickson et al., 2020). As AutoGluon builds up an internal leaderboard to rank it’s trained models, the leaderboards of each trainings were extracted and saved for later analysis.

To quantify the predictive performances, the common metrics shown in Table 3 were calculated on a held-out test set, while permutation feature importance scores were computed on the same test set to assess feature relevance. For further investigation into feature relevance, SHAP values were calculated using the TreeExplainer for XGBoost and RF, and the PermutationExplainer for kNN to provide deeper insights into feature contributions. Both explainers were taken from the SHAP Python library.

In summary, each experiment yielded performance metrics, feature permutation scores, and SHAP values for every model, grid, and task combination. This comprehensive approach resulted in 90 binary classification and 72 density regression runs per experiment.

The workflow allowed for the systematically variation of the input features to evaluate their impact on model performance. As a baseline, ERA5-Land quarterly variables were ensembled with and ArcticDEM topographic variables to reflect a broad coverage of traditional environment variables. The baseline dataset combination contains a large number of 179 variables, many of which were highly correlated, as illustrated in Figure 43 (Appendix C.1). To further refine the feature sets and reduce the amount of correlated features, two distinct feature selection methods were applied to the baseline combination. AlphaEarth satellite embeddings were employed in a fourth experiment as an example of embedding-based features to explore the potential benefits of vector representations. Both experiment variations are shown in Figure 28. Along the variable compositions some cell related statistics were also included in the datasets, such as the cells centroid position ("x" and "y" of the North Polar Stereographic projection EPSG:3413), cell area, land area, water area and land-to-water ratio. These additional variables were introduced to the dataset ensemble in the hope that the geographic locations could help the models to learn local connections. This process resulted in four distinct experimental setups, each utilizing a different subset of the available variables.

3.3.3 Feature selection

Feature selection methods were necessary to reduce the amount of highly correlated features present in the baseline dataset combination. Correlation was especially high for variables that are in their nature quite similar, such as effective snow depth and the naive snow isolation. This high degree of similarity is expected to lead to very heterogeneous

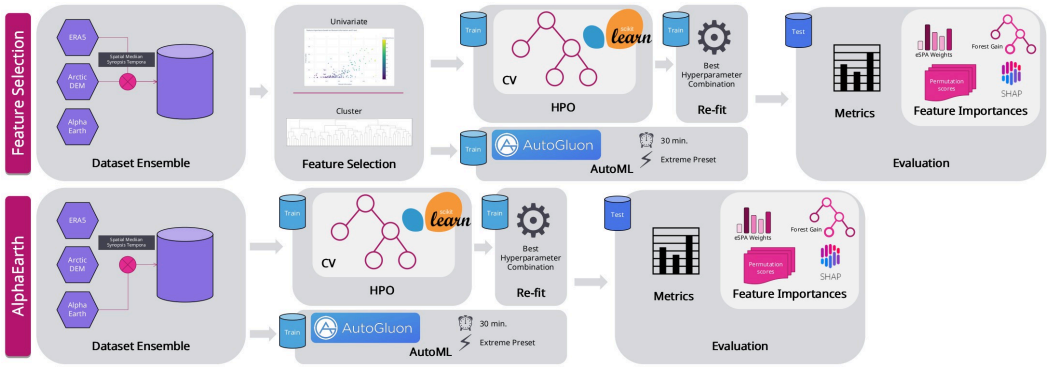

Figure 28 : Experiment variations. Top: Experiment with Feature Selection. Bottom: Experiment with AlphaEarth embeddings.

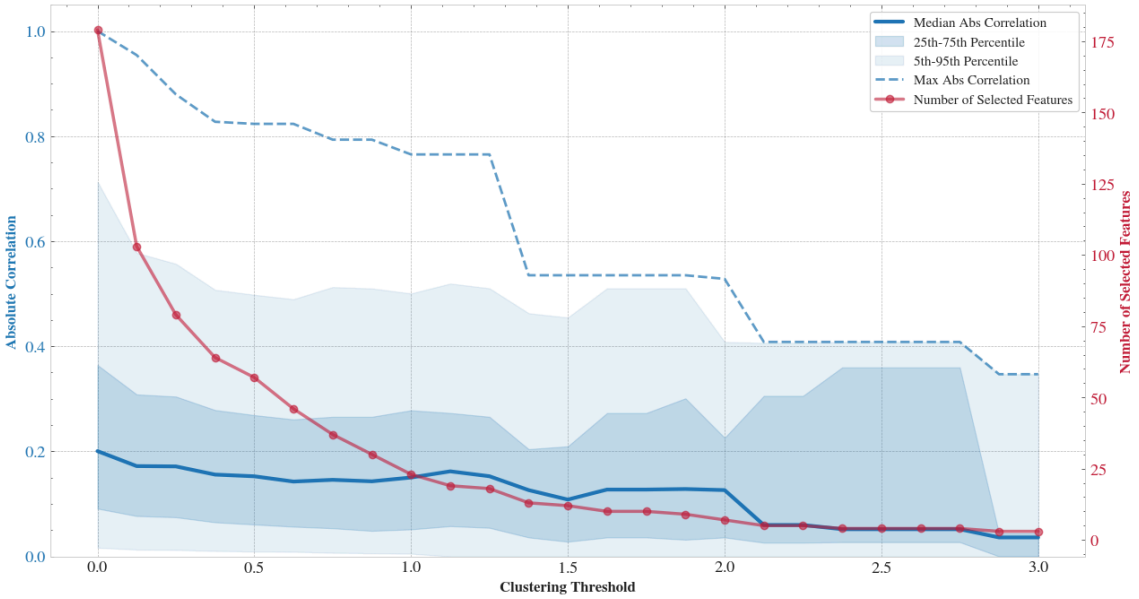

Figure 29 : Effect of the clustering threshold for feature selection on feature correlation and count.

feature importances between different training runs and the masking of relevant features. To address this, feature selection was performed based on two distinct methods, both utilizing data from the H3 level 6 grid because it contained the most samples of all used grids.

The first method employed dendrogram clustering with Ward linkage to group similar features, this dendrogram is also shown in Figure 43 (Appendix C.1). To select a suitable threshold for filtering the features based on their cluster linkage, the spearman correlations were calculated for each possible threshold, as shown in Figure 29. A threshold of 0.5 was selected to filter the features, as the 75th, 95th and maximum spearman correlations seem to plateau after that threshold until only a very small number of features is left. 57 feature, shown in Table 15 (Appendix B.4), clusters were left for this threshold, still enough to satisfy this works approach to let the models automatically learn which features are relevant and which are not. The spearman correlations of these 57 features are shown in Figure 44 (Appendix C.1), showing the success of this method to reduce the intra-similarity of the dataset, as only few high correlations are left.

The second method utilized univariate selection, calculating F-test p-values and MI for each feature against the target variable. An inverse geometric mean was computed as a mixed score to reflect both linear and non-linearities. The importance of this combination of both metrics becomes visible in Figure 30: some features, for example the two positional features “x” and “y”, showed a high MI and a relativly low linearity while others like the “freezing degree days summer trend” or “maximum spring temperature” behaved like a highly linear correlated feature without much non-linear MI. All features with a combined score of 0.425 were selected, as a clear regime shift was visible in the data between ≈ 0.42 and ≈ 0.43 for the ERA5-Land variables, also shown in Figure 30. This selection process yielded 47 features, shown in Table 14 (Appendix B.4).

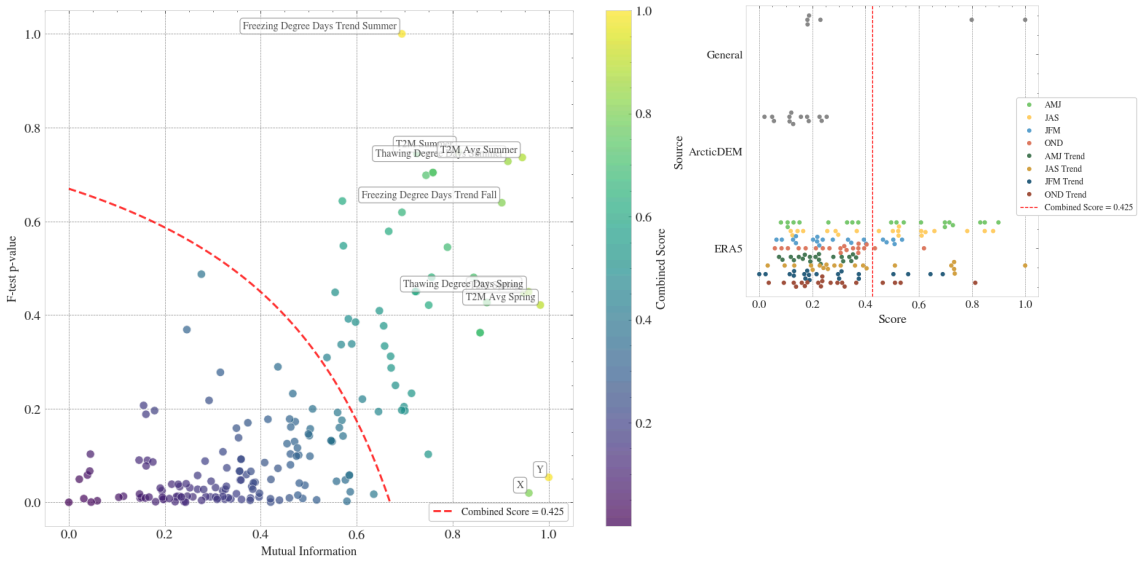

Figure 30 : Left: Mutual information (MI) vs p-value of the F-test measuring similarity. Right: Swarmplot of the combined scores categorized by source dataset and season.

Results

After a total of 192 452 model trainings for all 52 task-grid combinations across the four experiments, each combination evaluated with metrics, CV results of the HPO and feature importances. Each experiment took around 24 hours to complete, the HPO and calculation of the feature permutation score taken the longest of all steps. Despite setting the AutoGluon training time to 30 minutes, these models converged much faster usually already after 3 minutes. The amount of data generated by all experiments is around 100GB, however, the model states have also been saved and taking up the most space.

4.1 Model Performance across Resolutions and Scales

Figure 31 provides an overview of the predictive performances across all trainings, showing that all models performed almost equally well on the test split regarding the F1-score, with standard deviations between experiments and models ranging from 0.8% on low resolution grids to 6.8% on the highest resolutions. While still performing well, the eSPA models performed slightly less good than the baseline models XGBoost, kNN and RF models, especially at higher resolutions where the eSPA models achieved an F1-score of only $42.1\% \pm 6.2\%$ while the others achieved a score of $57.3 \pm 1.4\%$. This is further

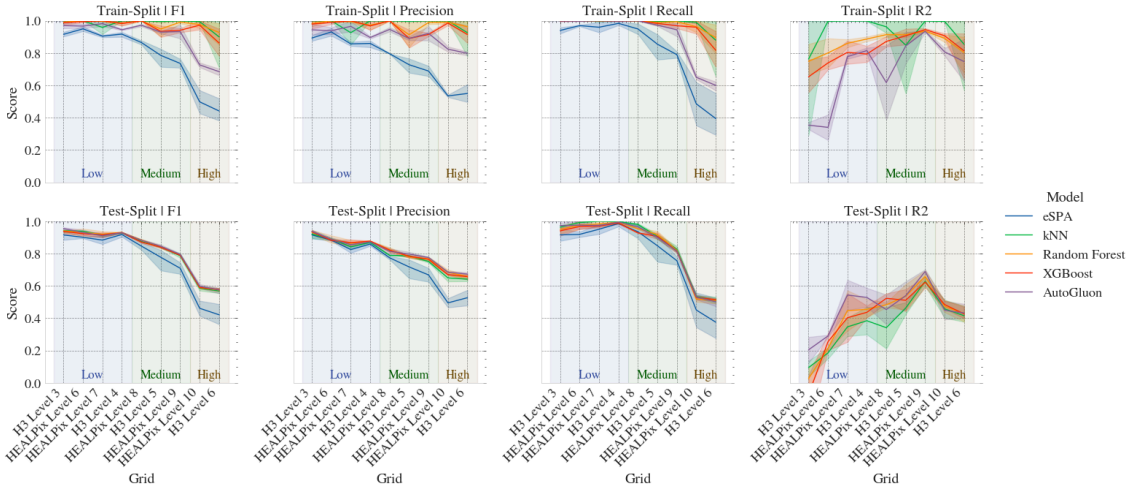

Figure 31 : Averaged metrics with value ranges across the experiments, calculated on the train- (top) and test-set (bottom). From left to right: F1 score, precision and recall from the binary RTS occurrence task and r^2 score from the RTS area-density regression task. Shows the metrics against the different grids, sorted by resolution, color-coded with the colors defined in Table 2. Colored by the model type used.

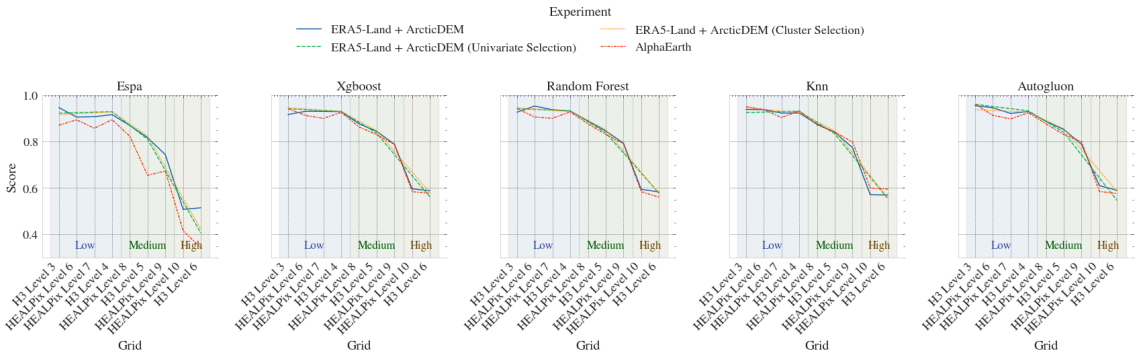

Figure 32 : F1 scores of all experiments (color) per model, calculated on the test-set. Shows the scores against the different grids, sorted by resolution.

Model	H3 Level 3	H3 Level 4	H3 Level 5	H3 Level 6
eSPA	0.915 ± 0.028	0.918 ± 0.014	0.777 ± 0.071	0.421 ± 0.062
XGBoost	0.937 ± 0.011	0.929 ± 0.002	0.840 ± 0.007	0.577 ± 0.010
Random Forest	0.940 ± 0.007	0.931 ± 0.002	0.842 ± 0.008	0.574 ± 0.008
kNN	0.941 ± 0.010	0.929 ± 0.004	0.841 ± 0.005	0.568 ± 0.017
AutoGluon	0.954 ± 0.010	0.929 ± 0.003	0.845 ± 0.010	0.574 ± 0.017
All	0.937 ± 0.020	0.927 ± 0.008	0.829 ± 0.042	0.543 ± 0.068
All (except eSPA)	0.943 ± 0.012	0.930 ± 0.003	0.842 ± 0.008	0.573 ± 0.014

Table 8 : F1-score average and standard deviation across the experiments.

shown in Table 8, a complete list of metrics can be found in Table 18 and Table 19 of the Appendix B.5.

Figure 32 shows the F1-scores of all methods across the different experiments. The application of feature selection methods did neither result in a performance loss nor gain, with F1-score differences to the baseline ranging between -3% and 2.9%. Contrary, the use of AlphaEarth embeddings resulted in a slight performance drop specifically for the eSPA model and for the tree based models on low-resolution grids. Otherwise the models trained on AlphaEarth embeddings performed similarly well as models trained on the environmental variables from ERA5-Land and the ArcticDEM, with differences ranging between an improvement of 0.6% to 2.5%. A list of F1-score differences per grid between the experiments can be found in Table 17 (Appendix B.5).

For the binary classification task, predictive performance generally decreases with higher resolution, though all models remain better than random chance with F1 scores exceeding well beyond 30%. Especially for the low and medium resolutions, the models perform very well with F1 scores exceeding even 93% on the test split. However, as the performance drops drastically for the high resolutions, the predictive performance of these models is not sufficient to justify a feature importance analysis at these resolutions. Additionally, the models across all resolutions show a higher recall than precision, implying a bias toward positive predictions and the potential for overestimation of RTS

occurrence. Baseline models strongly overfit, visible by their almost perfect ($\approx 99.9\%$) F1 and r^2 score on the training split, compared with the lower scores on the test split. While the AutoGluon models overfit less strongly than the baselines, the eSPA models do not overfit at all.

In contrast to the binary classification, the density estimation task shows improved performance with higher resolutions, peaking on the medium-resolution grids, also shown in Figure 32. All models suffer from severe overfitting, achieving excellent scores on training sets but underperforming on test data, indicating poor generalization capacity. As a result, the overall predictive quality for density estimation remains insufficient to support actionable or interpretative analysis of feature importances.

4.2 Maps and Spatial Patterns

The Figure 33 illustrates the RTS occurrence predictions of eSPA, XGBoost, and AutoGluon models from the baseline experiment across the four H3 grids. While every model produced distinct predictions, they were somewhat similar for the same grid resolution. However, predictions across different grid resolutions vastly differed: the low-resolution

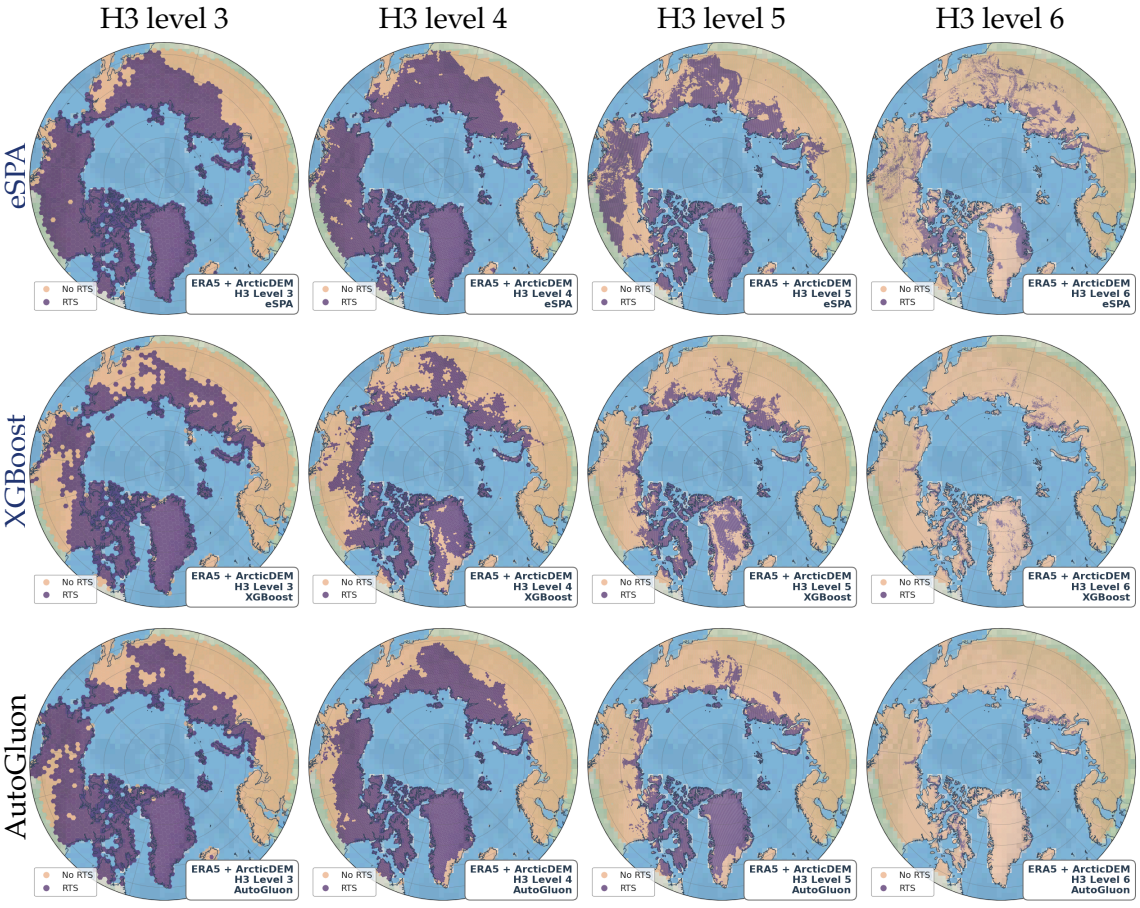

Figure 33 : Retrogressive Thaw Slump occurrence predictions across scales (columns) from different models (rows): eSPA (top row), XGBoost (middle row) and AutoGluon (bottom row). Models were trained with the baseline approach, meaning on all features from the ERA5-Land and ArcticDEM DGGS aligned datasets.

H3 level 3 grid showed for all models a vast overprediction across the pan-Arctic, particularly over Greenland and the American continent. As grid resolution increased, this overprediction declined. For the high-resolution H3 level 6 grid, AutoGluon even underpredicted RTS occurrence, leaving very few hotspots. These remaining hotspots were found mostly on coasts and large river deltas like the Mackenzie Delta, while the model predicted only very few RTS occurrences along the direct coast of Banks Island, a typical hotspot. Despite the F1 scores for the models across all grids, except H3 level 6, being very high, the maps reveal large errors, indicating that non of these models generalizes well on regions outside the training area.

In a similar fashion, Figure 46 (Appendix C.3) displays the RTS occurrence predictions of eSPA, XGBoost, and AutoGluon models from the experiment using AlphaEarth embeddings across the four H3 grids. While similar regarding prediction patterns across the grids and models compared to the baseline, two maps stand out: the eSPA predictions of the H3 level 5 and 6 grids. On these maps, RTS seem to be predicted in concentric circles. Comparing these circles with the area variance of the H3 DGGSs, this highlights a potential flaw in this experiments setup: because next to the embeddings, some grid related auxiliary data like position, but also cell area was ingested into the models. This lead to a strong overfitting on this cell-area feature in these two cases, resulting in such obscure maps.

Also in a similar fashion, but for RTS area-density instead of occurrence, Figure 47 (Appendix C.3) displays the predictions of the XGBoost and AutoGluon models from the baseline experiment and Figure 48 (Appendix C.3) of the experiment utilizing AlphaEarth embeddings. Similar overpredictions of RTS densities are visible across all maps, this time also effecting the coasts of Norway. In contrast to the overpredictive nature of these predictions, the two RTS hotspots in Taimyr and on Banks Island are underestimated compared to other regions. Compared to the binary RTS occurrence predictions of the baseline experiment, the spatial density patterns are somewhat more constant across the different resolutions. While these patterns show interesting features that remain somewhat stable across models and grid resolutions, these map predictions of RTS area densities are in general not very realistic highlighting the already in Section 4.1 mentioned performance gap of the density task.

4.3 Hyperparameter Optimization

The following analysis of the HPO was restricted to the binary occurrence predictions, as they showed better performance and more stable results as described in Section 4.1. Because HEALPix and H3 yielded comparable results the analysis is further reduced to the H3 grids to reduce complexity and noise.

As the configured hyperparameter space for the kNN models is small, Figure 34 shows the complete space. The figure displays the CV results from the baseline experiment on the H3 level 5 grid, plotting the the number of neighbors against the CV F1 score, colored by the weighting mechanism (uniform vs. distance). Contrary to the expectation, a performance decline is observed when using more neighbors, with distance weights slightly outperforming uniform weights. However, the difference between the worst and

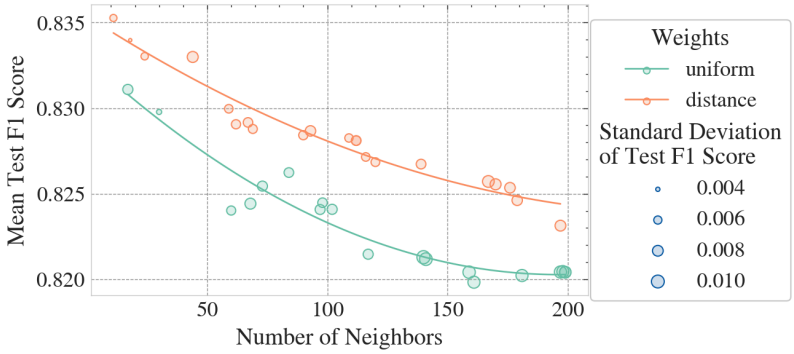

Figure 34 : Analysis of hyperparameter search results for kNN models on H3 level 5 grids. The plot shows the average CV F1 score across validation folds versus the number of neighbors as the hyperparameter, colored by the weighting metric used.

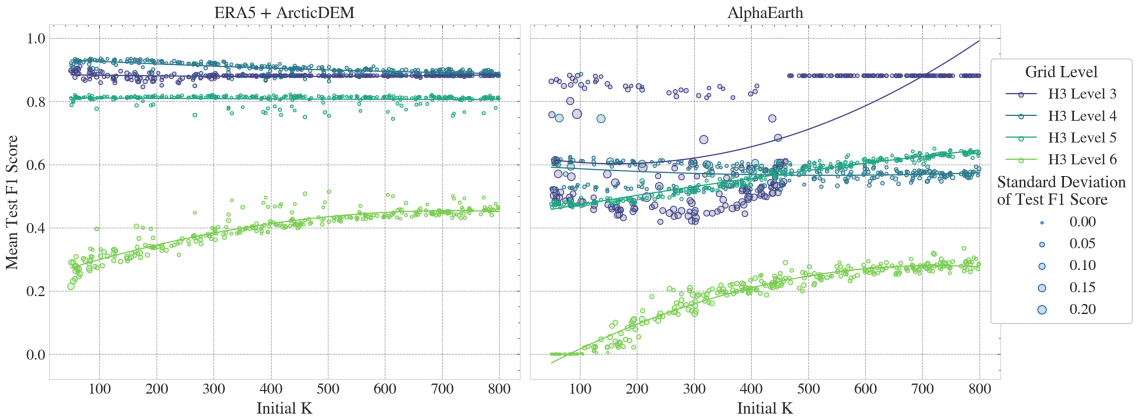

Figure 35 : Analysis of hyperparameter search results for eSPA models on the baseline (left) and AlphaEarth (right) experiments across all grids. Each plot shows the average CV macro F1 score across validation folds versus the initial number of clusters K_{initial} as the hyperparameter, colored by the used H3 grid.

best models is minimal ($\approx 1.6\%$), and the standard deviation of the F1 scores (up to $\approx 1\%$) is almost equal to this difference.

With three continuous parameters, the hyperparameter space for the eSPA models is more complex. As the space was already strongly filtered by the exploratory experiments, no clear trend or cluster was found regarding the relationship between the ε_{cl} and ε_{e} parameters. An interesting connection was found between K_{initial} and the CV F1 score, as shown in Figure 35. For the baseline experiment, predictive performance slightly dropped by $\approx 2\%$ for higher initial clusters K_{initial} values on low-resolution grids, whereas on high-resolution grids, higher K_{initial} resulted in a visible performance improvement of more than 20%. This is on par with the expectations, as a higher resolution suggest more complex relationships within the different environmental connections, thus needing more clusters to reflect this. For the AlphaEarth experiment, a similar pattern emerged except on the lowest resolution grid (H3 level 3), where a performance jump occurred for $K_{\text{initial}} > 450$. Beyond this point, performance remained consistent, suggesting that the embeddings can be clustered into a specific number of describing clusters for this resolution. This observation hints that a similar threshold might exist for higher resolutions ($K_{\text{initial}} >> 800$) outside the tested hyperparameter space. Further, this may indicate that utilizing embedding space in combination with such an entropy-

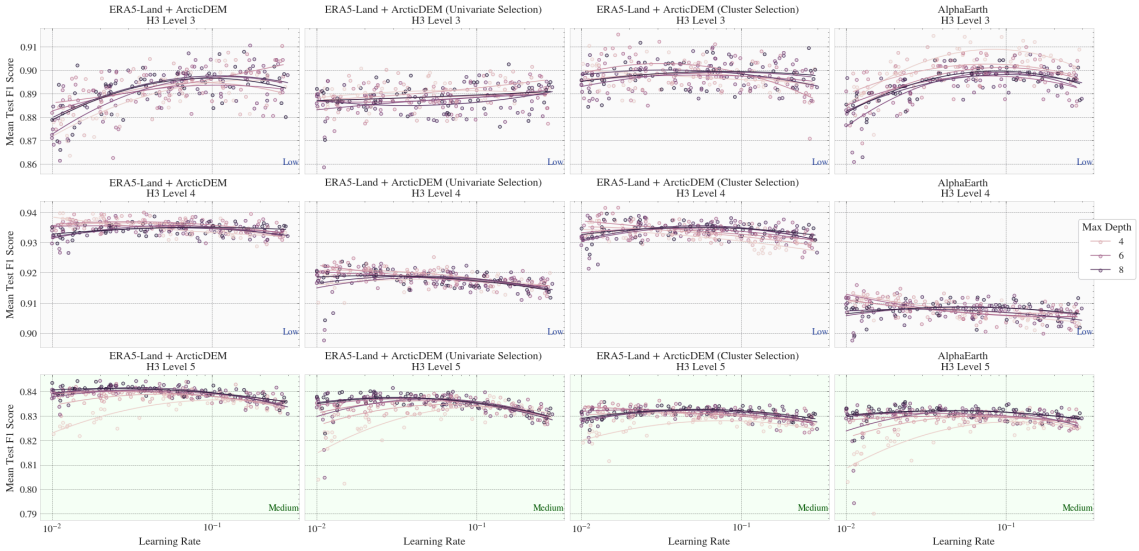

Figure 36 : Analysis of hyperparameter search results on all experiments (columns) and H3 grids (rows). Each plot shows the average CV F1 score across validation folds versus the learning rate, colored by the maximum tree depth.

reducing clustering approach could lead to optimal solutions. However, as outlined in Section 4.2, the eSPA models trained on the AlphaEarth embeddings overfitted on the cell-area for the medium and high resolution grids, indicating a potentially suboptimal configured training setup.

The extensive search of the large hyperparameter space for the XGBoost models did not result in an obvious and meaningful cluster or correlation between hyperparameter combinations and the CV F1 score. However, Figure 36 reveals some slight interactions between grid resolution, learning rate, and maximum tree depth. For medium-resolution grids (H3 Level 5), models with deeper trees performed better across all experiments, whereas for the lowest resolution grid (H3 Level 3), shallower trees performed equally well or better, particularly in the AlphaEarth experiment. Using the right learning rate can slightly improve predictive performance, with polynomial optima visible. Similar to the kNN analysis, the difference between the worst and best models is very small across all experiments and grid resolutions, leaving the performance increase from the HPO marginal, and discussed interpretations speculative.

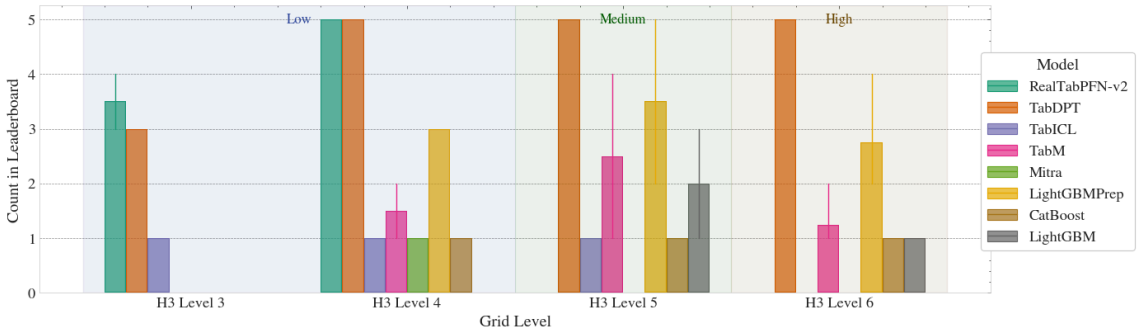

Figure 37 : Average number of model type occurrences in the AutoGluon ensemble leaderboards across all experiments for each H3 grid resolution. Error bars indicate the minimum and maximum counts per leaderboard entry. A single model may appear multiple times due to different predefined configurations selected by AutoGluon.

Since AutoGluon utilizes predefined configurations known to yield good results, no hyperparameter optimization was performed for these trainings, thus no hyperparameter space can be explored. Instead, the leaderboards of the ensembles trained by AutoGluon were analyzed across grid resolutions, as shown in Figure 37. A complete list of leaderboards can be found in Appendix B.6. For the lowest grid resolution (H3 Level 3), which contains only a few hundred training samples, only three model types were trained and ensembled: RealTabPFN-v2, TabDPT, and TabCL, all of which are tabular foundation model with a focus on small-sample size problems. For higher resolutions, more diverse model types were selected by AutoGluon, and the ensembles utilized a greater total number of models. Next to the foundation model, gradient boosting models such as LightGBM and CatBoost were also utilized for these grid resolutions.

4.4 Feature Importances

In total, four different metrics for feature importances were calculated in the evaluation part the experiments: the trained weights W from eSPA, the internal importance gains from XGBoost models, feature permutation scores, and SHAP values. To evaluate the consistency of feature importance rankings, these metrics were compared against each other by measuring the Spearman correlation across all grids, as shown in Figure 38.

As expected, the XGBoost total importance gains and the SHAP values were correlated, with most ρ values across all experiments and grids being above 0.5. The permutation feature scores showed no significant correlation with any other method. The correlation between the eSPA weights and the XGBoost total importance gains revealed the most patterns. For the cluster-based feature selection experiment, these two metrics were highly similar, especially for higher resolution grids. In the baseline and univariate selection experiments, the similarity between the metrics were not as high but clearly increased with higher grid resolutions. Oppositely, in the AlphaEarth experiment, the relationship became nearly anti-proportional at higher resolutions, a counter-intuitive result likely stemming from the injection of the cell-area feature into the dataset, as discussed in Section 4.2.

Notably in Figure 38, the cluster selection experiment showed the largest correlations across almost all grids, suggesting that the metrics become noisier when applied to more correlated data. Furthermore, the observed increase in correlation between the eSPA weights and XGBoost total importance gains with higher grid resolution indicates that these metrics become more stable with higher sample counts, and being rather unstable for small sample sizes.

With SHAP values being the most insightful method to describe feature importances, they are used to compare the relevance of features across spatial scales. SHAP values are calculated on a per-sample basis, thus for every grid-cell of the test set feature importances are calculated for all features, thus spanning a matrix of $N_{\text{cells}} \times N_{\text{features}}$. Since the data from baseline experiment contained a high number of correlated features, the two feature selection experiments were also analyzed using SHAP values. Stronger signals in the distribution of SHAP values are expected, as fewer similar features are used which would otherwise share their relevance. Figure 41 illustrates beeswarm plots for all H3

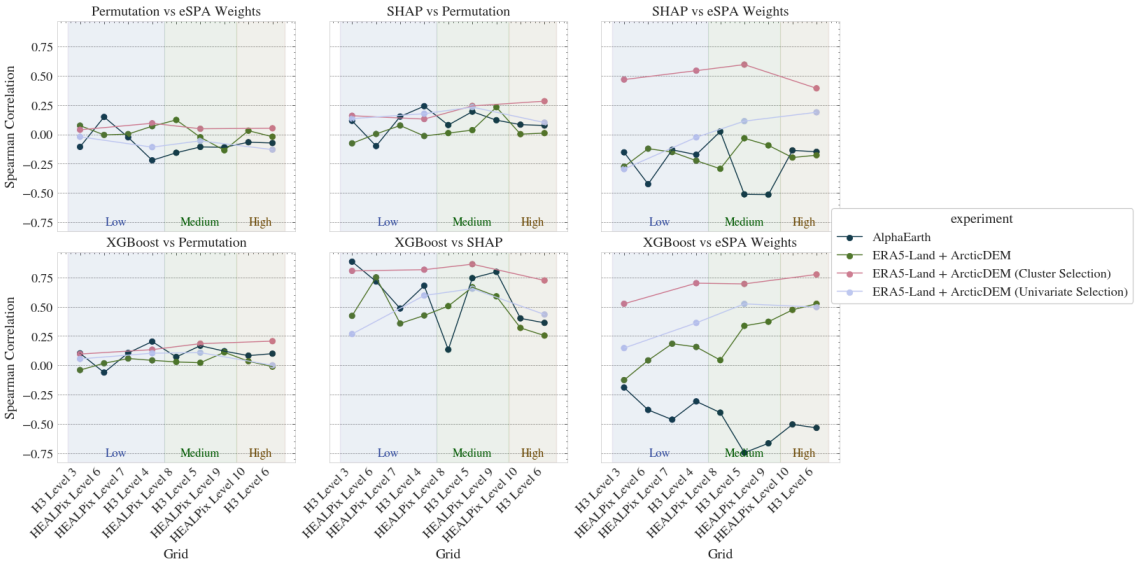

Figure 38 : Spearman rank correlation coefficients between the four different feature importance measurement methods.

grids (columns) and experiments (rows), where each pointcloud displays the SHAP values for one of the top nine most predictive features. In these visualizations, each point represents the SHAP value of a single sample from the test set, indicating how much that specific value has influenced the prediction. Additionally, the color of each point indicates whether the value is "high" or "low" relative to the distribution of all values for that feature, serving as a standardization that makes values from different features visually comparable.

A high degree of stability is indicated by the consistency of the most relevant features across grids, which is, especially for the baseline experiment, unexpected. This suggests that the SHAP method remains effective even when dealing with highly correlated features. Figure 39 and Figure 40 show the average SHAP importance ranks of the seasons and variables. The average ranks were calculated by grouping all features of a season, or variable respectively, for each experiment and grid-level. For the season, each group consists of up to 40 features, because each of the 20 variables have a mean and a trend variant for each season. The number of features for a single variables, however, is much smaller, as at maximum eight features can be grouped for a variable: two variant times four seasons. Of course, both numbers are lower for each group of the feature selection experiments, as features are filtered out. For the seasons, there is no clear most important season visible, however, the Fall gets increasingly more relevant with higher grid resolutions, especially in the the cluster selection experiment. Looking at the average ranks of individual variables, some variables seem to be clearly more relevant than others. This, however, strongly depends on the feature selection method used as some variables were completely filtered out by the selection methods. For the baseline experiment the *Naive Snow Isolation*, *Snow Depth* ("*Sde*") and *Surface sensible heat flux* ("*Sshf*") are the two highest ranked features. This changes for the selection experiments, as both variables lack any feature present in these experiments. *Thawing Days*, *Thawing Degree Days*, *Snowfall Occurrences* and *Snow Cover* ("*Snowc*") are the highest ranked variables across all grids, whereas *Fall Lake Bottom-Temperature* ("*Lblt Max Fall*"), *Snow fall* ("*Sf*") and the *Snow*

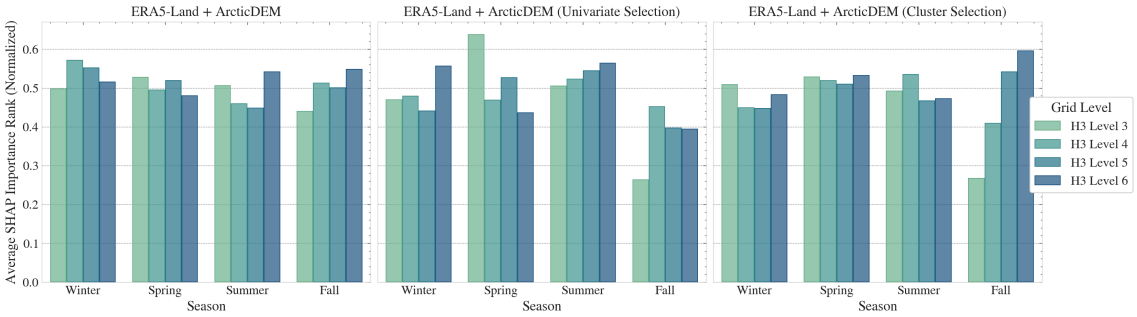

Figure 39 : Normalized average Shapley additive explanations (SHAP) importance ranks per season. Grouped by the experiment (different facets) and the H3 grid levels (color).

Cover rank above others. Of course, these ranks can not be compared across the different experiments, as the number of features and presence of variables widely differs.

Certain features consistently dominate the top 9 explanations visible in the beeswarm plots of Figure 41 across the experimental setups. The *Fall Lake Bottom-Temperature* appears in all four baseline grids, though its importance diminishes with increasing grid resolution and it is absent in other experiments due to filtering. Similarly, *Thawing* and *Freezing Days* are present in all experiments and grid resolutions, with summer trends being especially frequent with 14 occurrences. Summer temperature features were relevant across all grids in the baseline and univariate feature selection experiment. Winter and spring *Precipitation Occurrences* are also frequently in the top 9, similar to the *Effective Snow Depth Trend* in the cluster selection experiment. Notably, the relevance of Y and X coordinates only showed at higher grid resolutions, suggesting that the model is rather memorizing the locations of RTS occurrence than engaging in learning to understand the patterns behind. No topographic variables were found to have a significant impact.

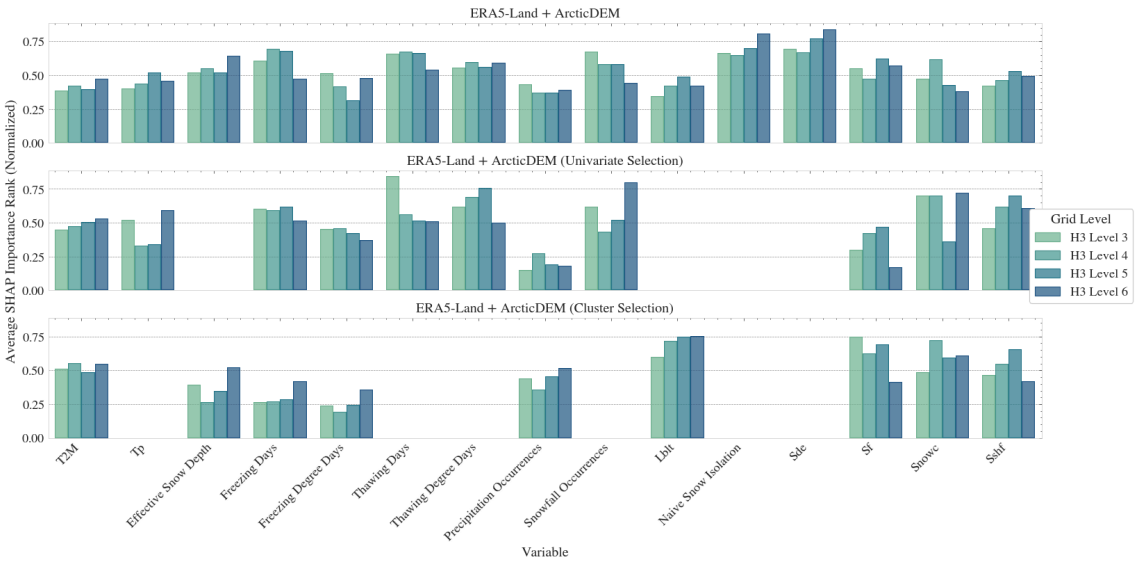

Figure 40 : Normalized average Shapley additive explanations (SHAP) importance ranks per season. Grouped by the experiment (different facets) and the H3 grid levels (color).

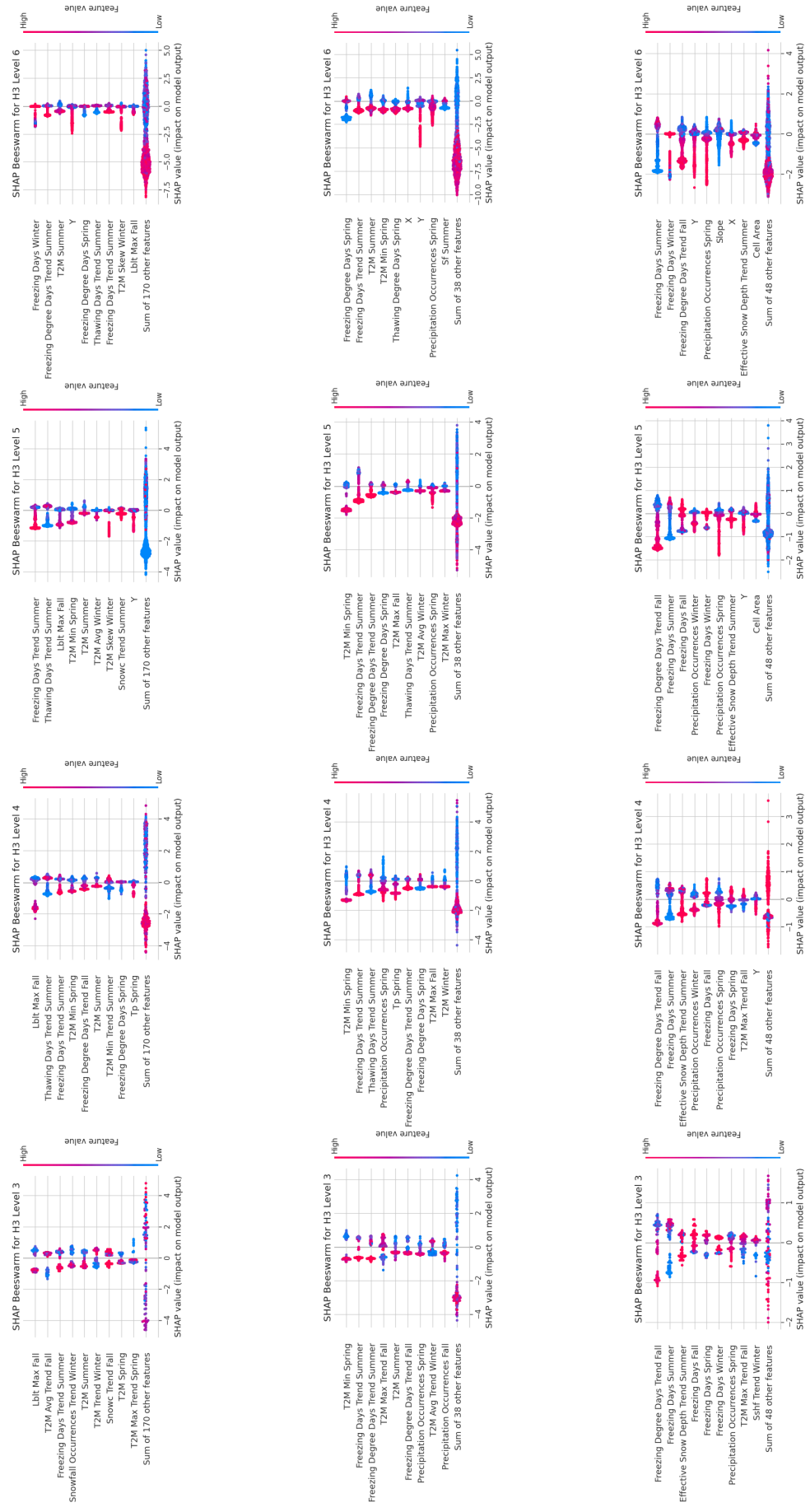

Figure 41 : SHAP values for the three experiments which used only environmental variables. From top to bottom: Baseline, Univariate Selection, Cluster Selection. Each grid has it's own beeswarm plot to analyze the feature importances over scales, ordered ascending from left to right. Only the 9 most relevant features are shown, while the Shapley additive explanations (SHAP) values of the other features are accumulated and jointly shown.

Discussion

The ensemble of DGGS aligned datasets and their creation method developed in this work is seen as a core asset of this research. The DGGS structure enables a wide range of novel experimental designs in spatial statistics, pattern analysis but also ML, especially in the Arctic. The ensembling method applicable on various DGGS grids and levels proved scalable and efficient, however, the alignment process of the original datasets onto the DGGS grids did not. While the aggregation of the original dataset onto DGGS is conceptually feasible, several limitations remain, particularly in terms of computational scalability and performance under global-scale processing. Not only is the lack of efficient software for such spatial alignments a major bottleneck in processing such global-scale environmental datasets, it also shows a research gap regarding the specific aggregation and interpolation methods that are optimal for spatial statistics and ML tasks.

For the binary RTS occurrence classification task, all models performed well on the test-set and remained stable across various hyperparameter settings, with F1 score well above 90% on low and medium resolution grids. Although predictive performance decreased as grid resolution increased, even the models trained on the highest resolutions yielded with around 60% F1 scores better than random results on the training area. Of course, the, especially in the case of high resolution grids, difference to a perfect predictive performance shows the need for further method refinement. Still, these results underscore the general feasibility of the approach to train models on a vast amount of different variables without having extensive prior knowledge about the processes behind RTSs.

Despite these strengths, critical shortcomings limit the methodological completeness of this work. Most notably, the temporal dimension is with the temporal averages and trends only in compressed form present in the experiments, even though RTS can be polycyclic and triggering is suspected to be driven by events like extreme rainfall episodes or prolonged warm winters (Nesterova et al., 2025a). Because the dataset ensembling method specifically allows for the creation of such event driven datasets, these shortcomings can be solved by adding event data, e.g. through different, event driven temporal aggregation methods of the ERA5-Land variables, and by changing the overall problem type from a tabular learning approach to a timeseries or attention based approach. Switching to a timeseries approach would open the need for further investigation in the feasibility of different timeseries models regarding large feature spaces and following analysis of feature importances.

Additionally, the methodology does not adequately reflect the strong non-linearity present within the data. This especially holds true for the spatial aggregations used to create DGGS aligned dataset. Similar to the smoothed out events on the temporal dimension by the temporal aggregations utilized, similar peaks and non-standard distribution

patterns across the spatial domain are smoothed out by the utilized aggregation methods. This opens up further research into different spatial aggregation and interpolation methods and how they can be best incorporated into model architectures. Also the ML models chosen do not properly reflect such non-linearities, as the interpretation of the RTS occurrences as a tabular data problem without any specialized feature engineering interprets features individually one-by-one. With advanced recurrence and attention based mechanisms used in advanced DL models such non-linearities, recurrence patterns or teleconnections could be reflected and learned along both spatial and temporal dimensions.

Looking also outside of the training and test areas revealed that none of the models were able to generalize well on unseen regions. This supports the statement above, that the model used do not reflect non-linearities very well, however, making proper conclusions from this would need further experiments and extensive analysis on more out-of-distribution data. It was also shown, that the models which were not prone to overfitting, eSPA and the AutoGluon ensembles, still weren't able to generalize well on unseen data.

The enrichment of the datasets with positional metadata, such as the cell-area, turned out to be a big methodological mistake. It partially prevented the models from learning patterns within the environmental data and instead optimizing on these meta features, leading even to a visible overfit onto the cell-area feature in two cases.

5.1 Dataset errors

With the use of target variables derived from DARTS v1.2, the use of DL model outputs as learning target for other ML models can probably be seen as the biggest weakness of this works approach. The equally performing models absent from a perfect score indicate problems within the data used for training, especially the target variables. Probably, the target dataset exhibits high noise regarding both RTS occurrence and RTS density. This noise would also explain the drop of predictive performance on higher resolutions for the binary RTS occurrence classifications, as lower resolutions tend to have less false labels thanks to their aggregations over larger areas. Further, the low performance on the RTS area-density regressions indicates a problem on the RTS segments provided by DARTS v1.2, as shown in Figure 13. This problem was also recognized by the authors of DARTS, as 63% of on bare ground classified RTS were false positives (Nitze et al., 2025). The spectroscopic similarity between ALD and RTS could also play a role here, as the DARTS v1.2 dataset also contained some misclassified ALDs, in one region up to 8% according to the authors. (Nitze et al., 2025). Another explanation for the underperformance could be delivered by the missing temporal information, as described before. Further, the smoothing applied by the median aggregations, as they were the only spatial aggregations used in the experiments, potentially eliminated extreme values, which could be relevant for RTS occurrence. For the density regression, a large bias within the target variable could also have played a role, both Banks island and the Taimyr peninsula stand out as hotspots for RTS activity. These two locations exhibit significantly higher observed densities compared to other Arctic regions, which may not be representative of the

broader pan-Arctic pattern. Such outliers could distorted model learning, as no special outlier treatment happened in the preprocessing steps. The extreme density values found in these regions suggest that RTS occurrence might follow a log-scale distribution across the pan-Arctic.

5.2 Novel ML techniques

AutoGluon demonstrates desirable behavior in terms of both model performance and overfitting mitigation, offering a robust pipeline with minimal need for advanced ML expertise. From a researcher's perspective, it offers the advantage of being the easiest to implement, requiring less specialized experience advanced ML techniques such as HPO, CV, bagging or stacking. The fact that AutoGluon achieves strong results without significant overfitting suggests reliable generalization across spatial scales and diverse environmental conditions, if part of the training data. A big drawback of AutoGluon is the lack of a robust method to determine feature importance scores. At time of writing this thesis, the only method available in AutoGluon for measuring feature importance is permutation feature importance, which has been shown to be highly unstable when applied to high-feature-count datasets.

eSPA shows this property of generalization even stronger, as the models almost did not overfit at all. However, the predictive performance of the eSPA models were slightly less good than the performances of the baselines or AutoGluon. The learned feature weights, in contrary to the expectations, did not correlated with the permutation feature importance scores as the Spearman's ρ never exceeded 0.2 for any experiment or grid resolution. This not only opens up the question which metric measures feature relevance the best, but also whether the eSPA feature weights are meaningful when trained on a lot of potential relevant features. The creators of eSPA only tested their algorithm against datasets with a handful of predictive features and many un-predictive features, not with a large amount of predictive features (Horenko, 2020). On the other hand, the feature weights did correlate well with XGBoost total importance gains and SHAP values, especially for high resolution grids. This could be a hint, that the feature permutation method does not work well on a large amount of features no matter their correlation, resulting in meaningless feature importance scores.

The relationship between the initial cluster counts K_{initial} of the eSPA models and their performance is non-linear, with optimal values varying by resolutions. This indicates that the environmental patterns behind RTS occurrences are less complex on larger scales than on high resolution grids. Such a finding shows that the interpretability features of the eSPA models can indeed extract useful information from complex environmental data.

Despite a slight performance drop at low resolutions, AlphaEarth embeddings proved to be a viable approach for the estimation of RTS occurrences. These embeddings function effectively with robust, well-studied models such as XGBoost, RF, and kNN, as well as with modern AutoML techniques like AutoGluon. The observed performance degradation at lower resolutions can likely be attributed to the original 10-meter resolution of AlphaEarth, which may not be accurately preserved when downscaling via Earth Engine through the aggregation of a limited subset of maximum 10 000 pixels.

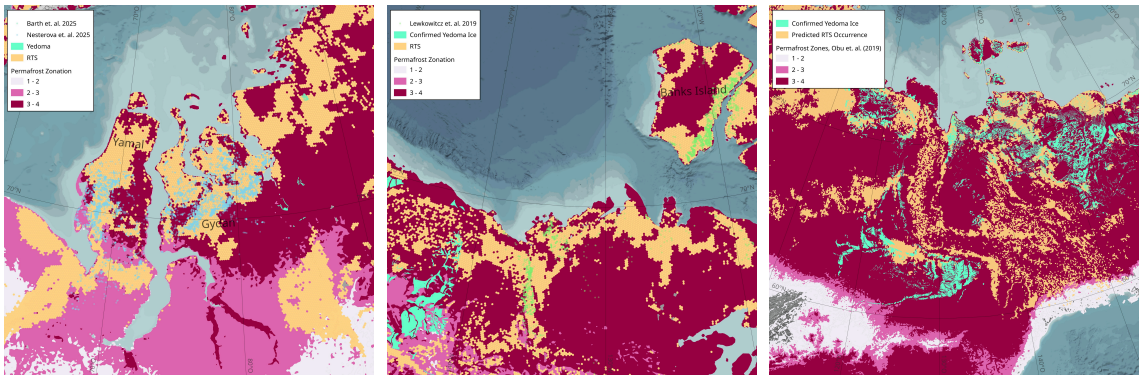

Figure 42 : Close-up on the predictions from the entropy-optimal Scalable Probabilistic Approximations algorithm (eSPA) model of the H3 level 6 grid trained with the baseline features. Left: Gydan and Yamal peninsulas, North-West Siberia, Russia, with the manual mappings from Nesterova et al. (2025a), Barth et al. (2025) and Lewkowicz (2025). Yedoma coverage from Strauss et al. (2022), Permafrost zonation from Obu et al. (2019). Middle: Mackanzie Delta and Bank Island, North Canada. Right: Central and North East Russia, showing regions with Yedoma ice. RTopo basemap from " maps@awi 2.0".

5.3 Validation

The RTS occurrence predictions of the best eSPA model, trained on all ERA5-Land and ArcticDEM features on the highest H3 grid, is presented in Figure 42. These predictions appear optically superior and generally align better with known RTS distributions compared with other predictions on the H3 level 6. In Yamal and Gydan, the RTS occurrences overlaps well with most manual mappings, with only a few sparse regions missed. Similarly, the Mackenzie Delta is mostly correctly identified, as are the South, North, and East coasts of Banks Island. However, in both regions a lot of false positives are visible. It predicts RTS occurrence in the north of Yamal, where no RTS are present. The West coast of Banks Island, were only very few to no RTS at all can be found, is also incorrectly marked as region with RTS occurrences.

The third map shows central and east Russia, were many regions are underlain by ancient Yedoma ice. Interestingly, in these regions the models RTS occurrence predictions seem to avoid most Yedoma regions. This is counter intuitive and probably incorrect, as especially the ice-richness of these regions yields a high potential for RTS in case of thaw. Considering that Yedoma ice is ancient ice formed millennia ago under completely different climate conditions, however, this may indicate an interesting observation (Strauss et al., 2017; Veremeeva et al., 2021). Because the method used by this thesis only utilized today's climate conditions to estimate RTS occurrences, the understanding of the models also only accounted for todays climate. The resulting mismatch could thus be accounted to the changed climate in these regions. As discussed before, however, the predictive performance of the model is not good enough to confidently support this theory.

5.4 Feature selection and importance

The feature selection results demonstrate that the employed methods are viable, as predictive performance remained consistent with the baseline experiments. This further shows that the baseline dataset contained redundant information, with certain features likely serving as direct proxies for others. Regarding importance stability, the selection methods generally improved alignment across metrics, with the exception of permutation scores. Consequently, permutation importance appears to be an unreliable metric for identifying relevant variables in this setup, particularly when the number of predictive features is that high. Cluster selection methods resulted in the best alignment between metrics, which is expected given that this method focussed on reducing the number of correlated, thus similar, features, finally preventing similar variables from "sharing" importance scores. Additionally, higher grid resolutions with higher sample counts resulted in more aligned importances, likely because larger datasets provide a more robust statistical representation of the real world.

A core idea of this works experiment setup, to isolate relevant variables from a large set using explainability methods, is supported by the high correlations between the eSPA feature weights and SHAP values calculated on the XGBoost models. These two ML methods are very different and so are their methods to extract feature importances, yet both methods tend to assign similar relevances to the large amount of features. This indicates that the extracted feature importances are indeed relevant for RTS occurrence, highlighting that at least a correlation exists between the most relevant features and RTS occurrences. Of course, to prove that also causalities exist between these features and RTS occurrences, extensive field measurements and modelling is required. Leveraging SHAP and eSPA weights, indications can be drawn about which variables would be important to measure in the field and which new features could be incorporated into earth-system or process-models.

The differences between the by the SHAP values highest ranked variables of Figure 40 and the number of occurrences of specific features in the 9 most relevant features of Figure 41 indicate that the seasonality of the variables plays an important role. This is especially visible for the *Lake Bottom-Temperatures* where the Fall feature was present in the 9 most relevant features for all grids of the baseline experiment, yet, the all other seasons were not seen as relevant by SHAP resulting in a poor average rank for this variable. Apart from this, the large relevance of the *Fall Lake Bottom-Temperature* is generally interesting as there exist no known connection between this variable and RTS. Because the variable is derived from ERA5-Land, the correctness of the variable values is highly uncertain, making the found relevance even more unusual. Potentially, the *Fall Lake Bottom-Temperature* is correlated within the ERA5-Land models with another sub-surface hydrological variable. Because the majority of the permafrost heat-transfer happens below ground through subsurface hydrology, this relevance may hint to a potential hydrological influence on RTS occurrence. Another possibility is the correlation with fall ground temperatures and thus summer temperatures, as the ground temperatures usually lag behind air temperatures due to a slow heat propagation. As the two features *Fall Lake Bottom-Temperature* and *Summer Mean Air Temperature* seem to correlate indicated by a Pearson's r of 0.61 and a Spearman's ρ of 0.83, this second option is more plausible.

The importance of *Summer Precipitation* and *Winter Temperature* regarding RTS occurrence, as proposed by Nesterova et al. (2025b), is reflected in the derived SHAP feature importances. However, this alignment is nuanced by the temporal definitions used, as the authors utilized a 5-month summer interval, whereas this work employed a 3-month interval. Consequently, both the summer and spring values in this study map to the authors summer period. With only *Spring Precipitation* was found to be relevant by the SHAP values, it appears that spring or early summer precipitation is more dominant than mid- or late-summer precipitation. Similarly, *Winter Temperature* is only indirectly reflected, manifesting primarily through features such as *Freezing Days* and *Temperature Skew* rather than *Mean Winter Temperature*.

Beyond these primary drivers, the SHAP analysis revealed that variables related to *Thawing* and *Freezing Days* are also significant. A particularly counter-intuitive finding is that the models ranked the occurrence counts of *Precipitation* and *Thawing*/*Freezing Days* higher than the respective total amounts of *Precipitation* and *Thawing*/*Freezing Degree Days*. This probably happens because occurrence data is statistically easier to separate, as the total amounts of especially *Thawing* or *Freezing Degree Days* may introduce noise in regions where temperatures hover near zero.

However, it is important to stress that the feature importances measured by SHAP still contained a lot of noise and the underlying models were not able to perfectly estimate RTS occurrence nor RTS area-density values. This especially holds true for high grid resolutions, were the best models barely achieved an F1-score of 57%, but also the models trained on lower resolutions showed a large discrepancy between predicted and expected RTS occurrences. Thus, only a qualitative analysis of the feature importances was made instead of a quantitative.

Conclusion

This thesis addresses the two primary research questions regarding the occurrence of RTS: identifying the critical environmental variables across spatial scales and finding the most suitable ML methods for their prediction. To answer these questions over 190000 model trainings were analyzed across several experiments, mixing different ML methods, DGGS grids on different resolutions, datasets and feature selection methods. Two novel ML methods and one novel satellite embedding dataset were tested against traditional ML methods used in environmental science and remote sensing.

A key contribution of this work is the development of a novel framework for ensembling different data sources into DGGS aligned grids, working even on consumer grade hardware. This work was able to show that it is possible to successfully train various ML algorithms on various DGGS resolutions even on large amounts of features. Further, it also showed that using certain methods for estimating feature relevance of such trained models can be a viable approach to find potential relevant variables regarding RTS occurrence.

6.1 Environmental Variables

To answer the primary research question **Q1**, four distinct feature importance metrics were applied to the best-performing models trained on a large set of environmental variables. An extensive SHAP analysis revealed a handful of key variables driving RTS occurrence. Notably, winter and spring precipitation were identified as critical, which aligns regarding spring precipitation with previous research. Additionally, the number of thawing and freezing days proved significant, especially the summer trends of both variables seemed to have a statistical impact. This was partially expected, as winter temperatures were also previously found to have an impact. A particularly interesting discovery was the relevance of fall lake bottom temperatures, which likely serves as a proxy for a hidden hydrological variable or is just a lagged heat propagation into the ground. In contrast, topographic variables showed no significant relevance across any spatial scale analyzed. Regarding spatial scale, feature relevances were observed to shift slightly, a result consistent with expectations, though a precise quantification of these changes was not feasible due to noisy relevance scores.

6.2 Machine Learning

To address the two secondary research questions Q2 an Q3, a variety of ML methods were tested against the specific remote sensing topic of estimating RTS occurrence. Three novel approaches were evaluated: (1) eSPA, a cheap and scalable cluster-based learner with great interpretability, (2) AutoGluon, an AutoML framework and (3) Alpha Earth satellite embeddings, a novel form of environmental dataset created by a large DL foundation model.

The results indicated that eSPA and AutoGluon performed well compared with well studied baseline algorithms, such as XGBoost. Although eSPA showed a slight performance decrease compared with the other algorithms, the learned feature weights lead to similar feature importances like the SHAP values of XGBoost showing the strong explainability of this cheap algorithm. While AutoGluon was effective, it lacked a suitable method for estimating feature importances given the high feature count, as permutation scores were not meaningful in these regimes. Finally, the Alpha Earth embeddings showed promising results, matching the performance of training on ordinary data, yet they did not excel beyond such standard approaches.

6.3 Outlook

While already showing promising results, the methods used in this thesis can be further improved in various aspects by future work. A probable key component of future work can be the incorporation of temporal data into the training process. This would include new temporal aggregations and adjusted feature engineering for finding and summarizing events, such as extreme rainfall or wild fires, but also adjusted model and ML mechanisms capable of correctly learning from such data. Further, more advanced statistical methods and tricks should be utilized to also reflect the non-linearity of the data, for example advanced recurrence analyses, both on the spatial and temporal dimension.

With the new and improved DARTS v2 dataset, covering the complete pan-Arctic without gaps over a timespan of nine years, more and potential better targets can be used to train models. Another possibility would be the usage of a mix of more diverse sets of target datasets, potentially improving the quality of the targets before training models on them. This may result in better predictions also for higher resolutions, opening another potential improvement: the use of even higher DGGS levels to really incorporate local dynamics and processes into the data. As this would be very compute demanding for the data engineering and DGGS alignment steps, more research must be conducted on efficient DGGS alignment methods. With improved compute efficiency, more climate variables and other datasets could be added to the dataset ensembling framework, finding even more potentially relevant variables and proxies.

Finally, the scope of application can be broadened to encompass a wider variety of permafrost features. Beyond the current targets of RTS occurrence and area-density, future studies could investigate other landscape phenomena such as lakes, taliks, ice-

wedges, gullies, pingos, or palsas. Extending the methodology to these diverse landforms could provide further insights in the drivers behind the changing cryosphere.

Glossary with Index

AL – **Active Layer**: The uppermost layer of permafrost landscape, which is thaws and become biological active in the summer. 3, 4, 9

ALD – **Active Layer Detachment**: Landslides of the active layer. 10, 11, 66

ARTS – **Arctic Retrogressive Thaw Slumps dataset** 14, 102

AUC-ROC – **Area Under the Receiver Operating Characteristic Curve** 35, 36

CNN – **Convolutional Neural-Network** 14

CV – **Cross-Validation** 37, 39, 49, 50, 55, 58, 59, 60, 67

DARTS – **Database of AI-detected Arctic RTS footprints** 14, 24, 25, 26, 29, 48, 49, 66, 72, 100, 103

DGGS – **Discrete Global Grid System**: A mosaic that covers the entire Earth's surface, based on a mathematical structure. 1, 2, 19, 20, 21, 22, 24, 26, 28, 29, 30, 31, 32, 33, 48, 57, 58, 65, 71, 72, 100, 115, 116, 117

DL – **Deep-Learning**: Similar to Machine-Learning, but with more complex (deep) models, often multi-layered neural-networks. 1, 2, 13, 14, 15, 18, 19, 33, 34, 35, 36, 37, 38, 41, 42, 49, 66, 72

DNN – **Deep Neural-Network** 38, 41, 49

DSM – **Digital Surface Model** 31

ECMWF – **European Centre for Medium-Range Weather Forecasts** 26

ECV – **Essential Climate Variable**: "Essential Climate Variables (ECVs) are a defined set of physical, chemical, and biological variables identified by the Global Climate Observing System (GCOS) as being critical for characterizing the state and evolution of the Earth's climate system" (Bojinski et al., 2014) 8

EO – **Earth-Observation** 17, 20

ERA5-Land – **ECMWF Reanalysis v5 (Land dataset)**: Climate dataset from the ERA5 reanalysis products: "ERA5 is the fifth generation ECMWF atmospheric reanalysis of the global climate covering the period from January 1940 to present. ERA5 is produced by the Copernicus Climate Change Service (C3S) at ECMWF. ERA5 provides hourly estimates of a large number of atmospheric, land and oceanic climate variables. The data cover the Earth on a 31km grid and resolve the atmosphere using 137 levels from the surface up to a height of 80km. ERA5 includes information about uncertainties for all variables at reduced spatial and temporal resolutions." (Hersbach et al., 2020) 26, 27, 31, 32, 39, 51, 52, 56, 57, 65, 68, 69, 99, 116

ERA-Interim/Land: "ERA-Interim/Land is a global land surface reanalysis data set covering the period 1979–2010." (Balsamo et al., 2015) It is the predecessor of the ERA5 dataset. 6

***ESA* – European Space Agency** 26

***ESA CCI* – ESA Climate Change Initiative** 6

***eSPA* – entropy-optimal Scalable Probabilistic Approximations algorithm** 16, 17, 19, 42, 44, 47, 48, 49, 50, 55, 56, 57, 58, 59, 60, 61, 66, 67, 68, 69, 72, 108, 115

***GBM* – Gradient boosting model** 40, 41, 42

***GCOS* – Global Climate Observing System**: "The Global Climate Observing System (GCOS) is co-sponsored by the World Meteorological Organization (WMO), the Intergovernmental Oceanographic Commission of the United Nations Educational, Scientific and Cultural Organization (IOC-UNESCO), the United Nations Environment Programme (UN Environment), and the International Science Council (ISC). It regularly assesses the states of global climate observations of the atmosphere, land and ocean and produces guidance for its improvement.

GCOS expert panels maintain definitions of Essential Climate Variables (ECVs) required to observe Earth's changing climate systematically.

The observations supported by GCOS contribute to solving challenges in climate research and also underpin climate services and adaptation measures." (WMO, 2026) 8

***GTN-P* – Global Terrestrial Network for Permafrost**: "The Global Terrestrial Network for Permafrost (GTN-P) is the leading international programme for long-term permafrost monitoring. Its mission is to sustain a comprehensive global network that delivers consistent, high-quality, long-term observations of key permafrost indicators (Permafrost Essential Climate Variable quantities). These data support the assessment of the current state of permafrost and its changes over time." (Irrgang & Lübker, 2026) 8

***HPO* – Hyperparameter-Optimization** 37, 38, 39, 40, 41, 42, 49, 50, 55, 58, 60, 67

***IoU* – Intersection over Union** 12

***KDE* – Kernel Density Estimate** 21

***kNN* – k-Nearest-Neighbors** 19, 42, 44, 49, 50, 51, 55, 56, 58, 59, 60, 67, 108, 109

***LandTrendr* – Landsat-based detection of Trends in Disturbance and Recovery** 14, 101

***MAAT* – Mean annual air temperature** 6

***MAE* – Mean Absolute Error** 36

***MAGT* – Mean annual ground temperature** 6

***MI* – Mutual information**: Measure to estimate the amount of information one random variable contains about another. 30, 36, 43, 44, 52, 53

***ML* – Machine-Learning**: Group of methods to numerically and heuristically "fit" observed data to a target variable in order to "learn" complex statistical relationship between the data and the target variable. 1, 2, 13, 14, 15, 16, 17, 18, 19, 20, 25, 27, 29, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 47, 49, 65, 66, 67, 69, 71, 72

MODIS – **Moderate Resolution Imaging Spectroradiometer**: Instrument aboard the Terra and Aqua satellites, launched by NASA. 6

MSE – **Mean Squared Error** 36

NDVI – **Normalized Difference Vegetation Index** 14, 103

PCA – **Principle Component Analysis**: Method to reduce the number of features before training a machine-learning model. 44

PZI – **Permafrost Zonation Index**: Index which describes how favourable the conditions for permafrost occurance at a positions are. (Gruber, 2012) 6

RF – **Random Forest**: An ensemble of learned decision trees. 19, 39, 40, 41, 42, 49, 50, 51, 55, 56, 67, 108, 109

RMSE – **Root Mean Squared Error** 36

RSFM – **Remote-Sensing Foundation Model** 15, 16, 17, 19

RTS – **Retrogressive Thaw Slump**: A permafrost thermo-erosion landscape feature. 2, 4, 9, 10, 11, 12, 13, 14, 15, 17, 18, 19, 24, 25, 26, 33, 47, 48, 49, 55, 56, 57, 58, 63, 65, 66, 67, 68, 69, 70, 71, 72, 100, 101, 115, 116, 117

SHAP – **Shapley additive explanations**: An unified framework for interpreting predictions through feature importances. 46, 47, 51, 61, 62, 63, 64, 67, 69, 70, 71, 72

SOC – **Soil Organic Carbon**: Carbon which is stored in organic material in the soil 6, 7

SPARTAn – **Sparse Probabilistic Approximation for Regression Task Analysis** 16, 17, 42, 50

SVM – **Support Vector Machine** 14

TPI – **Topographic Position Index**: Elevation compared to it's neighborhood. Also known as relative elevation. 14, 32, 103

TRI – **Terrain Ruggedness Index** 31

TTOP – **Temperature at the top of the permafrost** 6

UC – **Uncertainty Quantification** 34

VRM – **Vector Ruggedness Measure** 31

XGBoost – **eXtreme Gradient Boosting** 19, 40, 49, 50, 51, 55, 56, 57, 58, 60, 61, 67, 69, 72, 108, 109, 115, 116, 117

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Code & Data availability

The code created for this thesis can be found on my private Forgejo instance: https://forgejo.tobiashoelzer.de/tobias/entropice.
A mirror repository which contains a snapshot at the date of this thesis' submission is available on Github: https://github.com/relativityhd/entropice. This mirror is further available under the following DOI: https://doi.org/10.5281/zenodo.18827838
Data can be provided on request.

Tables

B.1 Used and derived variables from ERA5-Land

Variable	Kind	Unit	Formula
Total Precipitation tp	Accum	m	
Surface sensible heat flux sshf	Accum	$\frac{J}{m^2}$	
Snowfall sf	Accum	m	
Snowcover $\text{snowc}_{\text{mean}}$	Instant		
Lake bottom temperature lblt_{max}	Instant	K	
Snow depth sde_{mean}	Instant	m	
Mean Air temperature t2m_{mean}	Instant	K	
Minimum Air temperature t2m_{min}	Instant	K	
Maximum Air temperature t2m_{max}	Instant	K	
Air temperature range $\text{t2m}_{\text{range}}$	Instant	K	$\text{t2m}_{\text{max}} - \text{t2m}_{\text{min}}$
Average Air temperature t2m_{avg}	Instant	K	$\frac{\text{t2m}_{\text{range}}}{2}$
Air temperature skew t2m_{skew}	Instant		$\frac{\text{t2m}_{\text{mean}} - \text{t2m}_{\text{min}}}{\text{t2m}_{\text{range}}}$
Naive snow isolation	Accum	m	$\text{snowc}_{\text{mean}} \cdot \text{sde}_{\text{mean}}$
Effective snow depth $S_{\text{depth,eff}}$	Accum	m	$\frac{\sum_{m=1}^{M}(S_m \cdot (M+1-m))}{\sum_{m=1}^{M} m}$
Thawing days	Accum	d	$\sum_{d}^{D} \begin{cases} 1 \text{ if } \text{t2m}_{\text{avg}} > 0 \\ 0 \text{ else} \end{cases}$
Freezing days	Accum	d	$\sum_{d}^{D} \begin{cases} 1 \text{ if } \text{t2m}_{\text{avg}} < 0 \\ 0 \text{ else} \end{cases}$
Thawing degree days	Accum	Kd	$\sum_{d}^{D} \begin{cases} \text{t2m}_{\text{avg}} \text{ if } \text{t2m}_{\text{avg}} > 0 \\ 0 \text{ else} \end{cases}$
Freezing degree days	Accum	Kd	$\sum_{d}^{D} \begin{cases} \text{t2m}_{\text{avg}} \text{ if } \text{t2m}_{\text{avg}} < 0 \\ 0 \text{ else} \end{cases}$
Precipitation occurrences	Accum	d	$\sum_{d}^{D} \begin{cases} 1 \text{ if tp } > 0.001 \\ 0 \text{ else} \end{cases}$
Snowfall occurrences	Accum	d	$\sum_{d}^{D} \begin{cases} 1 \text{ if sf } > 0.001 \\ 0 \text{ else} \end{cases}$

Table 9 : Used and derived ERA5-Land variables.

B.2 DGGS Cell counts

Grid	labels	v1.2	v1.2 (Synopsis)
H3 Level 3	37 / 191	706 / 4 380	238 / 730
H3 Level 4	57 / 357	2 585 / 24 342	999 / 4 056
H3 Level 5	103 / 902	9 573 / 154 050	4 360 / 25 671
H3 Level 6	240 / 3 420	23 420 / 1 028 028	12 787 / 171 329
HEALPix Level 6	33 / 223	857 / 5 112	296 / 852
HEALPix Level 7	47 / 340	2 091 / 16 956	790 / 2 826
HEALPix Level 8	74 / 611	5 474 / 59 928	2 299 / 9 988
HEALPix Level 9	126 / 1 313	12 678 / 223 428	6 090 / 37 236
HEALPix Level 10	248 / 3 403	23 015 / 861 072	12 527 / 143 505

Table 10 : Number of cells containing RTS and number of cells (rts/total) of the different DGGS realisations of the different DARTS target datasets.

Dataset / Author	Resolution	Coverage / Region	Method & Data Sources	Constrains
Sluijs & Kokelj (2023)		Two patches in Northern Canada and Alaska	Manual delineationion	Limited spatial extent; subjective delineations; manual effort; short time span (2011-2016)
Lewkowicz (2025)	Centroids	Several patches in Northern Canada and Alaska	Manually mapped centroids derived from 30 m Landsat time-series (1984–2016/2018)	Limited spatial extent; manual effort; no delineations
Nesterova et al. (2025a); Nesterova et al. (2025b)	Centroids; ~3m/px delineations	Gydan and Yamal peninsulas (West Siberian Arctic)	Manual filtering of published datasets with ESRI Wayback (~1m/px); Manual delineation on declassified spy imagery	Limited to two peninsulas; large manual effort
Runge et al. (2022)	30m/px delineations	Russian Arctic	Change-detection using LandTrendr; filtering steps	Many false-positives despite filtering; low resolution
Bernhard et al. (2020); Bernhard et al. (2022)	Delineations (2m ArcticDEM)	Northwest Territories (Northern Canada); follow-up several study sites across the Arctic	Linear threshold models, Random Forests, SVM; elevation change from TanDEM-X (2011–2016)	Limited spatial extent; short time span (2011-2016); limited delineations (only elevation-difference is delineated)

Table 11 : Comparison of Retrogressive Thaw Slump (RTS) dataset and inventories.

Dataset / Author	Resolution	Coverage / Region	Method & Data Sources	Constrains
Huang et al. (2023)	Delineations (2m/px)	Arctic (using ArcticDEM)	Multi-step DL setup; heavy feature engineering; YOLOv4 object detection	short time span (2011-2016); need for heavy manual filtering
Dai et al. (2025)	2m/px delineations	Arctic (using ArcticDEM)	Threshold classification on elevation change; followed by CNN-based filtering, delineation and volume estimation	Limited delineations (only elevation-difference is delineated)
Maier et al. (2025a); Maier et al. (2025b)	10m/px delineations	Several study sites across the Arctic	TanDEM-X elevation data; UNet	Limited spatial extent; limited delineations (only elevation-difference is delineated)
ARTS Yang et al. (2023); Yang et al. (2025)	4m/px delineations	collection of study-sides across the Arctic	Augmented U-Net trained on Maxar, Sentinel-2, ArcticDEM; Multiple Arctic datasets (standardized collection)	Incomplete spatial coverage;
Lewkowicz & Way (2019)	Centroids	Banks Island, Northern Canada	Manually mapped centroids on Google Earth Engine Timelapse	Limited spatial extent; manual effort; no delineations

Table 12 : Continuation of Table 11

Dataset / Author	Resolution	Coverage / Region	Method & Data Sources	Constrains
Barth et al. (2025)	High resolution delineations	Russian Arctic	Manually mapped on several satellite optical imageries	Limited spatial extent; manual effort
DARTS Nitze et al. (2021); Nitze et al. (2025)	3m/px yearly delineations	Several sites across the pan-Arctic	U-Net with Planetscope optical imagery, ArcticDEM, Landsat Trends (Tasseled Cap Indices), engineered featues (TPI, Slope, NDVI); ensemble model	Noisy; poor temporal continuity; incomplete spatial coverage
Makopoulou et al. (2024)	$1km^2$	pan-Arctic	Spatio-statistic modelling based on CHELSA Climate, geomorphology and soil-grid data	Limited data source variability; limited spacial resolution.

Table 13 : Continuation of Table 11

B.4 List of selected feature variables

Variable	Winter	Spring	Summer	Fall	Winter Trend	Spring Trend	Summer Trend	Fall Trend	General
X									X
Y									X
Freezing Days		X	X				X		
Freezing Degree Days	X	X	X		X		X	X	
Precipitation Occurrences		X		X					
Sf			X						
Snowc							X		
Snowfall Occurrences			X				X		
Sshf	X			X					
T2M Avg	X	X	X		X			X	
T2M Max	X	X	X	X				X	
T2M	X	X	X		X			X	
T2M Min		X			X		X		
T2M Range		X	X						
T2M Skew			X						
Thawing Days		X	X				X		
Thawing Degree Days		X	X						
Tp		X	X						

Table 14 : Selected features based on the univariate selection.

Variable	Winter	Spring	Summer	Fall	Winter Trend	Spring Trend	Summer Trend	Fall Trend	General
Cell Area									X
Water Area									X
X									X
Y									X
Aspect									X
Curvature									X
Dem									X
Slope									X
Effective Snow Depth					X	X	X	X	
Freezing Days	X	X	X	X		X		X	
Freezing Degree Days					X	X		X	
Lblt Max			X		X	X	X	X	
Precipitation Occurrences	X	X			X	X	X	X	
Sf		X				X	X		
Snowc	X					X		X	
Sshf	X	X			X	X			
T2M Avg						X	X		
T2M Max		X			X		X	X	
T2M Min	X							X	
T2M Range	X				X	X			
T2M Skew					X	X	X	X	

Table 15 : Selected features based on the cluster selection.

Selected Feature	Other Features in Cluster
Slope	Tri Large, Tri Medium, Tri Small, Vrm Large, Vrm Medium, Vrm Small
Dem	—
Water Area	Land Ratio, Effective Snow Depth Spring, Effective Snow Depth Summer, Effective Snow Depth Winter, Effective Snow Depth Fall, Naive Snow Isolation Spring, Naive Snow Isolation Summer, Naive Snow Isolation Winter, Naive Snow Isolation Fall, Naive Snow Isolation Trend Summer, Sde Spring, Sde Summer, Sde Winter, Sde Fall, Sde Trend Summer, Snowc Summer, Snowc Trend Winter
Snowc Winter	Snowc Fall
Freezing Days Winter	Freezing Days Trend Winter, Thawing Days Winter, Thawing Days Trend Winter, Thawing Degree Days Winter, Thawing Degree Days Trend Winter
Snowc Trend Fall	—
Effective Snow Depth Trend Spring	Naive Snow Isolation Trend Spring, Sde Trend Spring
Precipitation Occurrences Trend Fall	Snowfall Occurrences Trend Fall, Tp Trend Fall
Freezing Days Trend Fall	Thawing Days Trend Fall, Thawing Degree Days Trend Fall
Lblt Max Trend Fall	—
Effective Snow Depth Trend Fall	Naive Snow Isolation Trend Fall, Sde Trend Fall
Effective Snow Depth Trend Winter	Naive Snow Isolation Trend Winter, Sde Trend Winter, Sf Trend Fall
Precipitation Occurrences Trend Winter	Sf Trend Winter, Snowfall Occurrences Trend Winter, Tp Trend Winter
Curvature	Tpi Large, Tpi Medium, Tpi Small
Sf Trend Spring	Snowfall Occurrences Trend Spring
T2M Range Trend Spring	—
Precipitation Occurrences Trend Spring	Tp Trend Spring
T2M Skew Trend Spring	—
T2M Avg Trend Spring	T2M Trend Spring
Freezing Days Trend Spring	T2M Max Trend Spring, Thawing Days Trend Spring, Thawing Degree Days Trend Spring
Sshf Trend Winter	Sshf Trend Fall
T2M Max Trend Summer	—
Cell Area	Land Area
T2M Max Trend Winter	—
T2M Skew Trend Winter	—
Aspect	—
Lblt Max Trend Winter	—
Lblt Max Summer	Lblt Max Winter
Lblt Max Trend Spring	—
Lblt Max Trend Summer	—
T2M Avg Trend Summer	T2M Trend Summer, Thawing Degree Days Trend Summer
T2M Range Trend Winter	—
Precipitation Occurrences Trend Summer	Sshf Trend Summer, Tp Trend Summer
Freezing Degree Days Trend Spring	T2M Min Trend Spring
X	T2M Range Trend Summer, T2M Range Trend Fall, T2M Skew Spring, T2M Skew Summer, T2M Skew Winter, T2M Skew Fall
T2M Min Winter	T2M Min Fall
Freezing Days Fall	Freezing Degree Days Winter, Freezing Degree Days Fall, Lblt Max Fall, T2M Avg Winter, T2M Avg Fall, T2M Winter, T2M Fall, Thawing Days Fall, Thawing Degree Days Fall
Precipitation Occurrences Winter	Precipitation Occurrences Fall, Sf Winter, Sf Fall, Snowfall Occurrences Winter, Snowfall Occurrences Fall, Tp Winter, Tp Fall
Precipitation Occurrences Spring	Precipitation Occurrences Summer, Tp Spring, Tp Summer
Freezing Days Spring	Freezing Days Trend Summer, Freezing Degree Days Spring, Lblt Max Spring, Snowc Spring, T2M Avg Spring, T2M Avg Summer, T2M Max Winter, T2M Max Fall, T2M Spring, T2M Summer, T2M Min Spring, Thawing Days Spring, Thawing Days Trend Summer, Thawing Degree Days Spring, Thawing Degree Days Summer
Freezing Days Summer	Freezing Degree Days Summer, Freezing Degree Days Trend Summer, Sf Summer, Snowfall Occurrences Summer, T2M Min Summer, Thawing Days Summer
Effective Snow Depth Trend Summer	Snowc Trend Summer
Sf Spring	Snowfall Occurrences Spring
T2M Range Winter	T2M Range Fall
T2M Max Spring	T2M Max Summer, T2M Range Spring, T2M Range Summer
Sshf Spring	Sshf Summer
Y	—
T2M Max Trend Fall	T2M Min Trend Summer
T2M Skew Trend Summer	—
Snowc Trend Spring	—
Sf Trend Summer	Snowfall Occurrences Trend Summer
T2M Skew Trend Fall	—
Freezing Degree Days Trend Winter	T2M Avg Trend Winter, T2M Trend Winter
Freezing Degree Days Trend Fall	T2M Avg Trend Fall, T2M Trend Fall, T2M Min Trend Winter
Sshf Winter	Sshf Fall
Sshf Trend Spring	—
T2M Min Trend Fall	—

Table 16 : Clusters of selected features.

B.5 Metrics

Grid	Min	Max	Mean ± Std
			ERA5-Land + ArcticDEM (Univariate Selection)
H3 Level 3	−0.030	0.029	0.002 ± 0.025
H3 Level 4	−0.002	0.011	0.002 ± 0.006
H3 Level 5	0.003	0.007	0.005 ± 0.002
H3 Level 6	−0.093	−0.005	-0.025 ± 0.038
			ERA5-Land + ArcticDEM (Cluster Selection)
H3 Level 3	−0.030	0.029	0.002 ± 0.025
H3 Level 4	−0.002	0.011	0.002 ± 0.006
H3 Level 5	0.003	0.007	0.005 ± 0.002
H3 Level 6	−0.093	−0.005	-0.025 ± 0.038
			AlphaEarth
H3 Level 3	0.007	0.025	0.015 ± 0.008
HEALPix Level 6	−0.048	0.000	-0.025 ± 0.020
HEALPix Level 7	−0.037	−0.019	-0.027 ± 0.008
H3 Level 4	−0.005	0.010	-0.001 ± 0.007
HEALPix Level 8	−0.014	−0.004	-0.010 ± 0.005
H3 Level 5	−0.022	0.000	-0.013 ± 0.009
HEALPix Level 9	−0.002	0.025	0.008 ± 0.013
HEALPix Level 10	−0.025	0.029	-0.005 ± 0.023
H3 Level 6	−0.022	0.025	-0.006 ± 0.021

Table 17 : Differences of F1-scores between the different experiments compared to the baseline. eSPA was filtered out for AlphaEarth due to a methological mistake.

Model	H3 Level 3	H3 Level 4	H3 Level 5	H3 Level 6
				ERA5-Land + ArcticDEM
eSPA	0.947	0.917	0.817	0.515
XGBoost	0.917	0.930	0.845	0.589
Random Forest	0.927	0.932	0.846	0.583
kNN	0.939	0.924	0.841	0.570
AutoGluon	0.956	0.930	0.852	0.590
All (except eSPA)	0.935 ± 0.014	0.929 ± 0.003	0.846 ± 0.004	0.583 ± 0.008
				ERA5-Land + ArcticDEM (Univariate Selection)
eSPA	0.923	0.930	0.810	0.405
XGBoost	0.942	0.931	0.836	0.561
Random Forest	0.942	0.933	0.839	0.577
kNN	0.926	0.932	0.834	0.553
AutoGluon	0.962	0.933	0.841	0.546
All (except eSPA)	0.943 ± 0.013	0.932 ± 0.001	0.837 ± 0.002	0.559 ± 0.012
				ERA5-Land + ArcticDEM (Cluster Selection)
eSPA	0.917	0.928	0.825	0.421
XGBoost	0.946	0.930	0.848	0.579
Random Forest	0.946	0.930	0.852	0.576
kNN	0.946	0.926	0.847	0.556
AutoGluon	0.938	0.928	0.856	0.585
All (except eSPA)	0.944 ± 0.004	0.929 ± 0.002	0.851 ± 0.004	0.574 ± 0.011
				AlphaEarth
eSPA*	0.871	0.895	0.655	0.342
XGBoost	0.942	0.927	0.830	0.577
Random Forest	0.943	0.929	0.831	0.561
kNN	0.951	0.934	0.841	0.595
AutoGluon	0.962	0.925	0.831	0.576
All (except eSPA)	0.950 ± 0.008	0.928 ± 0.003	0.833 ± 0.005	0.577 ± 0.012

Table 18 : F1-scores across H3 grids for each model of each experiment.
*: These values are not representatitve due to an methological error in the feature scaling.

Model	H3 Level 3	H3 Level 4	H3 Level 5	H3 Level 6
				ERA5-Land + ArcticDEM
XGBoost	0.023	0.498	0.557	0.409
Random Forest	−0.020	0.455	0.577	0.439
kNN	0.092	0.323	0.511	0.395
AutoGluon	0.228	0.556	0.562	0.404
All	0.937 ± 0.014	0.927 ± 0.006	0.840 ± 0.012	0.569 ± 0.028
				ERA5-Land + ArcticDEM (Univariate Selection)
XGBoost	0.024	0.438	0.439	0.396
Random Forest	0.068	0.548	0.515	0.358
kNN	0.150	0.493	0.450	0.375
AutoGluon	0.183	0.617	0.447	0.396
All	0.939 ± 0.014	0.932 ± 0.001	0.832 ± 0.011	0.528 ± 0.063
				ERA5-Land + ArcticDEM (Cluster Selection)
XGBoost	0.055	0.393	0.446	0.411
Random Forest	0.118	0.390	0.409	0.413
kNN	0.121	0.275	0.407	0.386
AutoGluon	0.302	0.448	0.466	0.411
All	0.939 ± 0.011	0.929 ± 0.002	0.846 ± 0.011	0.544 ± 0.062
				AlphaEarth
XGBoost	−0.395	0.414	0.601	0.490
Random Forest	−0.063	0.425	0.642	0.499
kNN	0.014	0.445	0.483	0.504
AutoGluon	0.104	0.490	0.682	0.517
All	0.934 ± 0.032	0.922 ± 0.014	0.798 ± 0.072	0.530 ± 0.094

Table 19 : r^2 across H3 grids for each model of each experiment.

B.6 AutoGluon Leaderboards

#	H3 Level 3	H3 Level 4	H3 Level 5	H3 Level 6
1	RealTabPFN-v2 r13 (0.959)	TabDPT r22 (0.934)	TabDPT r20 (0.903)	Ensemble (0.894)
2	RealTabPFN-v2 r11 (0.959)	TabDPT r1 (0.934)	Ensemble (0.903)	TabM r99 (0.894)
3	TabDPT r20 (0.949)	TabDPT c1 (0.934)	TabDPT r15 (0.901)	LightGBMPrep r13 (0.894)
4	TabDPT c1 (0.949)	TabM r99 (0.934)	TabICL c1 (0.901)	CatBoost c1 (0.891)
5	RealTabPFN-v2 r106 (0.949)	Ensemble (0.934)	TabDPT r1 (0.900)	TabDPT r1 (0.885)
6	Ensemble (0.949)	TabICL c1 (0.934)	TabDPT c1 (0.899)	TabDPT r15 (0.884)
7	TabICL c1 (0.949)	Mitra c1 (0.934)	CatBoost c1 (0.898)	TabDPT r20 (0.880)
8	TabDPT r1 (0.939)	TabDPT r15 (0.931)	TabDPT r22 (0.898)	TabDPT c1 (0.878)
9		TabDPT r20 (0.931)	TabM r99 (0.897)	TabDPT r22 (0.875)
10		RealTabPFN-v2 r196 (0.931)	LightGBMPrep r13 (0.894)	LightGBMPrep r41 (0.853)
11		RealTabPFN-v2 r13 (0.931)	LightGBMPrep r41 (0.689)	
12		RealTabPFN-v2 c1 (0.931)		
13		RealTabPFN-v2 r106 (0.931)		
14		RealTabPFN-v2 r11 (0.931)		

Table 20 : AutoGluon Leaderboards of the baseline experiment (ERA5-Land + ArcticDEM).

#	H3 Level 3	H3 Level 4	H3 Level 5	H3 Level 6
1	RealTabPFN-v2 r196 (0.961)	RealTabPFN-v2 r13 (0.973)	TabICL c1 (0.941)	Ensemble (0.944)
2	RealTabPFN-v2 c1 (0.961)	Ensemble (0.973)	TabM r99 (0.939)	TabM r99 (0.944)
3	RealTabPFN-v2 r106 (0.961)	RealTabPFN-v2 r11 (0.973)	CatBoost c1 (0.939)	CatBoost c1 (0.942)
4	RealTabPFN-v2 r11 (0.959)	RealTabPFN-v2 c1 (0.966)	Ensemble (0.939)	LightGBM r73 (0.925)
5	TabDPT r22 (0.958)	RealTabPFN-v2 r106 (0.966)	TabDPT r15 (0.938)	LightGBMPrep r13 (0.925)
6	TabDPT r20 (0.958)	TabICL c1 (0.966)	LightGBMPrep r13 (0.938)	LightGBMPrep r31 (0.925)
7	RealTabPFN-v2 r13 (0.958)	TabDPT r1 (0.952)	TabDPT r20 (0.938)	LightGBMPrep r41 (0.925)
8	TabICL c1 (0.958)	TabDPT r20 (0.952)	LightGBMPrep r21 (0.937)	
9	TabDPT r1 (0.957)	TabDPT c1 (0.952)	LightGBM r73 (0.937)	
10	TabDPT c1 (0.956)		TabDPT r1 (0.937)	
11	TabDPT r15 (0.954)		LightGBMPrep r31 (0.937)	
12	TabM r99 (0.954)		TabDPT r22 (0.936)	
13	Ensemble (0.954)		TabM r124 (0.936)	
14	TabM r124 (0.953)		TabDPT c1 (0.936)	
15	Mitra c1 (0.952)		TabM r69 (0.935)	
16	CatBoost c1 (0.951)		LightGBMPrep r41 (0.927)	
17	LightGBMPrep r31 (0.951)		LightGBM r37 (0.923)	
18	LightGBMPrep r13 (0.950)			
19	LightGBMPrep r41 (0.948)			

Table 21 : AutoGluon Leaderboards of the AlphaEarth experiment.

Figures

C.1 Feature Correlations

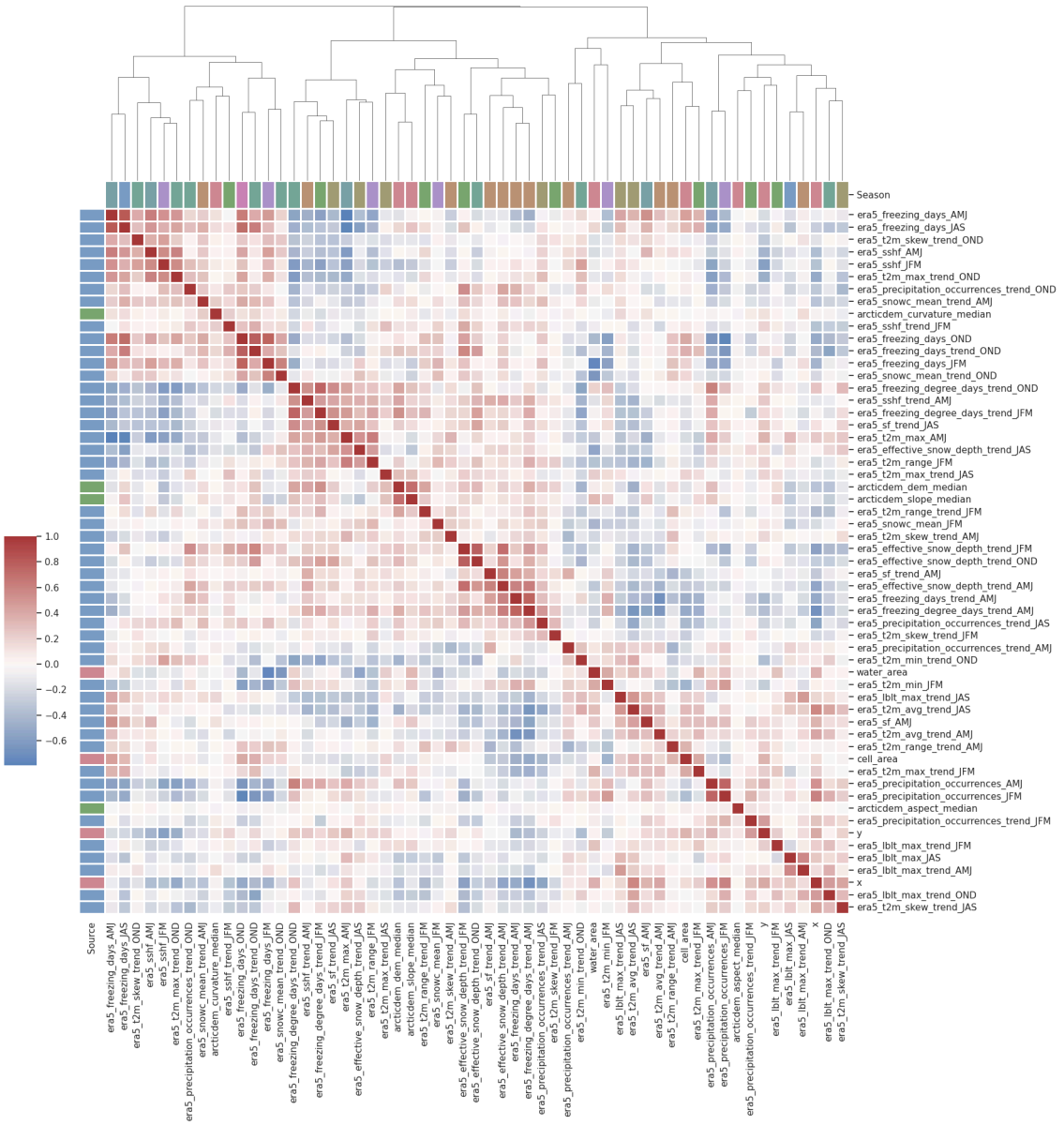

Figure 44 : Spearman correlations after the cluster-based feature selection.

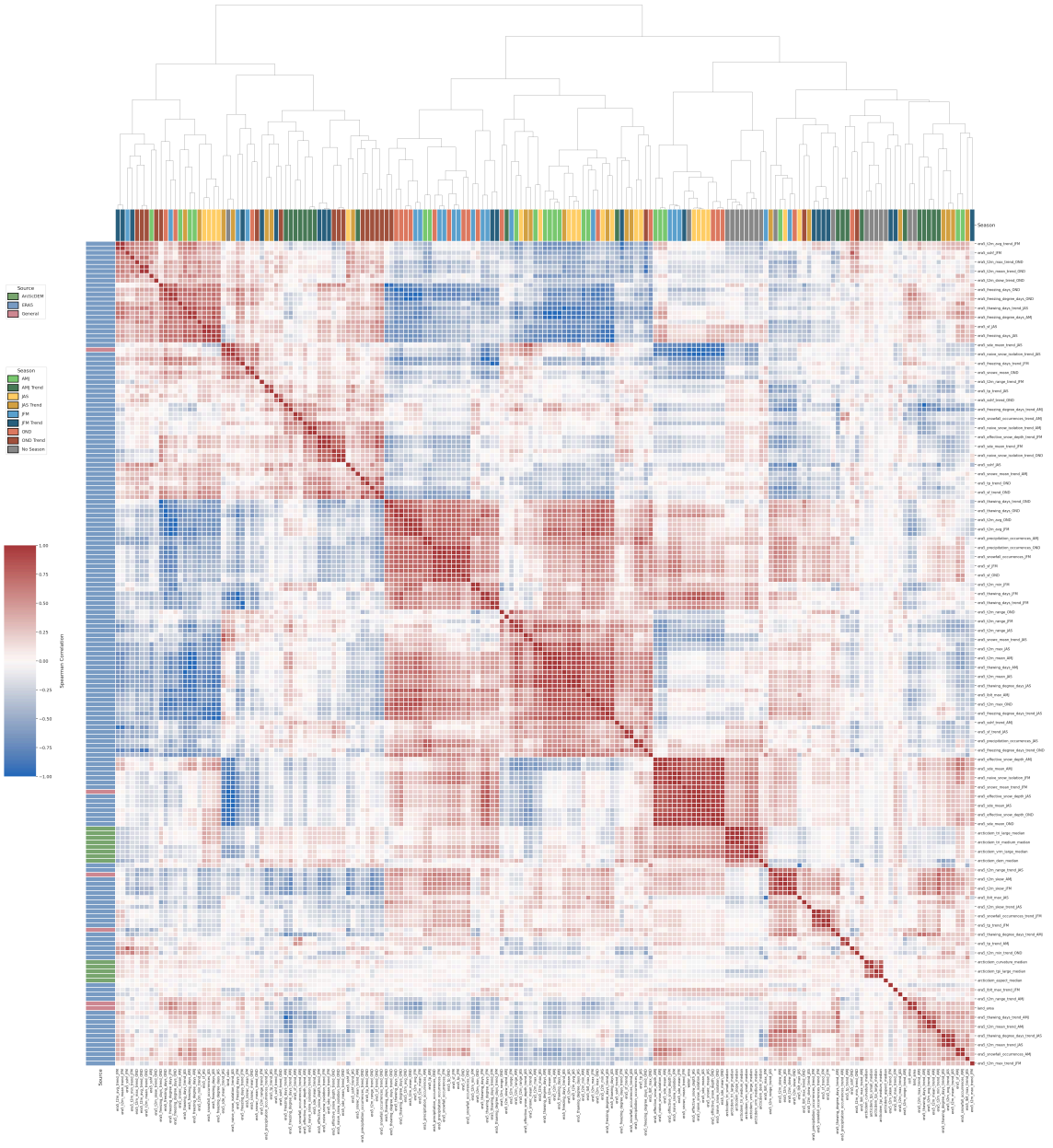

Figure 43 : Spearman correlations of all features from the ERA5-Land quarterly + ArcticDEM dataset ensemble. The colored row indices show the source dataset, with most variables coming from the quarterly ERA5-Land dataset (blue). The colored column indices show which seasons the ERA5-Land variables represent, with darker colors being trend variables and brighter colors being the temporal means. Above the graph a dendrogram based on ward linkage visualizes variable clusters which are strongly correlated. The correlations were calculated on the H3 level 6 training split of the dataset.

C.2 Feature breakdown

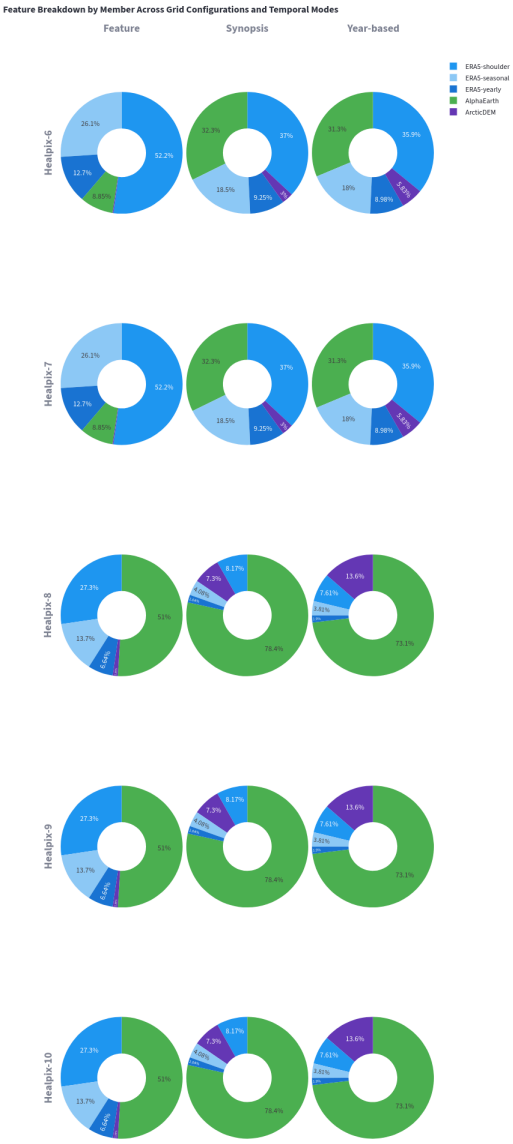

Figure 45 : Breakdown of share of features coming from the source datasets, across the different temporal modes and HEALPix grids. The legend still uses the development names for the ERA5-Land dataset: ERA5-shoulder describes the ERA5-Land quarterly variant, ERA5-seasons describes the ERA5-Land semiannual variant and ERA5-yearly describes the ERA5-Land yearly variant. Screenshot from the Streamlit Dashboard.

C.3 Maps

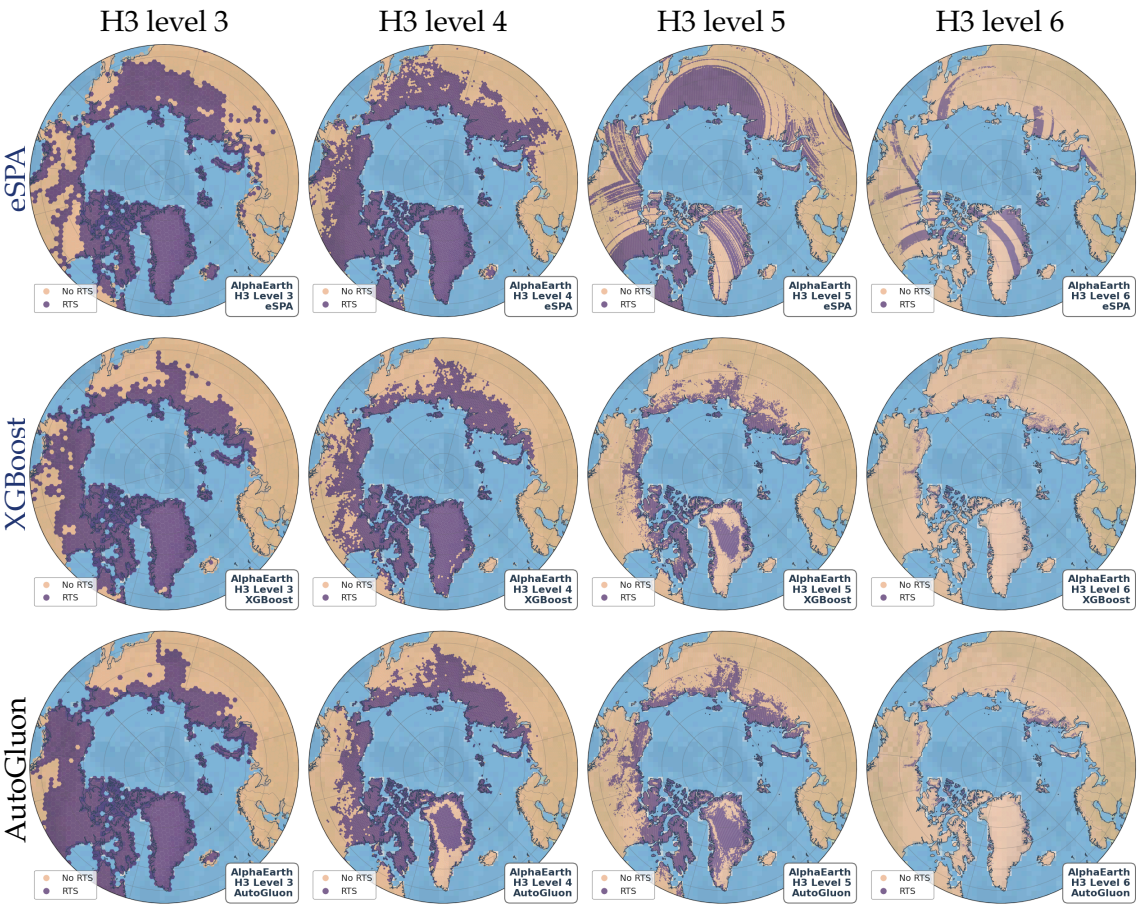

Figure 46 : Retrogressive Thaw Slump occurrence predictions across scales (columns) from different models (rows): eSPA (top row), XGBoost (middle row) and AutoGluon (bottom row). Models were trained on the DGGS aligned AlphaEarth satellite embeddings.

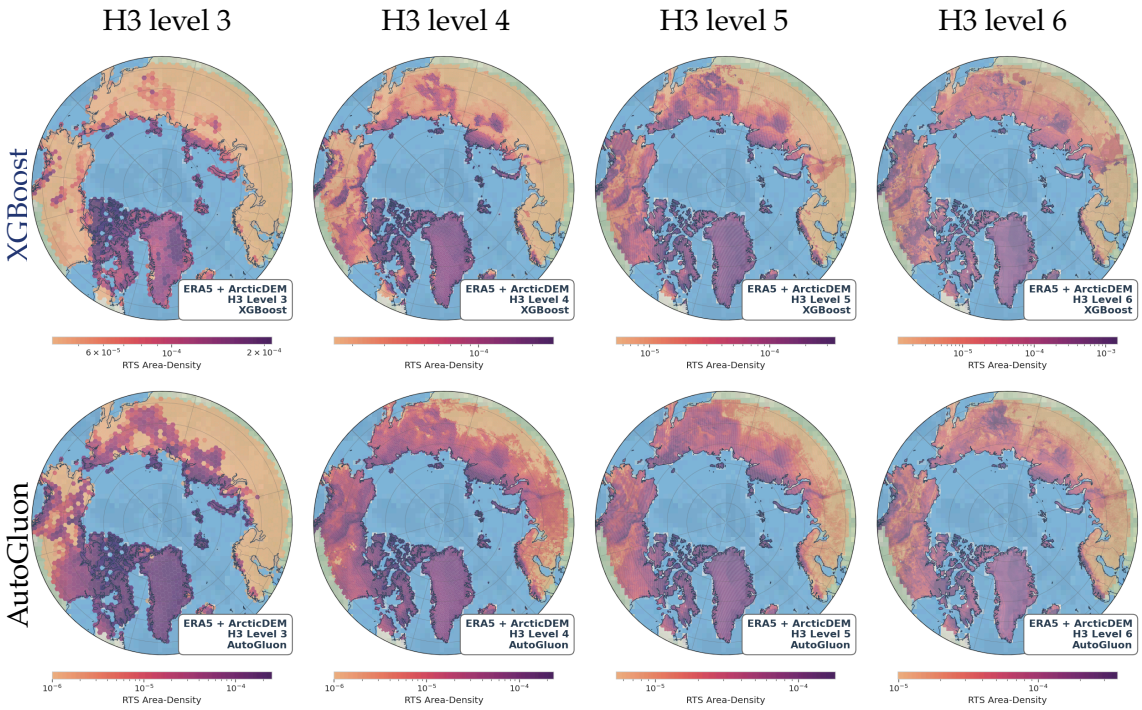

Figure 47 : Retrogressive Thaw Slump area-density predictions across scales (columns) from different models (rows): XGBoost (top row) and AutoGluon (bottom row). Models were trained with the baseline approach, meaning on all features from the ERA5-Land and ArcticDEM DGGS aligned datasets. Note that color scales are adapted to enhance the visibility of spatial features using individual logarithmic scales, making colors across different maps non-comparable.

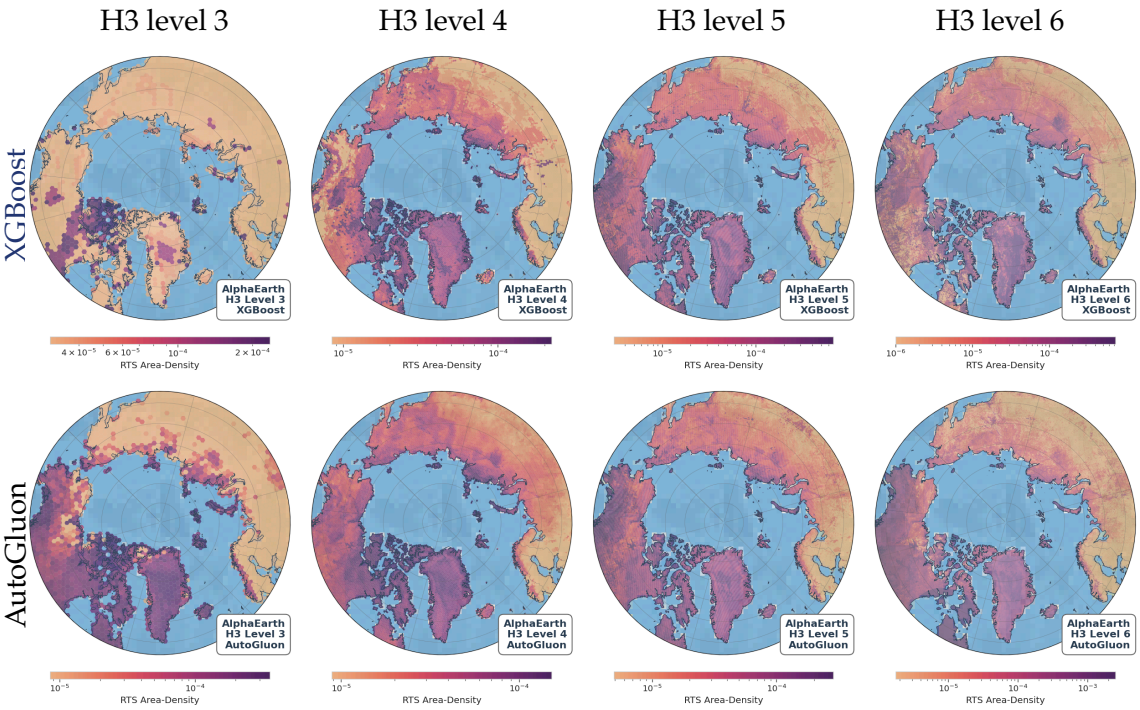

Figure 48 : Retrogressive Thaw Slump area-density predictions across scales (columns) from different models (rows): XGBoost (top row) and AutoGluon (bottom row). Models were trained on the DGGS aligned AlphaEarth satellite embeddings. Note that color scales are adapted to enhance the visibility of spatial features using individual logarithmic scales, making colors across different maps non-comparable.

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I would like to thank all the people who have helped we on my journey. Dear Ingmar, thank you so much for inviting me into this research group and giving me the possibility to prove myself as a scientist. Rarely has my life been influenced this positively. Dear Guido, thank you so much for creating a trustful work environment and being the fair leader that you are. This is not given. I further want to thank the team at AWI which helped be a lot during the time writing this thesis: Carl for your general advocacy and your support in statistics, Nina for our Café walks and showing me the world of thaw slumps, Cornelia for your drive to do things the right way and our long discussions about finding these ways, Jan for introducing me into the modelling aspects of permafrost, Sasha for explaining me Yedoma and especially Jonas for your support and having my back on our project so that I could focus on this thesis. Last but not least I want to thank my family and friends for their ongoing moral support and patience within the last months, especially Simon and Phia.

Statement of Originality

Several language tools were used in this work to improve readability and to reduce spelling and grammar mistakes. These include the "LanguageTool" and "DeepL" browser plugins and the use of several Large-Language-Models (LLM), hosted by Holmholtz BlaBlaDor and locally runned with LMStudio and JanAI. All thoughts in this paper are mine and do not originate by an LLM or any other computer software.

I hereby declare that this thesis is the product of my own work. All the assistance received in preparing this thesis and the sources used have been acknowledged.

Berlin, 1st March 2026

Signature Author